The Pandemic Trilemma

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Abstract

Pandemics confront governments with a trilemma between excess mortality, output stabilization, and fiscal sustainability. OECD responses to COVID-19 differed sharply in scale, composition, and timing—as did their outcomes. This paper asks what the optimal response would have been, and what deviating from it cost. I construct a database of 2,301 fiscal measures across 38 OECD economies, classified by transmission mechanism: demand injection, above-the-line capacity preservation, and below-the-line liquidity support. Exploiting this heterogeneity, I estimate a transition system linking instruments, containment, output, infections, and debt, and evaluate observed policies against a weight-free constrained policy frontier. The OECD response was strictly Pareto-dominated: policies within the observed menu would have reduced the output loss by 79 percent, excess mortality by 22 percent, or terminal debt by 6.3 percentage points of GDP; no economy operated on its own frontier. The gains come from composition and timing, not scale: front-loaded liquidity and capacity preservation dominate, while demand injection under binding restrictions adds debt without output.

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1 Introduction

What had appeared in late 2019 as a marginal news story turned, within weeks, into a worldwide shock as a rapidly spreading and deadly virus, soon known as COVID-19. By March 2020, governments responded with stay-at-home orders, business closures, restrictions on movement, and severe limits on international travel. Trade, mobility, and everyday economic activity were disrupted on a scale not seen since World War II (OECD, 2020; World Bank, 2020), with excess mortality rising sharply and output gaps reaching up to -25 percentage points in the most affected countries. Since the situation was unprecedented, governments had no common policy template to follow. Countries responded with highly heterogeneous combinations of containment measures and fiscal support. These strategies differed not only in size, but also fundamentally in composition: some governments concentrated support on preserving firms and employment relationships, while others relied heavily on transfers and demand stimulus. Understanding these choices and their impact requires a framework that captures the joint constraints faced by governments seeking simultaneously to protect public health, preserve economic stability, and maintain fiscal sustainability. To understand how these objectives interact, it is necessary to consider the particular structure of the pandemic environment. Four properties are central to this structure. First, once introduced exogenously, the viral prevalence of a lethal pathogen grows endogenously through human contact. Second, in the absence of an effective vaccine, the main available suppression instruments—lockdowns, social distancing mandates and other non-pharmaceutical interventions (NPIs)—protect public health by reducing interaction, but this same mechanism contracts economic activity. Third, fiscal instruments can compensate for the resulting output loss, but their effects are heterogeneous and they do so at the cost of public debt accumulation. Fourth, even without discretionary fiscal intervention, the output contraction itself erodes the fiscal position through automatic stabilizers, generating liabilities independently of policy choices.

Taken together, these properties make the three objectives mutually constrained by the structural architecture of the pandemic environment. This gives rise to a *trilemma*, in the economic sense of the Mundell-Fleming impossible trinity (Mundell, 1963): no feasible policy of a benevolent social planner can simultaneously minimize mortality, output losses, and debt accumulation. The planner's problem is therefore twofold: how strongly to suppress the virus and how to compose the accompanying fiscal intervention across instruments with fundamentally different transmission channels. Demand injection—for instance, direct transfers and cash payments—supports aggregate expenditure during containment and generates a demand-side output effect. Capacity preservation—such as wage subsidies that maintain employment relationships, loans, and guarantees that prevent firm insolvency—limits the persistence of the contraction by preventing the destruction of structures like organizational capital, supplier networks, and job-specific human capital.

These instruments differ in their mechanisms, fiscal costs, and dynamic profiles. The optimal fiscal response therefore depends not only on the aggregate size of support, but also on its composition and timing. As the pandemic state evolves, this composition should adjust accordingly, turning the social planner's problem into an intertemporal dynamic optimization problem.

A broad literature has formalized important parts of this problem. On the macroeconomic side, Atkeson (2020) introduce the baseline SIR structure into a macroeconomic setting, Alvarez et al. (2020) solve for the optimal lockdown trajectory of a planner trading off fatalities against output costs, and Eichenbaum et al. (2021) endogenize the behavioral recession by embedding epidemiological dynamics into macroeconomic equilibrium. Related work studies targeted containment policies (Acemoglu et al., 2021) and endogenous labor-supply responses (Jones et al., 2021). On the empirical side, a large literature estimates the effect of non-pharmaceutical interventions on economic activity across countries (Ashraf and Goodell, 2022; König and Winkler, 2020; Cross et al., 2020; Gordon et al., 2021), documenting the contractionary impact of containment but generally treating fiscal policy as absent, given, or secondary. A second strand studies the fiscal response to COVID-19 more directly. Chetty et al. (2020) show that U.S. direct transfers like demand injection had limited spending effects during containment; Autor et al. (2022) estimate the employment effects of the Paycheck Protection Program; Pappa et al. (2022) analyze the effects of on-budget fiscal measures on growth; and Faria-e Castro (2021) show in a nonlinear DSGE model that the pandemic shock changes the ranking of fiscal multipliers, with liquidity assistance dominating, The most comprehensive cross-country evidence is provided by Deb et al. (2021), who abstract from fiscal sustainability but find that what they call emergency-lifeline instruments outperform income-support measures during strict containment, while the ranking reverses upon reopening. The closest contribution to a trilemma formulation is Kaplan et al. (2020), who integrate SIR dynamics into a heterogeneous-agent model with both lockdown and fiscal transfers. Their analysis constructs a pandemic possibility frontier and shows how fiscal transfers affect the distributional incidence. Yet fiscal policy enters as undifferentiated lump-sum transfers, while government debt appears as a residual accounting outcome rather than as a state variable in the planner's objective. Two gaps emerge from this literature. First, existing work primarily studies a health–output dilemma, while fiscal sustainability is rarely treated as a co-equal objective in cross-country comparison. Second, fiscal policy is often represented as an aggregate instrument, although pandemic fiscal packages differed sharply in their transmission channels, timing, and budgetary pass-throughs. Therefore, the central research question is: *What is the model-implied optimal composition and timing of fiscal support between demand injection and capacity preservation over the course of a pandemic and what are the costs—in foregone output stabilization and excess debt—of deviating from this benchmark?*

So the paper makes four contributions. First, it develops a theoretical framework that formalizes pandemic policy as a dynamic trilemma between excess mortality, the output gap, and public debt. Second, it introduces a new measure-level database and a transmission-based classification of pandemic fiscal policy. Rather than treating fiscal support as a homogeneous aggregate, I distinguish between demand injection, capacity preservation, and health-related spending. Third, I use the measure-level database to empirically assess the transition channels and parameterize the system. Fourth, I use the estimated transition system to solve a finite-horizon planner problem. This converts the empirical transition parameters into a model-implied benchmark and allows me to evaluate and compare observed country strategies. The analysis is relevant beyond COVID-19 because pandemics differ fundamentally from standard demand-driven recessions. Understanding how fiscal instruments perform when economic activity is constrained by supply-side or mobility restrictions can inform policy design in future crises with similar characteristics. The findings also speak to current fiscal-policy debates, as the debt accumulated during the pandemic now interacts with higher borrowing costs and reduced fiscal space.

The analysis yields four sets of findings. First, the trilemma is a structural property of the pandemic environment, not an assumption. Within the estimated transition system, public debt is dynamically unstable and cannot be stabilized by any available instrument, containing the epidemic requires stringency high enough to depress output, and no fiscal instrument is self-financing; every instrument leaves a positive net debt footprint. Each stabilizing action on one dimension therefore feeds the instability of another.

Second, every leg of the trilemma is active in the data. The panel estimates identify all relevant parameters in the transition system with the expected signs: containment reduces infection transmission but depresses output; the output losses feed into public debt; and every fiscal instrument stabilizes output only at a positive net debt cost. Beyond their existence, the transmission channels differ sharply, and one of them is state-dependent in a way that proves central: the output effect of demand injection declines with containment and turns negative once stringency exceeds a threshold. Capacity-preserving support—above-the-line programs and, above all, liquidity instruments such as loans and guarantees, operating through persistent stocks—retains its effectiveness under containment. Pandemic fiscal policy is thus not a question of aggregate size: the same unit of support has fundamentally different output effects depending on the instrument and the containment state in which it is deployed.

Third, judged against the model-implied benchmark, the observed average OECD response was strictly Pareto-dominated: holding the other two dimensions at their observed levels, a feasible policy path—using no instrument, scale, or adjustment speed beyond what OECD governments demonstrably implemented—would have reduced the discounted output loss by 79 percent, excess mortality by 22 percent, or terminal debt by 6.3 percentage points of GDP. The dominating policies front-load fiscal support, shift it toward effective capacity preservation with loans, guarantees and wage subsidies, retime containment into the acute waves, and abandon demand injection entirely. So the gains come from composition and timing, not from additional fiscal volume.

Fourth, taking the benchmark to the country level shows that no OECD economy operated on its own frontier. The countries closest to their frontier pursued precisely the front-loaded, liquidity-oriented, low-demand-injection packages identified by the benchmark, whereas those furthest away relied on large, back-loaded, demand-injection-heavy programs whose scale yielded little at the margin. The realized fiscal exchange rate—the output stabilization gained per percentage point of fiscally induced debt—compresses the difference into one statistic: countries whose packages tracked the benchmark earned the highest returns (Norway 3.7, Switzerland 3.6, and Sweden 3.1 pp-quarters per point of debt, against a median of 2.8), while the United States (1.1), Isreal (-0.5) and Chile (-0.2) anchor the bottom. The return falls sharply in the demand-injection share of the observed package (correlation -0.89) and turns negative in exactly the four countries with the highest such shares, consistent with the model-implied ranking.

The paper proceeds as follows. Section 2 establishes the formal framework of the pandemic trilemma, introduces the relevant state variables, controls, and transition dynamics, and demonstrates the structural uniqueness. Section 3 introduces the measure-level database and the construction of the key variables. Section 4 estimates the transition parameters, and Section 4.3 calibrates and validates the resulting system. Section 5 formulates the planner's optimization problem, and Section 6 presents the normative results: the planner benchmark, the constrained policy frontier, and the cross-country assessment. Section 7 concludes.

2 The Pandemic Trilemma

2.1 Constitutive Properties

Property 1: Exogenous Shock with Endogenous Dynamics

A pandemic begins with the introduction of a novel pathogen into the population. I denote the pandemic pressure at time k by θ_k, which I refer to as the viral prevalence. The initial outbreak constitutes an exogenous shock—it originates outside the economic system and is not the result of endogenous market dynamics or policy failure. I represent this shock by a stochastic disturbance ε_k.

The defining characteristic of a pandemic shock is its endogenous growth dynamic: in the absence of intervention, each infected individual transmits the virus, causing the number of infections to grow over time. I capture this by a persistence parameter $\rho_\theta > 1$ (similar to the logic of the reproduction rate during the pandemic), implying that viral prevalence grows exponentially if left unchecked (Atkeson, 2020; Alvarez et al., 2021; Eichenbaum et al., 2021).[1] It is the lethality of the pathogen that transforms this explosive dynamic into a policy imperative: a non-lethal pathogen with $\rho_\theta > 1$ would spread but impose no mortality cost of sufficient magnitude to warrant contractionary public health interventions (Jones et al., 2021; Barro et al., 2020), thereby establishing the necessity of active policy intervention at the cost of economic contraction (Property 2).

This property distinguishes the pandemic shock qualitatively from an impulse shock—where the disturbance dissipates naturally over time ($\rho < 1$) and the system returns to equilibrium without active intervention (Jordà et al., 2022). These potential consequences are captured by the state variable: *excess mortality* (d_k), defined as the deviation of the periodic death rate from its historical seasonal average encompassing both direct COVID-19 deaths and indirect deaths attributable to avoidance of care and healthcare system overload (Msemburi et al., 2023; Mathieu et al., 2020). I denote the parameter governing the transmission from infections to deaths by δ_θ. Crucially, the biological lag between infection and death implies that mortality in period $k+1$ is already determined by the current viral prevalence θ_k and thus predetermined from the planner's perspective. Consequently, the planner's intervention S_k—which suppresses transmission to θ_{k+1}—only impacts mortality starting from period $k+2$. This relationship is given by:

$$d_{k+2} = \delta_\theta \, \theta_{k+1}, \qquad \text{with} \quad \theta_{k+1} = \rho_\theta \theta_k + \varepsilon_k \tag{1}$$

The interaction between the lethality parameter δ_θ and the transmission potential ρ_θ shows the decisive role described above. If $\delta_\theta = 0$—that is, if the virus were non-lethal—or

[1] The linear specification abstracts from the susceptible-depletion mechanism of the full SIR framework, where the effective reproduction rate declines endogenously as the susceptible population shrinks. This linearization is appropriate for the model horizon (2020–2021)

if the pathogen were not transmissible ($\rho_\theta = 0$), there would be no reason for the planner to suppress θ_{k+1}.

Property 2: Suppression Instrument

Given the necessity of intervention established by Property 1, the planner's primary tools to suppress viral transmission during a pandemic are non-pharmaceutical interventions (NPIs): measures including business closures, restrictions on gatherings, school closings, restrictions on internal movement as well as international travel controls and stay-at-home orders. I denote the intensity of these measures at time k by S_k, where $S_k = 0$ represents the absence of restrictions and $S_k = 100$ represents a complete lockdown in every aspect.

As NPIs suppress viral prevalence by reducing human contact (Flaxman et al., 2020), I denote the effectiveness of this suppression by the parameter $\phi_S > 0$. Crucially, NPIs do not remove a fixed number of infections from the population; rather, they reduce the rate at which existing infections generate new ones. Each unit of suppression S_k reduces the transmission potential ρ_θ by the fraction ϕ_S, yielding an effective reproduction rate $\rho_\theta^{\text{eff}}(S_k) = \rho_\theta(1 - \phi_S S_k)$. Combining this with the mortality transmission from Property 1, the full two-period channel from policy action to health outcome is:

$$d_{k+2} = \delta_\theta \cdot \underbrace{\left[\rho_\theta(1 - \phi_S S_k)\,\theta_k + \varepsilon_k\right]}_{\theta_{k+1}} \tag{2}$$

This equation encodes two features that are central to the planner's problem. First, the suppression effect is multiplicative in the current level of prevalence: at high θ_k, each unit of lockdown prevents more infections—and therefore more deaths—than at low θ_k. The marginal health benefit of suppression, $\partial d_{k+2}/\partial S_k = -\delta_\theta \rho_\theta \phi_S \theta_k$, is state-dependent and vanishes as the pandemic recedes. Second, the planner does not control the infection level or the death rate directly but only the growth rate of the epidemic. When $\phi_S S_k$ is sufficient to push ρ_θ^{eff} below unity, infections decline; otherwise, they continue to grow despite the intervention. This threshold property—$\rho_\theta^{\text{eff}} < 1$ requires $S_k > \frac{1-1/\rho_\theta}{\phi_S}$—implies a minimum lockdown intensity below which suppression is ineffective at reversing the epidemic trajectory.

The central feature of Property 2 is that the channel through which NPIs suppress viral transmission simultaneously reduces the volume of economic activity. Containment measures close businesses, idle labor, and curtail the flow of goods, services, and persons that constitutes aggregate economic output. This duality is inherent to the nature of any instrument that operates through contact reduction.

To formalize this, I introduce the next state variable *output gap* (y_k): the percentage deviation of real GDP from its potential trend, where $y_k = 0$ represents normal economic conditions and $y_k < 0$ represents a recession. The contractionary effect of NPIs on output is governed by the parameter $\alpha_S > 0$. Every unit of suppression $S_k > 0$ therefore simultaneously implies some values of:

$$\text{Epidemiological benefit:} \qquad \Delta\theta = -\phi_S\, S_k < 0 \tag{3}$$

$$\text{Economic cost:} \qquad \Delta y = -\alpha_S\, S_k < 0 \tag{4}$$

The suppression instrument cannot control the epidemiological shock without simultaneously affecting the economic state. Beyond the direct contractionary effect of mandated containment, the pandemic generates an additional output cost through the behavioral response. When individuals observe rising death tolls, they voluntarily reduce economic activity independently of any government-mandated restrictions (Goolsbee and Syverson, 2021) making it a do or die strategy for the social planner. I capture this channel through the parameter $\beta_d < 0$, which governs the output response to the mortality flow (d_k) and is commonly referred to as the "fear term".

A natural objection to Property 2 is that alternative instruments, such as testing, might control the pandemic without suppressing economic activity, thereby dissolving the contractionary link. However, testing capacity constitutes a complement to, rather than a substitute for, NPIs (Eichenbaum et al., 2020). Critically, testing does not alter the biological parameters driving the pandemic: it reduces neither the inherent transmissibility (ρ_θ) nor the lethality (δ_θ) of the virus. Instead, it operates exclusively through the suppression channel S_k by identifying infected agents. Ideally, this improves the ratio of epidemiological benefit (ϕ_S) to economic cost (α_S) by replacing broad lockdowns with specific quarantines. However, testing cannot eliminate the contractionary effect since even targeted isolation disrupts economic transactions. Moreover, the net effect on the efficiency ratio is ambiguous as higher detection rates may intensify responses.

Vaccination, by contrast, is the instrument that dissolves the trilemma by invalidating Property 2. It operates through both channels, reducing viral transmission (lowering ρ_θ) and reducing the lethality of breakthrough infections (lowering δ_θ), which, as shown above, is the critical mechanism. Crucially, it achieves it without operating through the suppression channel S_k—and thus, no restriction of economic activity (Giordano et al., 2021), thereby eliminating the contractionary link. However, effective vaccines were not widely available until mid-2021, and the critical vaccination thresholds required to sustainably decouple mortality from viral prevalence were not reached in most OECD economies until the end of 2021 (OECD, 2021a). Consequently, for the main phase of the pandemic (Q1 2020–Q4 2021)—the period during which critical policy decisions were taken (Hale et al., 2021)—NPIs remained the only available suppression instrument, and were thus inherently contractionary.

Figure 1 illustrates on the OECD average this mechanism. During Waves 1–3, infection prevalence and excess mortality move in parallel, and governments maintain elevated containment. With the onset of Wave 4 (Omicron, late 2021), the relationship breaks: infection prevalence reaches its pandemic peak, yet excess mortality declines sharply—reflecting the reduction in δ_θ achieved by widespread vaccination. Governments respond by withdrawing containment to near-zero levels despite record case counts. This decoupling is precisely the condition $\delta_\theta \to 0$ formalized above. This observation defines the planner's horizon: the trilemma is operative from Q1 2020 through Q4 2021. Beyond the terminal date, the planner deals exclusively with the fiscal legacy of the crisis.

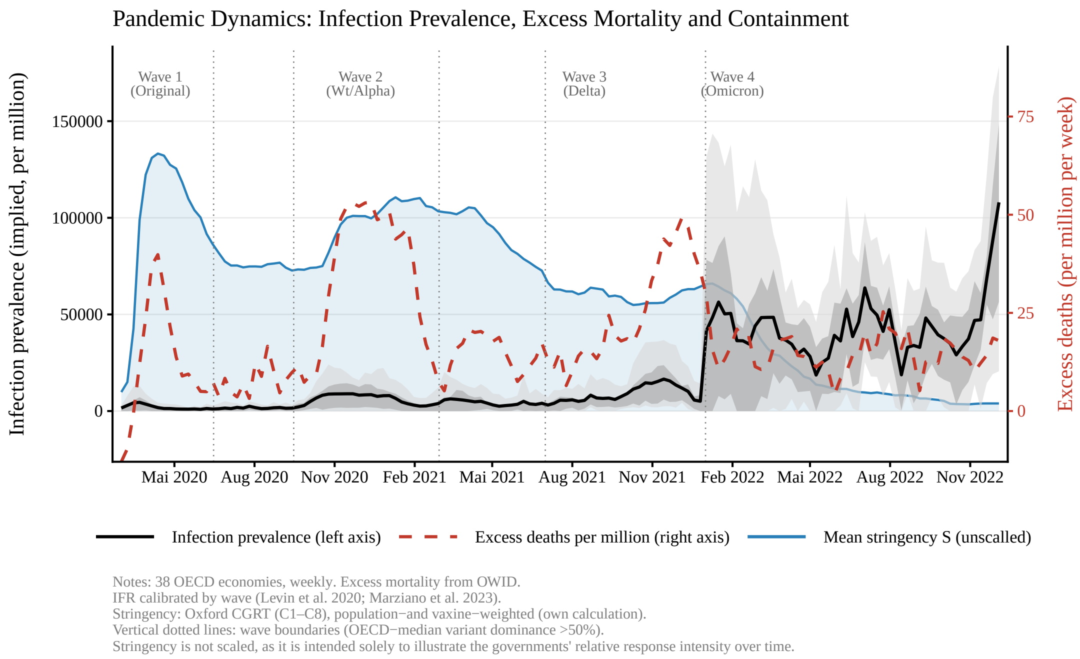


Figure 1: Pandemic Dynamics: Infection Prevalence, Excess Mortality, and Containment

Property 3: Fiscal Compensation Instruments

Given that pandemic suppression is inherently contractionary (Property 2), the social planner requires compensating instruments to stabilize the economic state. The pandemic damages the economy through two main channels. Containment and behavioral retrenchment directly reduce the level of output by restricting economic activity and suppressing expenditure. Simultaneously, the prolonged disruption erodes the economy's capacity to recover—through firm failures, dissolution of employment relationships or disruption of supply chains—prolonging the contraction beyond the duration of the shock itself. To address these two channels, the social planner has to target these channels separately.

The first category, I frame it as demand injection (F_k^{DI}). Those comprises measures that directly increase the flow of disposable income in the economy (level): stimulus checks (such as the U.S. Economic Impact Payments under the CARES Act), direct transfers to households (such as the Australian JobSeeker supplement or the Japanese universal cash transfer of ¥100,000 per resident), and tax reductions that raise disposable income (such as the German temporary VAT reduction from 19% to 16% in H2 2020). These instruments transmit through the standard Keynesian multiplier channel and are widely used, shifting the output gap by $\alpha_{DI} > 0$ per unit deployed. [2]

However, applying these instrument to a pandemic environment is questionable on both theoretical and practical grounds. The pandemic recession is not a demand shock but a supply disruption induced by containment (Guerrieri et al., 2022): businesses are closed by government order, not by insufficient demand. Injecting purchasing power into an economy where the supply of goods and services is physically restricted cannot translate into expenditure at the usual rate (Chetty et al., 2020; Coibion et al., 2020). This raises the question whether α_{DI} under pandemic conditions retains a meaningful magnitude, is merely delayed, or vanishes entirely.

The second category, capacity preservation (F_k^{CP}), comprises measures that maintain the economy's productive structure rather than directly raising contemporaneous demand: short-time work schemes, such as the Swiss *Kurzarbeitsentschädigung* or the UK Coronavirus Job Retention Scheme; payroll subsidies; direct grants; and government-backed liquidity support, such as the French *Prêts Garantis par l'État* or the German KfW-Sonderprogramm. When containment closes firms, CP does not immediately restore output; instead, it prevents temporary disruptions from becoming persistent losses by preserving firm-worker matches, supplier relationships, and organizational capital—the amplification mechanism emphasized by Guerrieri et al. (2022).

Capacity preservation can be further decomposed into above-the-line (direct budget outlay) and below-the-line instruments (balance-sheet transactions). Above-the-line measures, such as short-time work schemes, payroll subsidies, and direct grants, operate as fiscal flows that replace lost income or revenue. Below-the-line measures and contingent liabilities, such as loans and guarantees, operate through liquidity provision; their effect arises from their interaction with above-the-line measures and the cumulative stock of available support.

Combining the pandemic shocks from Property 2 with the fiscal instruments above, the output evolves according to:

$$y_{k+1} = \rho_y\, y_k + \alpha_S\, S_k + \alpha_{\mathrm{above}}\, F_{k-2}^{\mathrm{above}} + \alpha_{\mathrm{below}}\, \bar{F}_k^{\mathrm{below}} + \alpha_{\mathrm{DI}}\, F_{k-1}^{\mathrm{DI}} + \alpha_{S,\mathrm{DI}}\, S_k F_{k-1}^{\mathrm{DI}} - \beta_d\, d_k \tag{5}$$

where $\bar{F}_k^{\mathrm{below}} = \sum_{s \le k}(\tilde{F}_s^{\mathrm{loans}} + \tilde{F}_s^{\mathrm{guar}})$ denotes the cumulative below-the-line stock at quarter k.

Equation (5) captures several distinct mechanisms.

First, output dynamics are governed by the autoregressive coefficient ρ_y, which captures persistence in the output gap. Previously implemented stringency is therefore ab-

[2]Pre-pandemic estimates place expenditure multipliers in a range of approximately 0.5 to 1.5 (Ramey and Zubairy, 2018; Blanchard and Perotti, 2002).

sorbed into this term through the lagged state y_k. Containment policy S_k enters additively with a contemporaneous level effect $\alpha_S < 0$.

Second, capacity-preservation measures enter through two distinct channels. Above-the-line measures enter with a two-period lag through F_{k-2}^{above} and are governed by the coefficient α_{above}. These measures substitute income or revenue streams, help maintain income, and reduce the persistence of the output shock. Below-the-line measures enter through the cumulative stock variable $\bar{F}_k^{\text{below}}$, with coefficient α_{below}. This reflects the fact that loans and guarantees operate less through contemporaneous expenditure flows than through the availability of liquidity. The relevant mechanism is therefore the accumulated amount of loans and guarantees available up to quarter k, which helps reduce liquidity shortages and thereby limits structural damage.

Third, demand-injection measures enter with a one-period lag through F_{k-1}^{DI} and operate through the usual multiplier effect, governed by α_{DI}. Their effectiveness, however, depends on the current restrictiveness of the economy. This state dependency is captured by the interaction term $S_k F_{k-1}^{\text{DI}}$, with coefficient $\alpha_{S,\text{DI}}$.

Finally, mortality losses enter through d_k with coefficient $-\beta_d$, capturing the adverse effect of excess deaths on economic activity.

The preceding decomposition might suggest a straightforward resolution of the output–health trade-off: deploy fiscal support to offset the lockdown-induced contraction, then exit. However, this reasoning abstracts from the budgetary consequences of the fiscal responses. Each instrument that stabilizes the output gap carries fiscal consequences for the government's balance sheet—and the magnitude of this fiscal cost is heterogeneous across categories and across several dimensions that is not yet established in the literature. To formalize this constraint, I introduce the third state variable: the *pandemic debt gap* (b_k), defined as the deviation of real gross public debt from its country-specific pre-pandemic level, normalized by 2019 GDP.[3]

$$b_{k+1} = (1+r)\, b_k - \gamma_y\, y_k + \kappa_{\text{above}}\, F_k^{\text{above}} + \kappa_{\text{loans}}\, F_k^{\text{loans}} + \kappa_{\text{guar}}\, F_k^{\text{guar}} + \kappa_{\text{DI}}\, F_{k-1}^{\text{DI}} + \kappa_H F_{k-1}^{H} \tag{6}$$

where r is the quarterly sovereign interest rate, $\gamma_y > 0$ captures the automatic stabilizer channel, and the κ coefficients are instrument-specific budgetary pass-throughs into the pandemic debt gap. Since $y_k < 0$ represents a negative output gap, the term $-\gamma_y y_k$ increases b_{k+1} through lower revenues and higher automatic transfers.

The split of capacity-preservation measures in the debt equation reflects their different fiscal footprints. Above-the-line measures, such as short-time work schemes, payroll subsi-

[3] I normalize by 2019 GDP rather than contemporaneous GDP to avoid mechanical denominator effects from the recession. The state b_k therefore captures changes in the real stock of government liabilities relative to the country's own pre-pandemic debt position, rather than cross-country differences in inherited debt levels.

dies, and direct grants, generate immediate budgetary outlays, although these may partly operate through existing fiscal structures. Loans also create immediate fiscal exposure because the state extends new credit and must finance the lending operation, even though it receives a financial claim on the borrower in return. Guarantees, by contrast, do not generate an immediate disbursement but create an immediate contingent liability whose realized fiscal cost depends on the default rate among supported borrowers (International Monetary Fund, 2022).

Demand injection and health spending enter with a one-period lag because their budgetary impact may arise only after transfers are paid, tax relief affects revenues, procurement is settled, or reimbursement claims are recorded. For demand injection, this timing also aligns the debt cost with the period in which the measure becomes economically effective. Since the marginal output effect in equation (5) is $\alpha_{\mathrm{DI}} + \alpha_{S,\mathrm{DI}} S_k$ with $\alpha_{S,\mathrm{DI}} < 0$, the output return to demand injection falls with containment intensity, while its debt cost remains positive. Demand injection is therefore less attractive when containment is high. Health spending, F^H_{k-1}, follows the same lagged fiscal-pass-through logic, but enters only as an exogenous debt component because it is not part of the planner's optimized fiscal choice set (assumed to be binding).

The optimal fiscal mix therefore depends jointly on three margins: the prevailing level of stringency, which determines the effectiveness of demand injection; the composition of the capacity-preservation package, which determines the fiscal cost and the cumulative fiscal legacy of past decisions, which affects the debt state b_k. The problem is therefore genuinely dynamic: the planner does not simply choose the volume of fiscal support, but must allocate support across instruments whose output effects and fiscal costs differ across states and over time.

Property 4: Endogenous Fiscal Feedback via Automatic Stabilizers

While Property 3 introduced the direct budgetary costs of discretionary fiscal instruments, Property 4 captures the endogenous fiscal feedback generated by the recession itself. When NPIs contract economic output (Property 2), the resulting negative output gap triggers automatic stabilizer effects. The parameter $\gamma_y > 0$ in the debt transition (6) captures the aggregate sensitivity of the budget to the output gap.[4]

Because this term operates symmetrically, it has a direct implication for instrument choice: a fiscal instrument whose output effect is large relative to its fiscal cost generates a double dividend—it stabilizes output directly and attenuates the automatic fiscal deterioration indirectly, as the improved y_k feeds back through γ_y to slow debt accumulation.

The central contribution of this property is to close the trilemma by establishing that fiscal abstinence is not a feasible strategy (Fornaro and Wolf, 2020). Setting all

[4]For OECD economies, estimates of the budget semi-elasticity range from approximately 0.35 to 0.65, with a cross-country average around 0.5 (Price et al., 2015).

discretionary instruments to zero ($F_k^{DI} = F_k^{CP} = 0$) in (6) yields:

$$b_{k+1} = (1+r)\, b_k - \gamma_y\, y_k \tag{7}$$

Since $y_k < 0$ during containment, $-\gamma_y y_k > 0$: the pandemic debt gap increases even without any discretionary fiscal expenditure. A lockdown without accompanying fiscal measures still erodes the fiscal position—the trilemma cannot be circumvented by fiscal abstinence. Combined with the necessity of suppression established in Property 1, this means that the planner's decision is not whether to deploy fiscal support but which composition to deploy and when—returning the problem to the instrument choice formalized in Property 3.

The four properties jointly characterize and formalized the structural environment of the pandemic trilemma that define this unique situation government faced.[5] The epidemiological block establishes the need for suppression; the suppression instrument improves the health state only by reducing economic activity; fiscal instruments can compensate the output loss but differ in their transmission channels and fiscal costs; and the debt state makes the problem dynamically constrained through both discretionary spending and automatic stabilizers. The planner therefore faces a joint transition problem over output, the pandemic debt gap, viral prevalence, and excess mortality. Therefore, the resulting transition system is:

$$\begin{aligned} y_{k+1} &= \rho_y\, y_k + \alpha_S\, S_k + \alpha_{\text{above}}\, F_{k-2}^{\text{above}} + \alpha_{\text{below}}\, \bar{F}_k^{\text{below}} \\ &\quad + \alpha_{\text{DI}}\, F_{k-1}^{\text{DI}} + \alpha_{S,\text{DI}}\, S_k F_{k-1}^{\text{DI}} - \beta_d\, d_k \end{aligned} \tag{8}$$

$$\begin{aligned} b_{k+1} &= (1+r)\, b_k - \gamma_y\, y_k + \kappa_{\text{above}}\, F_k^{\text{above}} + \kappa_{\text{loans}}\, F_k^{\text{loans}} \\ &\quad + \kappa_{\text{guar}}\, F_k^{\text{guar}} + \kappa_{\text{DI}}\, F_{k-1}^{\text{DI}} + \kappa_{\text{H}} F_{k-1}^{H} \end{aligned} \tag{9}$$

$$\theta_{k+1} = \rho_\theta\, (1 - \phi_S\, S_k)\, \theta_k + \varepsilon_k \tag{10}$$

$$d_{k+1} = \delta_\theta\, \theta_k \tag{11}$$

The transition system is deliberately reduced-form. Each channel has a microfounded counterpart in the pandemic macro literature, but the paper does not claim to identify policy-invariant deep parameters. Its purpose is to discipline a local retrospective benchmark within the realized OECD policy space. The reduced-form specification allows the transition parameters to be estimated directly and makes it possible to assess whether the state-transition structure implied by the theoretical framework is supported by the data.

[5] An application of the four constitutive properties to four alternative crisis types: armed conflict, natural disasters, financial crises, and climate change is showed in Appendix B.

The underlying mechanisms are well established in the pandemic-macro literature, although these mechanisms are typically studied separately. Guerrieri et al. (2022) and Bayer et al. (2023) show that a sectoral supply shock can generate an endogenous demand shortfall and that standard transfers lose effectiveness while supply constraints bind, corresponding to the state-dependent demand-injection multiplier and the persistence channel of capacity preservation. Baqaee and Farhi (2022) show that COVID-19 combined supply and demand shocks across sectors and that these shocks propagated, reinforcing the need to distinguish fiscal instruments by transmission channel rather than by aggregate size alone. Faria-e Castro (2021) finds in a nonlinear DSGE model calibrated to the U.S. labor market that liquidity assistance to firms is more effective than household transfers consistent with the predicted ranking of capacity preservation over demand injection under high stringency. Similarly, Céspedes et al. (2022) emphasize the role of firm-level financial frictions and productivity losses, providing a microfounded rationale for liquidity-preserving interventions. Eichenbaum et al. (2021) derive the behavioral recession from optimizing agents who reduce consumption and labor supply to lower infection risk, the microfounded counterpart of the fear term. Finally, Kaplan et al. (2020) show in a heterogeneous-agent SIR economy that containment policy determines the health–output trade-off, while fiscal transfers mainly shape the distributional and aggregate economic consequences of that trade-off. Therefore, the reduced-form system does not invent its transmission channels; rather, it combines mechanisms derived separately in the existing literature within a three-dimensional state-transition framework with explicit trade-offs. This makes the problem tractable and allows me to analyze which strategies stabilized output most effectively relative to their fiscal costs and the prevailing public-health environment.

3 Empirical Parametrization

The purpose of this section is to bring the transitions system to the data. The resulting estimates are not interpreted as policy-invariant causal effects, but as empirical transition parameters within the realized pandemic policy space. They provide the numerical inputs required to solve the planner problem and to evaluate observed country strategies against the resulting benchmark. Table 1 summarizes the target variables, state variables, and transition-system parameters that need to be measured or parameterized.

Symbol	Description	Block
A. Planner objective and target variables		
y_k	Output gap; measured as the percentage deviation of real GDP from its pre-pandemic potential trend	Target
d_k	Excess mortality; measured as the deviation of deaths from the historical seasonal baseline	Target
b_k	Pandemic debt-gap state; measured as the deviation of real gross public debt from the country-specific pre-pandemic level, normalized by 2019 GDP	Target
θ_k	Viral prevalence; epidemiological state that affects welfare only through future excess mortality	State
B. Transition-system parameters		
ρ_y	Persistence of the output gap	Output
α_S	Contemporaneous effect of stringency S_k on output	Output
α_{above}	Lagged effect of above-the-line fiscal spending on output	Output
α_{below}	Effect of the cumulative below-the-line liquidity stock on output	Output
α_{DI}	Lagged baseline effect of demand injection on output	Output
$\alpha_{S,\text{DI}}$	State-dependent interaction between stringency and demand injection	Output
β_d	Effect of observed excess mortality on output through the behavioral response	Output
r	Quarterly sovereign interest rate	Debt
γ_y	Budget semi-elasticity with respect to the output gap	Debt
κ_{above}	Fiscal pass-through of above-the-line spending into public debt	Debt
κ_{loans}	Fiscal pass-through of loan flows into public debt	Debt
κ_{guar}	Fiscal pass-through of guarantee flows into public debt	Debt
κ_{DI}	Fiscal pass-through of demand injection into public debt	Debt
ρ_θ	Persistence of viral prevalence in the absence of containment	Infection
ϕ_S	Effectiveness of stringency in reducing viral transmission	Infection
δ_θ	Transmission from viral prevalence to excess mortality	Mortality

Table 1: Target variables and transition-system parameters

3.1 Data and Measurement

I use a quarterly panel dataset covering all 38 OECD member states. The underlying data span Q1 2015 through Q4 2024, while the main estimation sample focuses on the pandemic period during which the trilemma was binding. The dataset combines macroeconomic outcomes, containment policy, epidemiological variables, and a newly constructed database of fiscal interventions.[6]

[6] The corresponding data, code, and supplementary materials are available in the project's GitHub repository: `https://github.com/pesentiA/Pandemic-Trilemma`.

Table 2: Variables and Data Sources

Variable	Description	Source
Dependent variables		
y_{ik}	Output gap: deviation of real GDP from HP-filtered trend (2015–2019, $\lambda = 1600$) (pp)	OECD (2024)
b_{ik}	Pandemic debt gap: deviation of real gross public debt from the country-specific pre-pandemic debt level, normalized by 2019 GDP (pp)	OECD (2024)
Policy variables		
S_{ik}	Stringency Index, population-weighted (0–100)	Own calc. (Appendix); Hale et al. (2021)
F_{ik}^{CP}	Capacity preservation (% of 2019 GDP)	Own database (Data Appendix)
F_{ik}^{DI}	Demand injection (% of 2019 GDP)	Own database (Data Appendix)
F_{ik}^{H}	Health spending (% of 2019 GDP)	Own database (Data Appendix)
Epidemiological variables		
d_{ik}	Excess mortality (P-score)	Msemburi et al. (2023); Mathieu et al. (2020)
θ_{ik}	Infection prevalence, estimated from excess mortality via country-specific IFRs	Own calc. (Appendix); Levin et al. (2020a)
Vax_{ik}	Cumulative vaccination doses per capita	Mathieu et al. (2021)

Notes: All absolute values are expressed in real values.

The output gap y_{ik} is measured as the percentage deviation of quarterly real GDP from its Hodrick–Prescott-filtered trend, estimated exclusively on pre-pandemic data (2015–2019, $\lambda = 1600$) and projected forward, yielding a country-specific counterfactual that is comparable across countries in percentage-point terms.[7] The debt variable b_{ik} is constructed as the deviation of real gross public debt from its country-specific pre-pandemic level, normalized by 2019 GDP. Thus, $b_{ik} = 0$ does not represent the absence of public debt, but the absence of additional debt accumulation relative to the country's pre-pandemic debt position. Specifically, I take the real gross public debt stock in 2019Q4 as the baseline and express subsequent deviations from this level in percentage points of 2019 GDP. This measurement isolates pandemic-related debt dynamics from cross-country differences in initial debt levels. For the measurement of the containment I use a modified version of the widely applied Oxford Stringency Index (Hale et al., 2021), corrected for the share of the population effectively affected by each measure: many restrictions applied heterogeneously—to the unvaccinated, to specific age groups, or exempting the recently recovered—so that the headline index overstates de facto stringency in periods of targeted policies. Weighting the index components by the affected population share reduces the composition and aggregation vulnerability inherent in any policy index. Appendix F.1 assesses sensitivity to this choice against alternative containment measures.

[7] Applying the HP filter to the full sample including 2020–2021 would pull the estimated trend downward, mechanically reducing the measured output gap. The pre-pandemic-only estimation ensures that counterfactual potential output reflects the economy's trajectory absent the pandemic shock.

The three core fiscal aggregates F_{ik}^{CP}, F_{ik}^{DI}, and F_{ik}^{H} are constructed from a novel micro-level database of 2,301 individual policy measures across all 38 OECD member states, building on and substantially extending an initial draft by Lacey et al. (2021). The database records each measure at daily frequency with its announcement date, description, primary source, and fiscal cost (in local currency and as % of 2019 GDP). The classification distinguishes instruments by their primary economic transmission mechanism required by the theoretical framework in Section 2. Table 3 details the composition of each channel while the full classification protocol and all further information for all the full dataset are documented in the separate codebook in the Data Appendix (Appendix A).

Table 3: Fiscal Instrument Classification

Channel	**Instruments**	**Measures**
Capacity Preservation (F^{CP})		
Above-the-line	Short-time work, payroll subsidies, direct grants	948
Below-the-line	Government loans, credit guarantees	400
Demand Injection (F^{DI})		
Transfers	Cash transfers, unemployment top-ups, ad-hoc benefits	346
Tax measures	Income tax relief, VAT reductions, tax credits	162
Demand stimulus	Infrastructure, green investment, tourism support	38
Health (F^{H})		
	Medical procurement, hospital capacity, vaccine rollout	232
Excluded: Tax deferrals (IMF Cat. 3)		154
Excluded: Unclassified and excluded policy codes		21
Total		**2,301**

Notes: Classification by primary transmission mechanism. Tax deferrals excluded as realized fiscal cost approaches zero. Full protocol in Data Appendix (Appendix A).

Excess mortality d_{ik}—the deviation of all-cause deaths from a projected seasonal baseline—is the preferred measure of pandemic health impact for cross-country comparison. Unlike reported COVID-19 deaths, which depend on testing capacity, health infrastructure, and cause-of-death attribution practices that varied widely across countries, excess mortality captures both direct viral fatalities and indirect deaths from healthcare-system overload, deferred care, and resource diversion (Msemburi et al., 2023). Excess mortality also enters the output equation as the fear term d_{ik}, capturing voluntary reductions in economic activity based on Goolsbee and Syverson (2021). Reported case counts are an unreliable measure of true infection prevalence, as testing capacity and reporting standards varied systematically across countries and over time. The infection prevalence variable θ_{ik} is therefore recovered indirectly. I first convert excess-mortality P-scores into excess-death counts using country-specific baseline mortality, and then infer infections by dividing excess deaths by country-specific age-adjusted infection fatality rates (Levin et al., 2020a).

Normalizing by population yields a cross-country comparable measure of pandemic pressure that is not contaminated by differential testing regimes.

3.2 Descriptives

Table 4 provides descriptive statistics for the key variables at the country-quarter level (Panel A) and the country level (Panel B).

Table 4: Summary Statistics

	Mean (SD)	Median	Min	P25	P75	Max
Panel A: Quarterly observations ($N = 380$ country-quarters)						
Outcome variables						
Output gap, pp of potential GDP	−4.36 (4.77)	−3.74	−23.47	−6.34	−1.74	11.13
Change in pandemic debt gap, Δb_{ik}, pp of 2019 GDP	0.85 (2.48)	0.46	−7.46	−0.61	2.00	14.20
Policy instruments						
Containment stringency index (0–100)	40.19 (18.70)	39.80	0.00	28.24	54.45	85.66
Capacity preservation F^{CP}, % of 2019 GDP	1.31 (3.03)	0.12	0.00	0.00	1.09	25.42
Demand injection F^{DI}, % of 2019 GDP	0.26 (0.71)	0.00	0.00	0.00	0.14	8.32
Health expenditures F^{H}, % of 2019 GDP[†]	0.12 (0.38)	0.00	0.00	0.00	0.00	4.25
Epidemiological controls						
Excess mortality, P-score (%)	10.23 (14.87)	5.51	−13.36	0.57	15.30	87.21
Infection prevalence, % of population	1.18 (1.89)	0.41	0.00	0.10	1.43	11.45
Cum. vaccination rate, % of pop.	31.35 (33.04)	12.46	0.00	0.00	66.52	89.68
Panel B: Cross-country totals ($N = 38$ countries)						
Outcome variables						
Avg. output gap, pp of potential GDP	−4.36 (3.05)	−4.22	−10.58	−5.96	−2.58	5.99
Cum. change in pandemic debt gap, pp of 2019 GDP	8.55 (5.44)	8.47	−2.21	5.31	12.31	20.64
Policy instruments						
Avg. containment stringency (0–100)	40.19 (6.22)	40.25	27.66	36.51	43.65	56.09
Cum. capacity preservation F^{CP}, % of 2019 GDP	13.11 (6.99)	11.02	0.88	8.01	17.86	29.79
Cum. demand injection F^{DI}, % of 2019 GDP	2.61 (2.54)	1.89	0.11	0.95	3.83	11.89
Cum. health expenditures F^{H}, % of 2019 GDP[†]	1.15 (1.24)	0.73	0.00	0.31	1.70	4.73
Epidemiological outcomes						
Cum. excess mortality, P-score (%)	102.25 (76.95)	88.74	−13.59	49.54	131.05	355.59
Cum. infection prevalence, % of pop.	11.82 (5.39)	10.88	3.33	8.13	15.11	31.28
Cum. vaccination rate, % of pop.	73.31 (9.47)	74.50	45.67	67.49	80.31	89.68

Notes: Standard deviations in parentheses. Panel A reports the distribution of quarterly country-level observations; Panel B reports the cross-country distribution of period averages (output gap, stringency) and cumulative totals (all other variables) over Q1 2020–Q2 2022. All variables defined in Section 3.1. [†]Health expenditures enter the debt equation only; excluded from the output equation.

The average OECD country experienced an output gap of −4.4 pp over the sample period, with the deepest contraction reaching −23.5 pp in a single country-quarter and the worst country-level average equal to −10.6 pp. Cumulative pandemic debt accumulation averaged 8.6 pp of 2019 GDP, ranging from −2.2 pp to 20.6 pp, indicating substantial cross-country heterogeneity in fiscal outcomes. Containment stringency was relatively synchronized across countries: the cross-country standard deviation of period-average stringency is 6.2, compared to a within-country quarterly standard deviation of 18.7. Fiscal composition exhibits the opposite pattern. The quarterly distributions of both F^{CP} and F^{DI} are strongly right-skewed: the mean of F^{CP} (1.31% of GDP) exceeds the median (0.12%) by an order of magnitude, with $F^{CP} = 0$ in 37.6% of country-quarters, reflecting the concentrated deployment of capacity preservation during the acute containment phase of 2020. Cumulatively, CP (13.1% of 2019 GDP) dominated DI (2.6%) by a factor of five, but the cross-country coefficient of variation is substantially larger for DI, indicating that countries that deployed demand injection did so at very different scales.

The epidemiological variables also display wide cross-country dispersion. Cumulative excess mortality ranges from -13.6 P-score points—reflecting a net reduction in non-COVID mortality in some countries—to 355.6 points, a cumulative mortality burden more than three times the projected baseline. Infection prevalence and vaccination rates follow similar patterns of cross-country divergence. This compositional heterogeneity, together with the weak association between fiscal support and containment intensity documented in Section 4, provides the empirical variation used to parameterize the transition system.

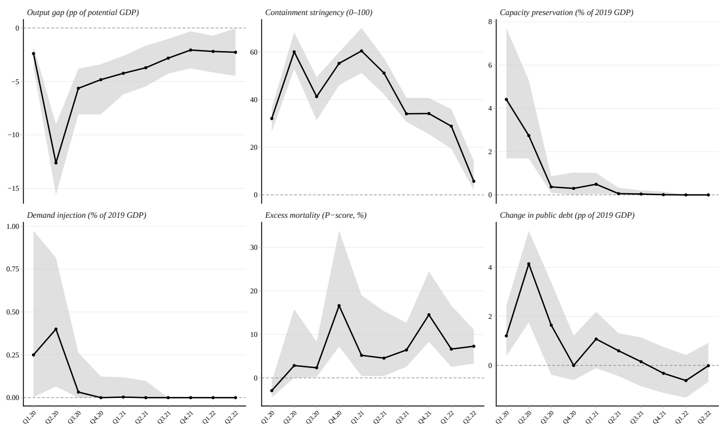


Figure 2: OECD Pandemic Trajectories by Quarter

Notes: Each panel plots the cross-country median (solid line) and interquartile range (P25–P75, shaded band) across 38 OECD countries per quarter, Q1 2020–Q2 2022.

Figure 2 shows the evolution of the key variables over time. The narrow interquartile range for containment stringency, compared to the much wider bands for capacity preservation and demand injection, illustrates that countries followed relatively similar containment trajectories but differed substantially in the composition and scale of fiscal support. For the full descriptive statistics, including further dynamics, see Appendix C

4 Parameterization Strategy

I parameterize the transition system equation by equation. The main text focuses on the output and debt transitions, where the paper's contribution lies. The health-side equations are estimated separately and reported in Appendix E.[8] The output estimating equation maps directly onto Eq. (8):

$$\begin{aligned} y_{ik} = \rho_y\, y_{i,k-1} + \alpha_S\, S_{ik} + \alpha_{\text{above}}\, F^{\text{above}}_{i,k-2} + \alpha_{\text{below}}\, K^{\text{below}}_{ik} \\ + \alpha_{\text{DI}}\, F^{\text{DI}}_{i,k-1} + \alpha_{S,\text{DI}}\, S_{ik} F^{\text{DI}}_{i,k-1} + \beta_d\, d_{ik} + \gamma_i + \varepsilon_{ik}. \end{aligned} \tag{12}$$

where $K^{\text{below}}ik = \sum s \leq k \left(F^{\text{loans}}is + F^{\text{guar}}is\right)$ denotes the cumulative below-the-line liquidity stock, and γ_i denotes country fixed effects. The baseline specification includes country fixed effects but omits unrestricted quarter fixed effects. This choice is deliberate. The objective is to parameterize the transition of output and debt over the pandemic trajectory where quarter fixed effects would absorb much of the common pandemic variation through which containment and fiscal deployment occurred. Since the model explicitly includes observed measures of the pandemic state and policy response, the baseline follows Deb et al. (2021) in controlling for the shock directly rather than absorbing it. Sensitivity checks with alternative time controls are reported in Section F.1.

The autoregressive coefficient ρ_y captures persistence in the output gap, while α_S measures the contemporaneous output effect of containment. The preferred timing specification follows the mechanisms derived in Section 2: above-the-line capacity preservation enters as a two-quarter-lagged flow, $F^{\text{above}}_{i,k-2}$; below-the-line support enters through the cumulative liquidity stock $K^{\text{below}}_{i,k}$; and demand injection enters with a one-quarter lag, $F^{\text{DI}}_{i,k-1}$. The interaction $S_{i,k} F^{\text{DI}}_{i,k-1}$ captures whether the demand-injection multiplier depends on the restrictiveness of containment.[9] A negative $\alpha_{S,\text{DI}}$ would indicate that demand injection becomes less effective under high containment. The fear channel remains part of the theoretical transition system as an exogenous control, but is omitted from the quarterly baseline specification because the excess-mortality proxy is weakly identified at the quarterly level and has no influence on the coefficients of interest.[8]

For the debt transition, I adjust the estimating equation so that the mechanical debt-service component is netted out before estimation rather than estimated directly:

$$\begin{aligned} \underbrace{\Delta b_{ik} - r_t b_{i,k-1}}_{\equiv \Delta \tilde{b}_{ik}} = \gamma_y, y_{i,k-1} + \kappa_{\text{above}}\, F^{\text{above}}_{ik} + \kappa_{\text{loans}}\, F^{\text{loans}}_{ik} + \kappa_{\text{guar}}\, F^{\text{guar}}_{ik} \\ + \kappa_{\text{DI}}\, F^{\text{DI}}_{ik-1} + \kappa_H\, F^{H}_{ik-1} + \mu_i + u_{ik}. \end{aligned} \tag{13}$$

[8]These equations are estimated at weekly frequency, because quarterly estimation would average over time frames too coarse for the underlying epidemiological dynamics. The estimates align with the existing literature; for the baseline specification I therefore adopt the corresponding literature values for the infection parameters and calibrate the wave-specific mortality mapping from the literature alone. My own estimates serve as a robustness check (see Table 7).

[9]Section F.1 reports alternative lag specifications.

where $\Delta b_{ik} = b_{ik} - b_{i,k-1}$ is the quarterly change in the pandemic debt-gap state, measured in percentage points of 2019 GDP, and $r_t b_{i,k-1}$ approximates the mechanical debt-service component.[10]

The lagged output gap $y_{i,k-1}$ is used because the debt equation describes the transition from $k-1$ to k. It therefore treats output as a predetermined state variable observed at the beginning of the transition. Therefore, the coefficient γ_y captures the automatic-stabilizer channel: a larger negative output gap increases debt accumulation through lower revenues and higher transfer payments.

The fiscal variables enter the debt equation according to their measurement and budgetary timing. Since all fiscal measures are assigned to their announcement date, the timing captures the pass-through from announced policy commitments to measured debt accumulation. Above-the-line capacity preservation enters contemporaneously because these measures were designed as immediate emergency support and typically created fiscal obligations within the same transition period. Loans and guarantees are also recorded as announced commitments, but are adjusted by take-up rates to approximate effective below-the-line fiscal exposure, following evidence that announced envelopes were only partially utilized during the pandemic (European Systemic Risk Board, 2021; European Parliament, 2022). Demand injection and health-related spending enter with a one-quarter lag. This accounts for delayed budgetary effects. The coefficients κ_{above}, κ_{loans}, κ_{guar}, κ_{DI}, and κ_H therefore measure instrument-specific debt-cost parameters in the transition system. Country fixed effects μ_i absorb time-invariant differences in fiscal institutions, debt-management practices, and accounting conventions. All relevant robustness checks with are reported in Section F.2.

4.1 Parameterization and Scope of Interpretation

Several features of the pandemic setting complicate causal interpretation. This is well recognized in the related literature, which explicitly aims to identify causal effects of pandemic policies. Although this paper does not pursue such a causal identification strategy, the interpretation of the estimated transition parameters still requires clarification. Containment and fiscal support were not randomly assigned: both may respond to pandemic severity, expected output losses, and country-specific policy capacity. Moreover, containment and fiscal support were implemented simultaneously, while treatment intensity was continuous, time-varying, and global. The baseline specification therefore relies on a conditional sequential-exogeneity assumption appropriate for a dynamic panel setting: after controlling for country fixed effects, lagged economic states, observed pandemic

[10] The estimates are not sensitive to whether the mechanical debt-service component is explicitly netted out. This is consistent with the low borrowing-cost environment during the main pandemic period: the average cost of new borrowing remained low, increasing only from around 1% in 2020 to 1.4% in 2021 before rising after the pandemic period (OECD, 2021b, 2023).

conditions, and the timing structure implied by the transition system in Section 2, the remaining innovations to output and debt are assumed not to be systematically correlated with the predetermined policy variables entering the transition equations. Country fixed effects absorb time-invariant differences in institutions, fiscal capacity, health systems, and policy preferences, while the lag structure reduces contemporaneous reverse causality from realized output to fiscal deployment. The weak correlation between containment intensity and fiscal deployment in the data ($r = 0.04$) further mitigates the concern that fiscal support was mechanically driven by stringency alone (see D.1 for the full discussion).

The lagged output gap follows directly from the state-transition structure of the model. Pandemic output losses were persistent, and omitting the lag would force the policy variables to absorb part of this persistence. Its inclusion turns the specification into a dynamic fixed-effects panel, so the estimates are interpreted under the conditional sequential-exogeneity assumption discussed above. The within estimate of ρ_y may therefore be subject to dynamic-panel bias.[11]

All baseline specifications report cluster-robust HC1 standard errors clustered at the country level. Since the number of country clusters is moderate ($N = 38$), I additionally report wild cluster bootstrap (p)-values clustered at the country level as a robustness.

The relevant empirical variation comes primarily from within-country and within-quarter differences, e.g., in the timing, scale, and composition of fiscal responses across countries facing a broadly common pandemic trajectory. In particular, countries differed substantially in their reliance on different fiscal instruments as seen in Figure 2. This compositional heterogeneity is used to discipline the channel-specific transition parameters. The estimates should therefore not be read as randomized treatment effects, but as conditional transition estimates that quantify whether the signs, timing, and relative magnitudes implied by the theoretical framework in Section 2 are consistent with the data. This allows me to assess whether the mechanisms underlying the pandemic trilemma were empirically present and binding in the realized OECD policy space. Further discusion, remaining concerns—including alternative time controls, vaccination dynamics, fiscal classification choices, dynamic-panel bias as well as a discussion about alternative estimation methods that could yield causal estimates—are addressed in the Appendix D and F.

4.2 Results

4.2.1 Output

Table 5 reports the output-transition estimates, gradually moving from a parsimonious state/shock specification to the preferred baseline transition equation.

[11] I interpret the persistence estimate as a lower bound. Bias-corrected dynamic-panel specifications show that the results are not sensitive to this concern; see Section D.1.

Table 5: Output Transition Estimates

	(1) State/shock only	(2) Aggregate fiscal support	(3) Decomposed fiscal channels	(4) Full baseline (preferred)
Persistence channel:				
$y_{i,k-1}$ (ρ_y)	$0.154***$	0.064	$0.221***$	$0.231***$
	(0.043)	(0.046)	(0.052)	(0.050)
Containment channel:				
S_{ik} (α_S)	$-0.101***$	$-0.087***$	$-0.101***$	$-0.095***$
	(0.011)	(0.010)	(0.011)	(0.013)
Aggregate fiscal channel:				
$F^{\text{total}}_{i,k-1}$		$-41.315***$		
		(9.556)		
Capacity-preservation channels:				
$F^{CP,\text{above}}_{i,k-2}$ $[\alpha_{\text{above}}]$			$0.588***$	$0.544***$
			(0.202)	(0.189)
$K^{CP,\text{below}}_{ik}$ $[\alpha_{\text{below}}]$			$0.304**$	$0.261*$
			(0.142)	(0.145)
Demand-injection channels:				
$F^{DI}_{i,k-1}$ (α_{DI})			-0.615	$1.470**$
			(0.419)	(0.569)
$F^{DI}_{i,k-1} \times S_{ik}$ $(\alpha_{S\cdot DI})$				$-0.041***$
				(0.014)
Country FE	✓	✓	✓	✓
Observations	418	418	418	418
R^2 (adj.)	0.565	0.638	0.584	0.590
Within R^2	0.395	0.498	0.426	0.436

Notes: Dependent variable: output gap y_{ik}, measured in percentage points of potential GDP. Sample: 38 OECD countries over the pandemic estimation window $t \in [4, 14]$. All specifications include country fixed effects. Standard errors clustered at the country level with small-sample correction are reported in parentheses. Column (1) includes only the lagged output gap and containment intensity. Column (2) adds aggregate fiscal support. Columns (3) and (4) decompose fiscal support by transmission channel. Column (4) is the preferred specification. The fear term is omitted from the baseline, as already stated in Section 4 see Appendix F.1. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Containment remains stable, negative, and highly significant across all specifications. Output persistence is positive throughout, but becomes insignificant in the aggregate fiscal specification that is shown in Column (2). This specification also shows why aggregate fiscal support is difficult to interpret: the coefficient has the wrong sign and is economically implausibly large. This suggests that an aggregate fiscal measure with a one-period lag mixes different instruments, timing structures, and potential reverse causality. The decomposition in column (3) therefore provides a more meaningful specification: the capacity-preservation coefficients become positive and plausible, while demand injection remains insignificant without the interaction term.[12]

In the preferred specification (4), the coefficients have the expected signs and are statistically significant at conventional levels. The persistence coefficient, at 0.231, is clearly lower than what would typically be expected in a normal business-cycle environment. This is consistent with the pandemic setting, where output dynamics were driven less by standard internal propagation and more by containment, fiscal interventions, and the timing of the shock. Containment has a statistically significant negative level effect on output, in line with the theoretical framework and with the empirical literature on the economic costs of NPIs (Deb et al., 2021; Ashraf and Goodell, 2022).

Turning to the fiscal channels, above-the-line capacity preservation increases the output gap by 0.54 percentage points per percentage point of 2019 GDP, while the below-the-line liquidity stock increases it by 0.26 percentage points. These estimates should not be interpreted as standard fiscal multipliers. They capture transition effects in a pandemic environment, where fiscal policy operates under binding health restrictions and supply constraints (Guerrieri et al., 2022; Chetty et al., 2020). Within this environment, the positive coefficients indicate that capacity-preserving measures limited persistent damage to firms, employment relationships, and liquidity conditions, consistent with the role of emergency lifelines emphasized in the theoretical model and in the literature by Deb et al. (2021); Faria-e Castro (2021).

Demand injection is the channel closest to a standard fiscal multiplier. Its baseline coefficient is larger than one, but the negative interaction with containment implies strong state dependence. The marginal effect is given by $\alpha_{DI} + \alpha_{S,\mathrm{DI}} S_{ik}$, which becomes zero at a stringency level of roughly $S^{*} \approx 36$. Above this threshold, demand injection no longer translates into higher output in the estimated transition equation. This is consistent with the underlying mechanism: demand-side measures rely on households and firms being able to spend the additional income. When containment restricts consumption opportunities and economic activity, the multiplier is attenuated or delayed (Chetty et al., 2020; Guerrieri et al., 2022; Coibion et al., 2020).

4.2.2 Debt

The debt specifications in Table 6 move from broad fiscal categories in column (1), to separate headline amounts for loans and guarantees in column (2), and finally to the preferred take-up-adjusted specification in column (3).

[12]This reflects aggregation, not irrelevance: the marginal effect $\alpha_{DI} + \alpha_{S\cdot DI}\, S$ is positive when the economy is open and vanishes at $S^{\star} \approx 36$. Since DI was deployed mostly during high-stringency quarters, the unconditional coefficient averages these regimes and is pulled below zero. A joint Wald test rejects $\alpha_{DI} = \alpha_{S\cdot DI} = 0$ ($F = 4.23$, $p = 0.022$); given the high correlation between the two terms, only the marginal effect at a given S is interpretable. See Appendix D.

Table 6: Debt Transition Estimates

	(1) Coarse categories	(2) Headline commitments	(3) Preferred take-up
Predetermined output state:			
$y_{i,k-1}$ $(-\gamma_y)$	$-0.120***$	$-0.117***$	$-0.117***$
	(0.039)	(0.039)	(0.039)
Above-the-line capacity preservation:			
$F_{ik}^{CP,\,\text{above}}$ $[\kappa_{\text{above}}]$	$0.639***$	$0.664***$	$0.664***$
	(0.202)	(0.203)	(0.203)
Below-the-line liquidity support:			
$F_{ik}^{CP,\,\text{below}}$ (aggregate)	$0.584***$		
	(0.114)		
$F_{ik}^{CP,\,\text{loans}}$ (headline)		$0.501***$	
		(0.096)	
$F_{ik}^{CP,\,\text{guar}}$ (headline)		$0.134*$	
		(0.071)	
$F_{ik}^{CP,\,\text{loans}}$ (0.60)			$0.836***$
			(0.160)
$F_{ik}^{CP,\,\text{guar}}$ (0.25)			$0.536*$
			(0.284)
Demand injection and exogenous health spending:			
$F_{i,k-1}^{DI}$ $[\kappa_{DI}]$	$0.540*$	$0.526*$	$0.526*$
	(0.296)	(0.297)	(0.297)
$F_{i,k-1}^{H}$ $[\kappa_{H}]$	$0.893**$	$0.908**$	$0.908**$
	(0.377)	(0.378)	(0.378)
Country FE	✓	✓	✓
Observations	494	494	494
R^2 (adj.)	0.185	0.188	0.188
Within R^2	0.232	0.236	0.236

Notes: Dependent variable: adjusted quarterly change in the pandemic debt-gap state, $\Delta\tilde{b}_{ik} = \Delta b_{ik} - r_t b_{i,k-1}$, measured in percentage points of 2019 GDP. Sample: 38 OECD countries over the longer pandemic debt-estimation window $t \in [4, 16]$ to capture the terminal liability. All specifications include country fixed effects. Standard errors clustered at the country level with small-sample correction are reported in parentheses. Column (1) uses broad fiscal categories. Column (2) uses headline commitments for loans and guarantees. Column (3) is the preferred specification, using a 60% take-up rate for loans and a 25% take-up rate for guarantees; based on (European Systemic Risk Board, 2021; European Parliament, 2022). Adjustments can be found in Appendix F.2. Demand injection enters with a one-quarter lag ($F_{i,k-1}^{DI}$), reflecting delayed budgetary pass-through.Health-related spending is included as an exogenous control with a one-quarter lag and is not part of the optimized fiscal instrument set. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

The coefficient on the lagged output gap is stable and statistically significant across all three specifications. A one percentage point higher lagged output gap reduces debt accumulation by about 0.12 percentage points of 2019 GDP. Conversely, a deeper recession increases the pandemic debt gap through lower revenues and higher automatic stabilizer payments. In the coarse specification, both above-the-line capacity preservation and below-the-line liquidity support increase debt accumulation by similar magnitudes. The coefficients are below one, which is plausible since fiscal measures do not translate mechanically one-for-one into the debt-gap variable: some measures are spread over several quarters, and some operate through existing budgetary frameworks (International Monetary Fund, 2022).[13]

Columns (2) and (3) further split below-the-line support into loans and guarantees, yielding three statistically significant distinct debt-cost coefficients for capacity preservation. Loans have the largest estimated debt pass-through (0.84), followed by above-the-line spending (0.64) and guarantees (0.536). This ranking is economically plausible. Loans imply an actual cash outflow once taken up and often required newly financed public credit facilities, even if they also create a financial claim on the borrower. Above-the-line measures also generate direct budgetary outlays, but their pass-through can be lower because many schemes operated through existing fiscal structures and were spread over several quarters. Guarantees, by contrast, mainly create contingent liabilities and therefore have the lowest immediate impact on measured public debt, since they affect debt only when guarantees are called. The demand-injection side also shows a lagged positive and significant debt pass-through (0.526), a magnitude similar to guarantees and close to above-the-line capacity preservation, which provides the closest direct comparison in terms of debt pass-through. Health-related spending enters as an exogenous control and has a lagged positive and significant effect on debt accumulation. The coefficient is close to one, suggesting an almost one-for-one pass-through into the pandemic debt gap. This is plausible because health measures mainly reflect newly authorized spending for medical procurement, hospital capacity, and vaccination. These main qualitative patterns are stable across the relevant robustness checks (see Section F).

Taken together, the output and debt estimates show the expected signs and economically plausible magnitudes. Containment reduces output, capacity preservation stabilizes output either with a delay or through cumulative liquidity provision, demand injection is state-dependent, and all fiscal instruments increase the pandemic debt gap. Combined with the health-side estimates, these results provide empirical support for the trilemma at the center of the model. Viral prevalence grows endogenously and becomes explosive without containment, making suppression necessary for health stabilization. Yet containment depresses output, while fiscal instruments that offset output losses widen the pandemic debt gap.[14]

Viewed in isolation, the estimates suggest a clear ranking of instruments. Demand

[13]Additional split-sample estimates by the strength of pre-existing social safety nets suggest some institutional heterogeneity in debt pass-through, but do not alter the baseline conclusion that output losses and fiscal support both increase the pandemic debt gap.

[14]The health-side dynamics are disciplined separately through the epidemiological parameters reported in Appendix E and G.

injection is attractive when containment is low, but loses effectiveness when the spending channel is constrained. Above-the-line capacity preservation is valuable for stabilization, but comes with relatively high fiscal costs. Loans and guarantees operate through the same liquidity-preservation channel yet differ sharply in their debt pass-through, making guarantees potentially the most cost-effective instrument when they are credible enough to stabilize firms and lenders—a differential that also mirrors the division of labor between state and financial system: banks retain screening and monitoring, while the government supplies only the insurance against the aggregate tail that private markets cannot provide in a systemic crisis. This isolated ranking, however, is not sufficient to evaluate actual country strategies. The instruments interact with the state of the economy and with each other. A country with high containment, large demand-injection measures, and little capacity preservation may generate debt without much effective output stabilization. Conversely, liquidity guarantees in an unrestricted economy may have limited additional value if firms no longer face binding liquidity constraints.

This turns the empirical findings into a dynamic allocation problem. The social planner must choose not only the total amount of fiscal support, but also its composition and timing, taking into account containment, output persistence, liquidity needs, and the accumulated debt state. The next section calibrates the model with the estimated transition parameters and assesses whether the proposed transition system can reproduce the observed pandemic dynamics.

4.3 Calibration

Using the transition system from Eqs. (8)–(11) and the parameters in Table 7—I evaluate the system's fit on the 38 OECD economies. The objective of this step is to assess whether the parameterized transition system can reproduce the observed pandemic dynamics before it is used for representative and country-specific counterfactual policy evaluation. I initialize the system at the pre-pandemic steady state, with an output gap of zero, excess mortality equal to zero, and public debt equal to its country-specific pre-pandemic level. Formally, the debt baseline is given by $B^{\mathrm{real}}_{i,2019\mathrm{Q}4}$, and the simulated pandemic debt gap measures deviations from this baseline, normalized by 2019 GDP. I take the observed paths of containment, fiscal instruments, and the epidemiological state as given and roll the output, debt and health blocks forward over the pandemic horizon while the initial health shock in 2020Q2 is treated as the initial exogenous shock. Table 8 summarizes the model fit. The output block performs well: the median output-gap RMSE is 1.85 percentage points, or 40% of the observed cross-country dispersion, so the model reproduces the main output dynamics. Debt dynamics are matched less tightly. The median RMSE of quarterly debt changes is 2.02 percentage points, 85% of the relevant dispersion.

Table 7: Model parameters: symbol, description, calibrated value, and equation block.

Symbol	Description	Value	Block
ρ_y	Persistence of output gap	+0.231	Output
α_S	Contemporaneous effect of stringency S	−0.095	Output
α_{above}	Lag-2 effect of above-the-line fiscal spending	+0.544	Output
α_{below}	Contemporaneous effect of below-the-line stock (take-up adjusted)	+0.261	Output
α_{DI}	Lag-1 effect of direct income support	+1.470	Output
$\alpha_{S,\text{DI}}$	Interaction of stringency with lag-1 direct income support	−0.041	Output
r	Quarterly sovereign interest rate	+0.001	Debt
γ_y	Automatic-stabiliser sensitivity of debt	−0.117	Debt
κ_{above}	Pass-through of above-the-line spending into debt	+0.664	Debt
κ_{loans}	Pass-through of take-up-adjusted loan flows	+0.836	Debt
κ_{guar}	Pass-through of take-up-adjusted guarantee flows	+0.536	Debt
κ_{DI}	Pass-through of lag-1 direct income support	+0.526	Debt
$\rho_{\theta,k}$	Infection persistence before/after vaccination break	1.50/ <1	Infection
ϕ_S	Stringency mitigation factor	0.80	Infection
$\delta_{\theta,k}$	Wave-specific mortality mapping based on effective population IFRs	Appendix E	Mortality

Notes: The health parameters are calibrated using external epidemiological evidence, see Appendix E and G for the full discussion.

The model thus captures the direction and magnitude of the fiscal pass-throughs but is weaker on part of the initial short-run debt movement, owing to accounting differences (see Appendix G for the full discussion). The overall trend, the growth rate, and especially the levels are nonetheless reproduced closely: the debt level attains a median RMSE of 2.53 percentage points and an RMSE/SD ratio of 0.06, so the model tracks debt dynamics appropriately with a final mean debt residual of around 1 pp.

Table 8: Calibration fit of the transition system under observed policy paths.

Moment	Horizon	Median RMSE	Mean RMSE	RMSE/SD	Interpretation
Output gap	Q4.19–Q2.22	1.85 pp	2.01 pp	0.40	Error is less than half of observed dispersion; dynamics are well reproduced.
Debt change	Q4.19–Q4.22	2.02 pp	2.00 pp	0.85	Primary debt metric; weakest fitted component.
Debt level	Q4.19–Q4.22	2.53 pp	3.02 pp	0.06	Debt levels are closely reproduced, partly because the initial debt stock is imposed.
Infection state θ	Q4.19–Q2.22	0.0000	0.0024	0.00	See text.
Excess mortality d	Q4.19–Q2.22	12.59	13.38	0.72	Mortality dynamics are reproduced reasonably well, conditional on θ.

Notes: RMSE/SD denotes the median RMSE relative to the cross-sectional standard deviation of the corresponding observed variable. Debt change is the primary debt validation metric because it tests the fiscal pass-through dynamics; debt levels are reported as a reference because the initial debt stock is imposed.

The infection state θ enters the transition system in the median forward roll as a calibrated input, not as a fitted moment. By construction the calibrated innovations ε_θ absorb the entire gap between the simulated and observed infection path, so the reported θ RMSE of zero reflects this identity rather than model fit. Conditional on the observed infection path, the mortality equation maps θ into excess deaths with a median RMSE of 12.59 deaths per million per week, or 72% of the observed dispersion, so the wave-specific IFR mapping reproduces the level and timing of mortality reasonably well (Figure 3).

Since in the first step of the main analysis the overall optimization is solved for the representative OECD economy, the mean trajectory is the object of interest, and here the planner faces a fully dynamic optimization problem. In this mean roll the infection state θ is no longer pinned to the observed path but enters as a state the planner can partly steer through containment, while the variant-driven innovations ε_θ remain exogenous. The exercise is therefore closer to a genuine optimization than a calibrated replay, yet the fit does not deteriorate: under observed mean inputs the representative economy reproduces the data at least as closely as the median roll—the mean output path with an RMSE of 1.24 percentage points, the mean mortality path with an RMSE of 2.87 deaths per million per week, better than the fixed, wave-adjusted median roll, and quarterly debt changes with an RMSE of 0.85 percentage points (see Appendix G for the full discussion).

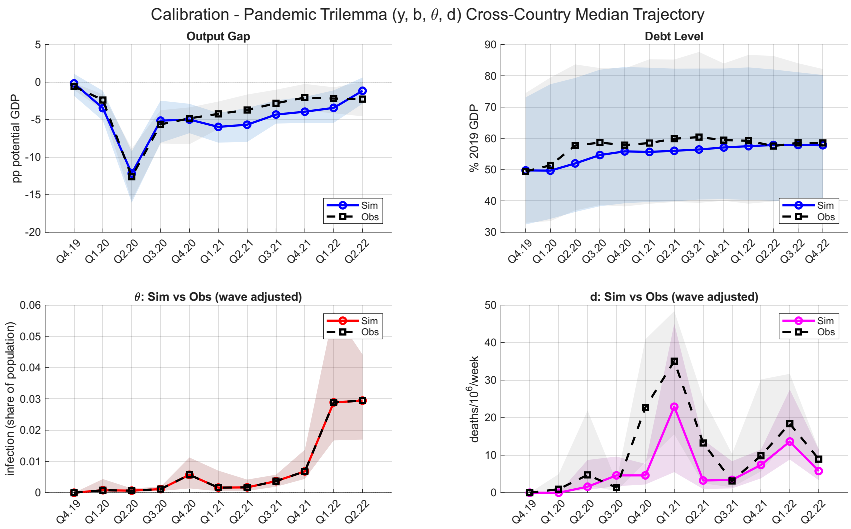


Figure 3: Calibration fit: OECD-median simulated and observed trajectories with IQR bands.

Because the model is also used to solve country-specific planner problems, Table 27 in Appendix G reports the same output, mortality, and debt fit statistics at the individual-country level; the cross-country dispersion of these endpoint residuals is consistent with the aggregate moments above, supporting country-level as well as representative-economy counterfactual analysis. Overall, the calibrated transition system is sufficiently accurate for counterfactual policy evaluation.

5 Optimization

The planner chooses the control vector $\boldsymbol{u}_k = (S_k,\ F_k^{\text{above}},\ F_k^{\text{loans}},\ F_k^{\text{guar}},\ F_k^{\text{DI}})'$ subject to the transition system in Eqs. (8)–(11) with the state vector $(y_k, b_k, \theta_k, d_k)$ is that is augmented with the lagged control and the two stocks to render the system first-order; Appendix H states the augmented system formally.

The two stocks are central to the model's economics. Below-the-line loans and guarantees accumulate into a liquidity stock L_k and above-the-line transfers into a capacity-preservation stock K_k, each decaying geometrically:[15]

$$L_{k+1} = (1-\delta_L)\, L_k + F_k^{\text{loans}} + F_k^{\text{guar}}, \qquad K_{k+1} = (1-\delta_K)\, K_k + F_k^{\text{above}}, \tag{14}$$

Because above- and below-the-line support operate through the same output channel but are distinguished in the debt dimension due to institutional differences, the model allows for an interaction between the two. Economically, the two measures are complementary: above-the-line support (wage subsidies, for example) mainly targets the preservation of the employment relationship, while below-the-line liquidity targets the solvency of the firm itself, so each is more effective when the other is in place, consistent with the argument made by International Monetary Fund (2022). The complementarity is captured by a bounded multiplier on the liquidity effect,

$$m(K_{k+1}) = 1 + \chi\, \frac{K_{k+1}}{K_{k+1} + \bar{K}}, \tag{15}$$

so that the output transition contains the term $\alpha_L\, L_{k+1}\, m(K_{k+1})$ in place of a simple additive liquidity effect, alongside a direct capacity effect $\alpha_K K_{k+1}$. The multiplier is bounded between 1 (no preserved capacity) and $1+\chi$ (capacity saturated relative to the empirical scale $\bar{K}$). The parameters χ, α_K, and the half-life governing δ_K are calibrated externally (Section 4.3).

The planner solves a finite-horizon discounted quadratic loss problem,

$$\min_{\{\boldsymbol{u}_k\}_{k=0}^{N-1}}\ \frac{1}{2}\left\{\sum_{k=0}^{N-1} \beta^{k+1}\left[\tilde{\boldsymbol{x}}_{k+1}'\boldsymbol{Q}\,\tilde{\boldsymbol{x}}_{k+1} + \boldsymbol{u}_k'\boldsymbol{R}\,\boldsymbol{u}_k\right] + \beta^N \tilde{\boldsymbol{x}}_N'\boldsymbol{Q}_N\,\tilde{\boldsymbol{x}}_N\right\}, \tag{16}$$

where $\tilde{\boldsymbol{x}}_k \equiv \boldsymbol{x}_k - \bar{\boldsymbol{x}}$ denotes the deviation from the target state and $\beta = 0.99$ is the quarterly discount factor. The running loss is evaluated on the states the policy brings about, $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N$; the predetermined initial state carries no loss. The model is solved for the main pandemic period, Q1 2020 through Q4 2021, where containment remained a relevant health instrument and it extends up to Q4 2022 to capture the fiscal legacy.

The problem is subject to two sets of policy constraints. First, all controls are non-

[15]The decay rates are calibrated in Section 4.3; robustness checks with alternative and zero decay rates leave the main results unchanged.

negative and bounded in their 99th percentile of the corresponding in-support observed values across the OECD panel. Second, containment is subject to a ramping constraint, $|S_k - S_{k-1}| \leq \Delta S_{\max}$, with $\Delta S_{\max}$ set to the 95th percentile of observed quarter-on-quarter changes in stringency: legal, administrative, and compliance frictions limited how fast restrictions could be tightened or lifted, and the constraint imposes this speed limit directly from the data. Because the transition system is nonlinear—through the complementarity multiplier and the interaction of stringency with infection dynamics and lagged income support—the problem has no closed-form solution; Appendix H describes the used algorithm, the treatment of the constraints, and the safeguards against local optima.

The running and terminal losses are evaluated at the deviation from the target state $\bar{\boldsymbol{x}}$: for output, infection prevalence, and excess mortality the target is the pre-pandemic benchmark of zero, while for public debt it is the pre-pandemic stock b_0—the fiscal objective is to avoid accumulating debt beyond the level inherited at the onset of the pandemic. The running state-cost matrix is diagonal,

$$\boldsymbol{Q} = \mathrm{diag}(w_y,\ \tau_b\, w_b,\ 0,\ \lambda_d\, w_d,\ 0,\ 0,\ 0)\,, \tag{17}$$

so the planner penalizes output-gap deviations, debt accumulation beyond b_0, and excess mortality. Infection prevalence receives zero direct weight because it is not a welfare object in itself; it enters the loss only through its role in generating future excess mortality via Eq. (11). The weights are normalized by economically interpretable reference scales—an output gap of 5 percentage points, a debt increase of 10 percentage points of GDP, and excess mortality of 100 deaths per million per week—so that the planner's preferences reduce to two free parameters: τ_b, the weight on fiscal sustainability relative to output stabilization, and λ_d, the weight on mortality relative to normalized output loss while assuming in the baseline specification that the health dimension is the main target. Because the loss is quadratic, λ_d is not itself the marginal mortality–output trade-off; I therefore anchor it to the value of a statistical life by setting λ_d such that the planner's marginal rate of substitution between deaths and output, evaluated at the observed sample means, equals a target VSL—following the approach of Hall et al. (2020) and the pandemic planner analyses of Alvarez et al. (2021) and Acemoglu et al. (2021).[16]

The terminal term weights public debt alone. By the end of the horizon, widespread vaccination has severed the link from infection to death and removed the output cost of containment, so the only state that persists as a welfare-relevant legacy is the accumulated fiscal burden. The terminal weight capitalizes the running debt loss into the present value

[16] The baseline $\lambda_d = 75$ corresponds to a VSL of approximately \$6.5 million—conservative within the standard range—and Section I reports sweeps over $\lambda_d \in \{58, 104, 127\}$, corresponding to VSLs of \$5, \$9, and \$11 million, together with alternative values of τ_b according to differences in age structure (Greenstone and Nigam, 2020; Robinson et al., 2021)

of a persistent post-horizon debt deviation (Appendix H.6). The control-cost matrix $\boldsymbol{R}$ plays no economic role: since fiscal costs are already captured through the debt transition, it is set to a small uniform penalty on normalized instrument use that keeps the problem well-posed while contributing negligibly (below 1 percent) to the objective at the reported solution.[17]

Because the model is not micro-founded, the weighted welfare solution serves as a benchmark rather than as the central welfare claim. The main analysis instead relies on the constrained policy frontier: fixing two of the three welfare-relevant dimensions at their model-implied observed level and minimizing the third, to test whether the observed policy is Pareto-dominated. These frontier problems are solved exactly, as nonlinearly constrained programs over the full control path, so no welfare weights enter the frontier results; Section 6.2 gives the formal statement and Appendix H.7 the computational details.

This problem is solved at two levels: for the representative OECD economy (Section 6.2) and independently for each of the 38 countries where the structural parameters are common across all applications; only the exogenous inputs vary—the country-specific epidemiological innovations ε_θ, the initial output shock, the country intercepts μ_y, μ_b, and the initial debt level b_0.

6 Results

This section presents the normative results in three steps. I first characterize the weighted planner benchmark—how the optimal policy differs from the observed average OECD response and what it delivers. I then turn to the central result, the constrained policy frontier, which tests whether the observed policy is Pareto-dominated without invoking welfare weights. Finally, I extend the frontier analysis to the country level: solving the frontier for each of the 38 economies under their own pandemic conditions identifies which countries, given their pandemic burden, came closest to the best attainable combination of output stabilization and debt containment

6.1 Planner Benchmark

Under the health oriented baseline preferences contrasted in Figure 4, the planner with perfect foresight[18]. sustains on average substantially more containment than observed until vaccination arrives (an average stringency of 48.7 against 34.1), building up at the ramping limit in the first two quarters and holding the acute-phase peak at the in-support cap.[19] To counter the output cost of containment, the planner deploys a slightly larger

[17]The target variables and transition parameters are summarized in Table 1.
[18]The value of this foresight is quantified in Section I
[19]Relaxing the containment cap to the full index scale is examined in (Appendix I.4).

and more front-loaded above-the-line envelope (6.0 against 5.1 percent of GDP). The sharpest difference lies in liquidity support: the planner follows a similar timing profile but scales the effective volume up (19.3 against 2.7 percent of GDP), mixing loans and guarantees and hitting the guarantee cap in two quarters.[20] This profile is in line with the framework of Section 2, where bridging liquidity over the restricted period is the key margin for keeping productive structures intact. Demand injection, by contrast, is never used despite its multiplier above one. The reason is the state-dependent multiplier: the output effect of direct income support turns negative once stringency exceeds $S^* \approx 36$—a threshold the planner's containment path exceeds throughout the acute phase, and which the observed OECD average, at 34.1, only barely undercuts. Once containment is unwound, the output gap has largely closed and demand injection has no remaining role.

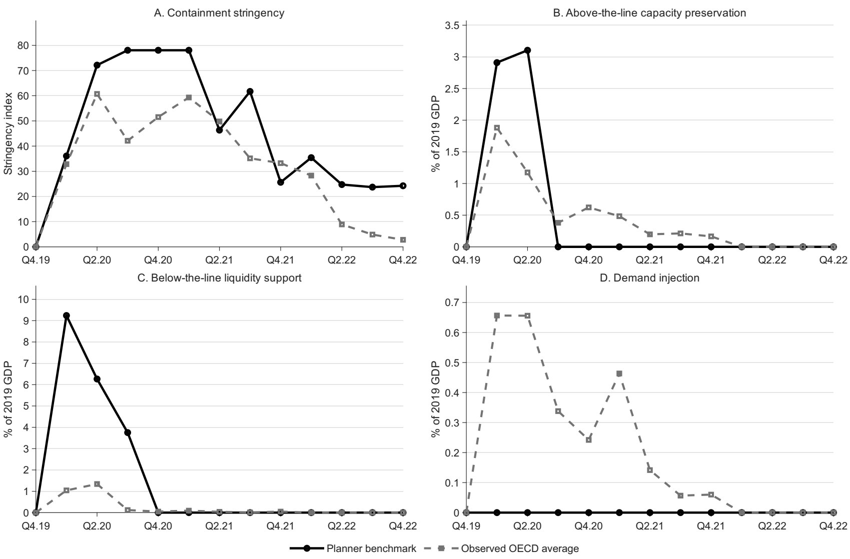


Figure 4: Planner benchmark and observed OECD average: policy paths. *Notes:* Solid lines show the optimal policy under the baseline preferences $\tau_b = 0.05$, $\lambda_d = 75$ (VSL $\approx$ \$6.5 million; Section 5); dashed lines show the observed OECD average policy. Fiscal instruments in percent of 2019 GDP; loans and guarantees are take-up-adjusted effective values. All instruments are subject to the in-support caps and the stringency ramping constraint of Section 5. The pre-pandemic quarter Q4.2019 is locked at zero.

Figure 5 shows the dynamics of these decisions against the model-implied benchmark. The planner achieves through the early ramp up a less severe trough and a faster recovery of the output gap, cutting the cumulative output loss by almost two thirds (-16.5 against -46.1 pp-quarters)—driven primarily by the front-loaded, larger liquidity envelope, complemented by a modestly larger but earlier above-the-line program. The price is fiscal: debt rises steeply during the acute phase and flattens quickly thereafter, ending 6.7 pp of GDP above the observed terminal level. This gap is the direct budgetary cost of the larger support package. The tighter containment, in turn, is what saves lives: cumulative excess mortality falls by 567 deaths per million (-23 percent).

[20]Absent the caps, the planner would allocate the entire liquidity envelope to guarantees: both instruments carry the same output coefficient per effective unit, but guarantees enter the budget at $\kappa_{\text{guar}} = 0.54$ against $\kappa_{\text{loans}} = 0.84$. This is also consistent with the view that, with a functioning money market, the government should guarantee rather than originate credit as discussed in 4.2.2.

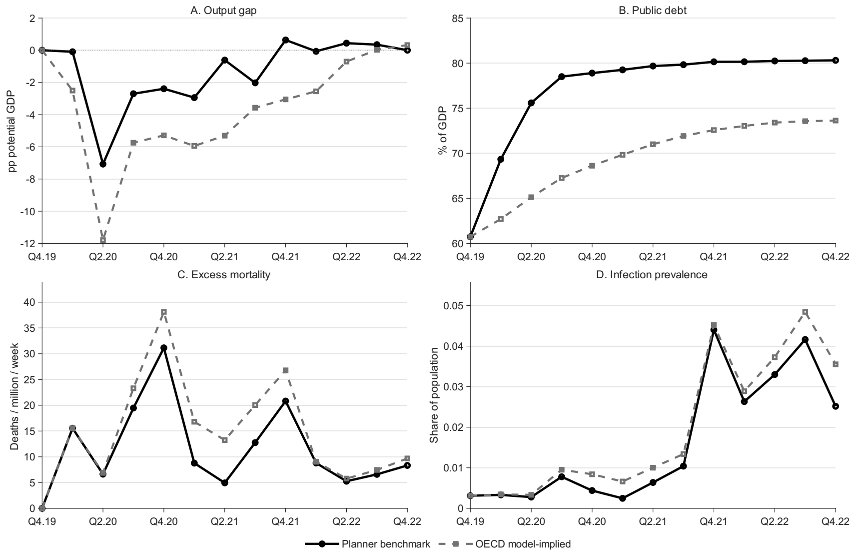


Figure 5: Planner benchmark and observed OECD policy: model-implied outcome paths. *Notes:* Solid lines show outcomes under the optimal policy; dashed lines show *model-implied* outcomes under the observed OECD average policy, i.e., the observed instruments fed through the calibrated transition system under identical shocks. Holding the model and shock paths fixed isolates the pure policy effect; the fit of the model-implied paths to the raw data is documented in Section 4.3. Both paths are initialized at the common pre-pandemic state in Q4.2019; the debt gap in Q1.2020 reflects the planner's front-loaded liquidity package. Baseline preferences $\tau_b = 0.05$, $\lambda_d = 75$ (VSL $\approx$ \$6.5 million).

Table 9: Planner benchmark versus observed OECD policy: aggregate comparison.

Outcome	**Observed (model-implied)**	**Planner**	**Difference**
Cumulative output gap (pp-quarters)	−46.1	−16.5	+29.6
Cumulative deaths per million	2,505	1,938	−567
Terminal debt (% of GDP)	73.6	80.3	+6.7
Average stringency	34.1	48.7	+14.5
Total above-the-line CP (% GDP)	5.1	6.0	+0.9
Total below-the-line liquidity	2.7	19.3	+16.5
Total demand injection	2.6	0.0	−2.6

Notes: Evaluation window Q1.2020–Q4.2022. Outcomes in the first column are model-implied under the observed OECD average policy. Below-the-line liquidity in take-up-adjusted effective terms; fiscal totals in percent of 2019 GDP. Baseline preferences $\tau_b = 0.05$, $\lambda_d = 75$ (VSL $\approx$ \$6.5 million).

Table 9 sums up the two margins behind the gains: the planner deploys a larger fiscal envelope—1.7 times the observed total when scaled by the containment intensity it compensates[21]—and shifts its composition from broad-based support toward effective and cheaper liquidity and capacity preservation, while demand injection is abandoned.

In sum, the planner benchmark displays the trilemma in action: it buys lives and output stabilization at the price of more debt, and it does so through an early ramp-up of fiscal support concentrated on capacity preservation combined with liquidity provision. These findings are suggestive rather than conclusive, since they rest on a particular choice of welfare weights; whether the observed policy can be improved upon without taking a stand on preferences is the subject of the constrained policy frontier in the next subsection.

[21] Total fiscal support per point of average stringency: 0.52 percent of GDP for the planner against 0.31 observed.

6.2 Constrained Policy Frontier

The frontier analysis dispenses with welfare weights entirely. Let $\mathcal{L}_y(\boldsymbol{u})$, $D(\boldsymbol{u})$, and $b_N(\boldsymbol{u})$ denote the discounted output loss, cumulative excess mortality, and terminal debt implied by a policy path $\boldsymbol{u}$ under the transition system, and let $(\mathcal{L}_y^{\text{obs}}, D^{\text{obs}}, b_N^{\text{obs}})$ be their values under the observed OECD policy. The three frontier programs each minimize one dimension subject to the other two being no worse than observed:

$$\min_{\boldsymbol{u}\in\mathcal{U}} \mathcal{L}_y(\boldsymbol{u}) \quad \text{s.t.} \quad D(\boldsymbol{u}) \leq D^{\text{obs}}, \;\; b_N(\boldsymbol{u}) \leq b_N^{\text{obs}}, \tag{18}$$

and analogously for mortality and debt. The observed policy is Pareto-dominated if any strategy attains a strictly lower objective. Appendix H.7 describes the computation. Table 10 reports the frontier results.

Table 10: Constrained policy frontier: observed policy versus frontier solutions.

	Observed (model-implied)	**A: min output**	**B: min deaths**	**C: min debt**
Discounted output loss $\mathcal{L}_y$	285.7	**60.8**	285.7	285.7
Cumulative output gap (pp-quarters)	−46.1	−15.2	−45.6	−36.6
Cumulative deaths per million	2,505	2,505	**1,947**	2,505
Terminal debt (% of GDP)	73.6	73.6	73.6	**67.3**
Improvement in target dimension (%)	—	78.7	22.3	8.6
Average stringency	34.1	33.0	45.1	24.0
Total above-the-line (% GDP)	5.1	4.1	1.5	0.0
Total below-the-line, effective (% GDP)	2.7	13.1	10.7	2.8
Total demand injection (% GDP)	2.6	0.0	0.0	0.0

Notes: Each scenario minimizes one dimension (boldface) subject to the other two being no worse than their model-implied values under the observed OECD policy, Eq. (18). The output constraint is imposed on the discounted quadratic loss $\mathcal{L}_y$; the cumulative output gap is reported as a memo item. In all three scenarios both constraints bind (satisfied with equality to numerical tolerance 10^{-8}). Evaluation window Q1.2020–Q4.2022; below-the-line liquidity in take-up-adjusted effective terms; controls subject to the in-support caps and the ramping constraint of Section 5. Computation and multistart: Appendix H.7.

The observed OECD policy is strictly Pareto-dominated in every direction: holding the other two dimensions at their model-implied observed levels, a feasible policy path reduces the discounted output loss by 78.7 percent (scenario A), cumulative excess mortality by 22.3 percent, or 558 deaths per million (scenario B) and terminal debt by 6.3 percentage points of GDP (scenario C). In all three scenarios both constraints bind and the observed policy lies strictly in the interior of the possibility set along each dimension.

The composition of the frontier policies identifies the source of the gains, and it is the same recomposition the weighted planner selects. Scenario A raises effective liquidity support from 2.7 to 13.1 percent of GDP at slightly lower average stringency, so the output gain is almost entirely fiscal recomposition rather than reduced containment—the frontier selects precisely the instrument with the smallest net debt footprint per unit of output identified analytically in Appendix H.8. Scenario B buys the mortality reduction with tighter containment (average stringency 45.1 against 34.1) and compensates its output cost through the same channels, yielding similar results to the health oriented weighted planner. Scenario C is the purest illustration of the timing margin: it abandons above-the-line support entirely, keeps the comparatively cheap effective liquidity at its observed level, and lowers average stringency to 24.0—retiming containment into the acute waves—which improves the cumulative output gap by 9.5 pp-quarters at identical $\mathcal{L}_y$ and identical deaths, and lets the automatic stabilizer deliver the debt reduction.

Demand injection is zero in all three solutions, for the same state-dependence reason as in the planner benchmark. None of the three solutions requires instruments, scales, or adjustment speeds outside what OECD governments demonstrably implemented.[22]

Taken together, these findings show that the weighted planner follows precisely the strategy the frontier identifies as dominant: early front-loading with a concentration on liquidity assistance and capacity preservation, buying output stabilization and lives at the price of debt. It is therefore no surprise that the weighted benchmark is robust across the preference scenarios and the VSL sweep, as documented in Appendix I.

Both results, however, are statements about the average OECD economy. In the data, countries chose markedly heterogeneous fiscal strategies: some implemented compositions and magnitudes close to what the average benchmark identifies as optimal—Japan's liquidity-heavy package is the most striking example—while others did not (Appendix C, Table 15). The final question is therefore whether these differences mattered: which countries, given their own pandemic burden, came closest to the best attainable combination of output stabilization and debt containment.

6.3 Cross-Country Results

The country-level analysis solves the three frontier programs of Eq. (18) separately for each of the 38 economies. The structural parameters and the policy menu are common across countries by design—holding the technology fixed isolates differences in policy from differences in transmission—while the country-specific ingredients (epidemiological innovations, onset shocks, intercepts, initial debt) and, crucially, the constraint thresholds are each country's own: every program conditions on the country's own model-implied realized outcomes, so improvements are measured at the country's own pandemic burden. Model fit enters as a continuous diagnostic as shown in Appendix J, the frontier distances are exactly invariant to country-specific mortality-level misfit, and the ranking is rank-stable to excluding the worst-fitting quartile and further robustness checks (see

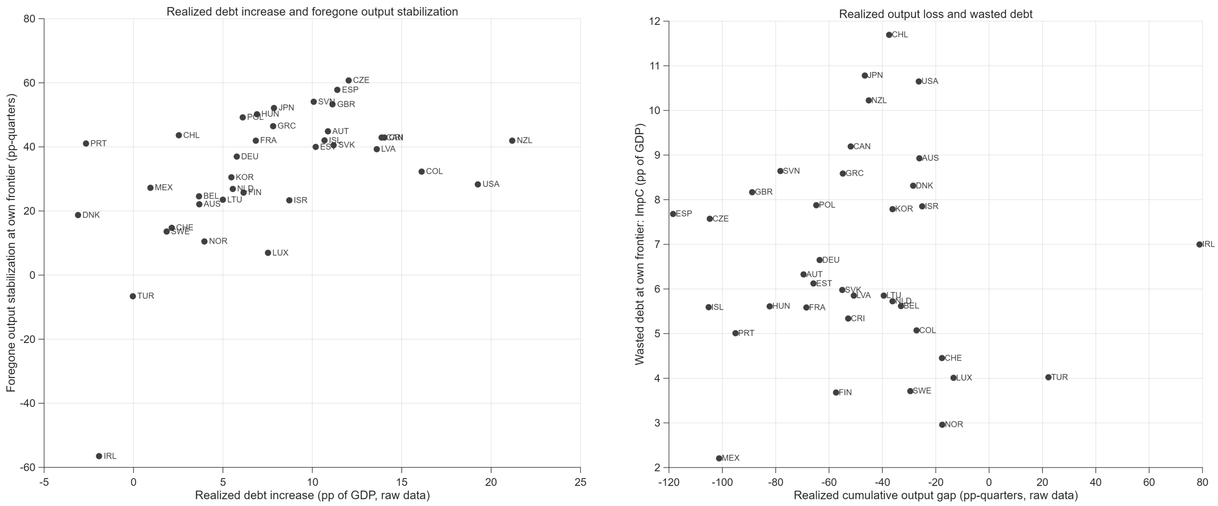
[22]The dominance result does not hinge on the calibrated capacity–liquidity complementarity. Shutting it down entirely ($\chi = 0$) leaves the improvements essentially unchanged, at 73.6, 22.3, and 9.0 percent; the complementarity accounts for roughly five percentage points of the output improvement and none of the qualitative pattern (see Appendix I). Its role is compositional: at $\chi = 0$ the frontier policies abandon above-the-line support altogether, so wage-subsidy-type instruments earn their place in the optimum through their interaction with liquidity support.

Appendix J.1).

The headline result extends the average finding to nearly every economy: no OECD country operated on its own frontier (Figure 17). In 37 of 38 cases, a feasible policy from the common menu improves each dimension at the country's own levels of the other two.

Figure 6, panel (a), displays the composition margin (Frontier A) in natural units, holding each country's realized mortality and epidemiological shock path fixed: what each country paid in debt against the output stabilization it left unexploited. New Zealand, the United States, and the United Kingdom combine large realized debt increases (+21, +19, and +11 pp of GDP) with large foregone stabilization (42, 28, and 53 pp-quarters); Switzerland, Sweden, and Norway occupy the opposite corner—small debt increases, little left unexploited.

Panel (b) prices the timing margin (Frontier C): at each country's realized output loss and mortality, how much debt was carried without corresponding returns. Denmark is the instructive case—inconspicuous in panel (a) because its debt ratio even declined at the end of the observation period, it nonetheless carried 8.3 pp of GDP that a retimed policy would have saved at identical output and deaths; Chile, Japan, the United States, and New Zealand top this dimension with 10 to 12 pp, while Türkiye and Norway wasted the least (4 and 3 pp) despite very different absolute outcomes.[23]



Panel A: Realized debt increase vs. foregone output stabilization

Panel B: Realized output loss vs. wasted debt

Figure 6: Frontier distances in natural units. *Notes:* Each point is one country; horizontal axes show realized outcomes from the raw data over Q1.2020–Q4.2022, vertical axes the unexploited improvement at the country's own frontier. Panel (a): realized change in the debt-to-GDP ratio against the output stabilization attainable at the country's own mortality and debt, in pp-quarters. Panel (b): realized cumulative output gap against the terminal debt reduction attainable at the country's own output loss and mortality (scenario C), in pp of GDP. Reading: countries toward the upper right paid much and left much unexploited; positions near the origin reflect little room to convert at the realized position, not necessarily favorable outcomes. Italy omitted (Appendix J.5).

[23] Türkiye and Ireland are special cases in the output dimension: both recorded positive cumulative output gaps over the evaluation window—Ireland's driven by the well-documented distortion of Irish GDP through multinational relocations, Türkiye's by its high-growth, high-inflation trajectory. Excluding both countries leaves the ranking and the cross-sectional results unchanged (Appendix F).

Combining the two margins yields the frontier distance: a single measure of how much a country left unexploited—in output stabilization and in debt—at its own realized position. It summarizes the quality of a strategy's composition and timing, conditional on the trilemma position the country chose, and it is the ranking of Table 11 (For a visual representation see Appendix J.1 Figure 17).

Ranking countries by their distance to their own frontier, Mexico, Sweden, Portugal, Finland, France, and Norway come closest: given their fiscal space and pandemic pressure, these countries chose the compositions and timing that left the least output stabilization and debt reduction unexploited. New Zealand, the United States, Australia, Japan, and Denmark are furthest (Table 11). The measure is conditional: it identifies the combination of policies that was least Pareto-improvable given the country's own realized trilemma position, not the best absolute outcomes. Mexico illustrates this exactly: its outcomes were poor, but its minimal fiscal deployment left almost no room for improvement at its own debt level—any material gain would have required accepting more debt, which the conditional measure holds fixed. Another special case is Italy. It exceeds the ramping limit pivotally: hit first and hardest, it tightened containment faster than any other OECD government, at a speed the common menu does not admit and is therefore treated as a special case. Raising the ramping limit to the 99th percentile of observed transitions renders Italy's programs feasible; under that relaxed menu, its improvements (47 percent of output gap, 5.4 pp of debt) would place it among the five countries closest to their frontier, consistent with its efficient, liquidity-led composition.

Relating the distances to the observed packages (Appendix J in Table 35) makes the ranking concrete. The frontier-near countries share a recognizable profile: liquidity-oriented, front-loaded packages with little demand injection with, on average, a liquidity share of 34 percent against 22 among the furthest, a demand-injection share of 24 percent against 35, and 84 against 75 percent of the package committed by Q3.2020. Sweden is the archetype: a package with a liquidity share of 34 percent, a demand-injection share of 13 percent, and 96 percent committed early. Norway pushed the composition furthest—half its package in liquidity, essentially no demand injection—and realized one of the highest returns in the panel. Switzerland shows that composition alone does not decide the ranking: with a nearly demand-injection-free, front-loaded package and one of the highest returns, it nonetheless sits in the midfield. Its distance is a scale margin, not a composition margin: although its realized debt increase was among the smallest—guarantees enter gross debt only upon being called, and the constitutional debt brake disciplined the envelope—the consolidation and stabilizer returns its policy left unused imply that four fifths of its (small) remaining loss was still removable at that same terminal debt. Low realized debt is not penalized; the measure holds it fixed and asks only whether it was fully converted.

Table 11: Distance to the country-specific frontier: closest and furthest countries.

	Frontier distance	Ranking dimensions		Context
	(avg. rank, 1–37)	**Output impr.** (% of own $\mathcal{L}_y$)	**Debt impr.** (pp of GDP)	**Deaths impr.** (% of own deaths)
Closest to own frontier				
Mexico	1.0	35.3	2.2	16.6
Sweden	4.0	58.5	3.7	21.3
Portugal	5.0	48.7	5.0	16.2
Finland	6.0	62.3	3.7	13.3
France	7.0	52.1	5.6	13.3
Norway	7.5	77.9	3.0	12.5
Furthest from own frontier				
Chile	31.5	92.9	11.7	27.4
Denmark	32.0	100.0	8.3	20.6
Australia	33.5	100.0	8.9	28.6
Japan	33.5	99.6	10.8	21.2
New Zealand	34.5	100.0	10.2	38.7
United States	35.0	100.0	10.6	34.6

Notes: Frontier distance is the average of the country's within-sample ranks in the output and debt improvements (37 ranked countries; lower = closer to the own frontier). Each improvement column reports a separate frontier program: the output and debt improvements hold the country's mortality (and the respective other dimension) at its observed level; the deaths improvement is from the program that instead minimizes mortality at the observed output loss and debt—it is contextual and does not enter the distance. Debt improvements are reported in absolute pp of GDP rather than in percent of the observed debt increase. Italy is excluded (observed policy not replicable within the menu, see text). Full 38-country table: Appendix J.6.

At the other end, there are two ways to be far from the frontier. The first is unfavorable composition and timing: the United States placed half its package in demand injection, committed only 54 percent of it by Q3.2020, and gained little output for the debt it added; Chile combined the highest demand-injection share of the panel (67 percent) with the most back-loaded profile (only 40 percent committed early) and realized a negative return; Australia and New Zealand concentrated their support in wage-subsidy commitments beyond the common menu while holding liquidity shares below a tenth of their packages. The second is unexploited fiscal space despite reasonable composition: Denmark and Japan ran liquidity-heavy, well-timed packages—Japan's is the most planner-like in the panel—but the fiscal space their realized debt paths left unused, combined with the output ceiling, implies that far more was attainable at their own debt levels than their scale delivered. The ranking is grounded in the realized magnitudes: Table 34 in Appendix J.1 juxtaposes each country's frontier distance with its raw outcomes and shows that the efficiency pattern is not an artifact of the model-implied thresholds—Sweden and the United States, for instance, suffered nearly identical raw output losses, but Sweden at one tenth of the debt increase and with fewer deaths.

While those frontier-distance findings rest on the constrained optimization and are conditional on each country's realized position, the realized fiscal exchange rate involves neither: it measures the cumulative output gained per percentage point of fiscally induced debt, computed against a counterfactual that keeps each country's observed containment path and sets the fiscal instruments to zero. The metric summarizes effectiveness in a single number; to the extent it identifies the same countries at the extremes, the efficiency pattern is a property of the realized data rather than an artifact of the frontier construction.

The median country earned 2.8 pp-quarters per point of debt; the high returns belong to Norway (3.7), Switzerland (3.6), and Sweden (3.1), all of which pursued liquidity-oriented strategies with limited demand injection. The lowest returns were observed in the United States (1.1), Israel (−0.5), and Chile (−0.2), which relied least on liquidity-oriented support. The extremes thus coincide with the frontier ranking—the efficiency pattern holds in the realized data alone, without the model implementation. Appendix J, Table 35 reports the full panel.

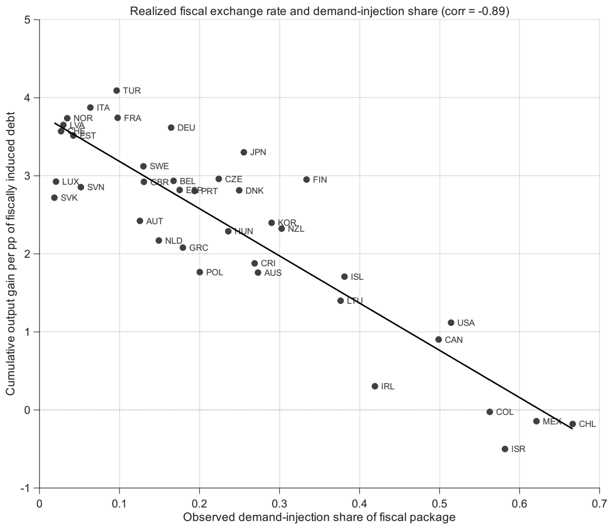


Figure 7: Realized fiscal exchange rate and demand-injection share. *Notes:* Each point is one country. The exchange rate is the cumulative output gain of the observed fiscal package per percentage point of fiscally induced debt, computed against a counterfactual that keeps the country's observed containment path and sets all fiscal instruments to zero. Solid line: linear fit; only selected countries are labeled. The four countries with a negative realized exchange rate (Chile, Colombia, Israel, Mexico) are those with the highest demand-injection shares.

The gradient across countries is almost entirely a demand-injection gradient: the exchange rate correlates at −0.89 with the observed demand-injection share, and it is negative in exactly the four countries with the highest such shares (Chile, Colombia, Israel, and Mexico, at 56 to 67 percent of their packages; Figure 7).

This is the cross-country footprint of the state-dependent multiplier of Section 4: demand injection deployed under binding containment adds debt without adding output—the same mechanism that drives the planner's abandonment of the instrument, visible here in the realized data rather than in the optimization.

Across all three levels of the analysis, the same answer emerges. The weighted planner, the weight-free frontier, and the cross-country comparison identify the identical dominant strategy for confronting the trilemma: fiscal support that is front-loaded and recomposed toward effective liquidity provision and capacity preservation, containment concentrated in the acute waves, and demand injection withheld while restrictions bind. No country's policy—whatever its size—reached its own frontier: the binding constraint on pandemic policy was not the fiscal envelope but its composition and timing. Countries whose packages came closest to this composition—Switzerland, Norway, and Sweden—realized the highest output return per unit of debt; those that relied on scale, late commitment, or demand injection paid for output they did not receive (see Appendix J.1 Table 35 for the full output). And because no instrument is self-financing and debt carries no stabilizing force of its own (Appendix H.8), that price does not expire with the pandemic: the debt accumulated without a corresponding return persists as a diminished fiscal position, narrowing the space available for the next shock. The cost of an inefficient pandemic response is thus paid twice—once in the crisis, and once in the starting conditions it leaves behind.

7 Conclusion

As the fiscal responses of OECD countries during COVID-19 were highly heterogeneous and conditional on pandemic pressure, this paper asked which strategy would have delivered the best attainable outcome in a setting with three mutually incompatible objectives. Within the estimated transition system, the costs of deviation were large: the observed OECD response was strictly Pareto-dominated—a feasible policy path, using no instrument, scale, or adjustment speed beyond what OECD governments demonstrably implemented, would have reduced the discounted output loss by 79 percent, excess mortality by 22 percent, or terminal debt by 6.3 percentage points of GDP, holding the other dimensions at their observed levels—and no individual economy operated on its own frontier. The trilemma itself, however, is binding: debt is dynamically unstable and cannot be stabilized by any instrument, no fiscal measure is self-financing, and containment protects lives only at an output cost. The gains therefore come not from escaping the trade-off but from navigating it efficiently. The composition that does so—the answer to the question's first part—is identical at every level of the analysis: front-loaded liquidity provision and capacity preservation, containment concentrated in the acute waves, and demand injection withheld while restrictions bind. The countries closest to their own frontier ran precisely such packages, and the output return per unit of debt falls sharply with the demand-injection share of the observed package.

For future crises that share the pandemic's structure, these results translate into a compositional heuristic rather than a spending target. A government facing such a shock should build the liquidity architecture first: guarantee and lending facilities, with a preference for guarantees, preserve productive structures through the restricted period at the lowest budgetary cost per unit of output, and their effect operates through the outstanding stock, which rewards anticipation. Wage-subsidy-type instruments earn their place primarily in combination with that liquidity channel, as they preserve capacity in the form of employer–employee matches. Demand injection, by contrast, should remain off while restrictions bind, because its output effect turns negative under tight containment—the instrument is not wasteful in general, but mistimed when deployed as crisis relief during lockdowns. Containment itself is best ramped early and concentrated in the waves, where its epidemiological return is highest, in line with the findings of **?**. None of this requires fiscal capacity beyond what OECD governments actually mobilized; it requires deploying it differently.

These conclusions carry three limitations. First, the transition parameters are reduced-form panel estimates, not causally identified deep parameters, and are therefore exposed to the Lucas critique when evaluated under counterfactual policies. The exposure is limited: the policy menu is restricted to the observed support, so the counterfactuals interpolate within—rather than extrapolate beyond—the variation that identifies the parameters, and the exercise is a retrospective, episode-specific evaluation. Above all, the two mechanisms that drive the normative results—the state-dependent demand-injection multiplier and the dominance of liquidity-type support under containment—emerge as equilibrium properties of the micro-founded models discussed in Section 2 (Guerrieri et al., 2022; **?**; Faria-e Castro, 2021), in which behavior is derived from optimization and the critique does not apply by construction. The transition system reproduces this structural policy ranking; what it adds is the joint quantification with the debt side of the trilemma. Second, the planner solves under perfect foresight of the realized shock paths. The no-anticipation variant, which fixes the pre-shock quarter at the observed policy, prices this foresight at roughly one sixth of the weighted planner's welfare gain and shows that the dominance result does not rest on it: all three frontier improvements remain strictly positive, with unchanged policy composition (Appendix I.3). Third, the model is estimated on aggregate country-quarter data. The common transition technology—the design choice that makes cross-country policy differences comparable in the first place—does not impose homogeneity in levels: the estimated country fixed effects absorb time-invariant differences in trend growth, fiscal positions, and health-system capacity, are carried into the country-specific planner problems, and ensure that the common coefficients are identified from within-country variation. What it does impose is equality of the marginal responses: genuine cross-country differences in fiscal multipliers and health systems remain outside the model, which the fit diagnostics flag rather than resolve.

The paper's contribution is a framework, a dataset, and a benchmark. The trilemma formalizes why pandemic policy cannot be understood as standard stabilization policy; the measure-level database of 2,301 fiscal interventions makes the composition of the pandemic fiscal response observable; and the estimated planner problem converts both into a model-implied standard against which realized strategies can be judged. The natural next steps follow from the limitations: micro-founding the capacity-preservation channel, extending the calibration to country-specific mortality and take-up profiles, and embedding genuine real-time uncertainty—learning about the shock as it unfolds, beyond the one-quarter no-anticipation benchmark. The retrospective answer, however, already stands: in a pandemic, the binding constraint on fiscal policy is not how much governments can spend, but how well what they spend matches the state of the crisis.

References

Abhijeet and Suman Chakravorty. A sequential quadratic programming perspective on optimal control. *arXiv preprint*, 2025.

Daron Acemoglu, Victor Chernozhukov, Iván Werníng, and Michael D. Whinston. Optimal targeted lockdowns in a multigroup SIR model. *American Economic Review: Insights*, 3(4):487–502, December 2021. doi: 10.1257/aeri.20200590.

Fernando Alvarez, David Argente, and Francesco Lippi. A simple planning problem for COVID-19 lock-down, testing, and tracing. *American Economic Review: Insights*, 3 (3):367–382, 2021.

Fernando E. Alvarez, David Argente, and Francesco Lippi. A simple planning problem for COVID-19 lockdown. Working Paper 26981, National Bureau of Economic Research, Cambridge, MA, April 2020. URL `https://www.nber.org/papers/w26981`.

Badar Nadeem Ashraf and John W. Goodell. Covid-19 social distancing measures and economic growth: Distinguishing short- and long-term effects. *Finance Research Letters*, 47:102639, 2022. doi: 10.1016/j.frl.2021.102639.

Andrew Atkeson. What will be the economic impact of COVID-19 in the US? Rough estimates of disease scenarios. NBER Working Paper 26867, National Bureau of Economic Research, 2020.

David Autor, David Cho, Leland D. Crane, Mita Goldar, Byron Lutz, Joshua Montes, William B. Peterman, David Ratner, Daniel Villar, and Ahu Yildirmaz. The $800 billion Paycheck Protection Program: Where did the money go and why did it go there? *Journal of Economic Perspectives*, 36(2):55–80, May 2022. doi: 10.1257/jep.36.2.55. URL `https://www.aeaweb.org/articles?id=10.1257/jep.36.2.55`.

David Baqaee and Emmanuel Farhi. Supply and demand in disaggregated keynesian economies with an application to the covid-19 crisis. *American Economic Review*, 112 (5):1397–1436, 2022. doi: 10.1257/aer.20201229.

Robert J Barro, José F Ursúa, and Joanna Weng. The coronavirus and the great influenza pandemic: Lessons from the "spanish flu" for the coronavirus's potential effects on mortality and economic activity. Working Paper 26866, National Bureau of Economic Research, March 2020. URL `http://www.nber.org/papers/w26866`.

Christian Bayer, Benjamin Born, Ralph Luetticke, and Gernot J. Müller. The coronavirus stimulus package: How large is the transfer multiplier. *The Economic Journal*, 133(652): 1318–1347, May 2023. doi: 10.1093/ej/uead003.

Dimitri P. Bertsekas. *Dynamic Programming and Optimal Control: Volume I.* Athena Scientific, Nashua, NH, 4th edition, 2017.

Olivier Blanchard and Roberto Perotti. An empirical characterization of the dynamic effects of changes in government spending and taxes on output. *The Quarterly Journal of Economics*, 117(4):1329–1368, 2002. doi: 10.1162/003355302320935043.

Jan M. Brauner, Sören Mindermann, Mrinank Sharma, David Johnston, John Salvatier, Tomáš Gavenčiak, Anna B. Stephenson, Gavin Leech, George Altman, Vladimir Mikulik, Alexander John Norman, Joshua Teperowski Monrad, Tamay Besiroglu, Hong Ge, Meghan A. Hartwick, Yee Whye Teh, Leonid Chindelevitch, Yarin Gal, and Jan Kulveit. Inferring the effectiveness of government interventions against COVID-19. *Science*, 371(6531):eabd9338, 2021. doi: 10.1126/science.abd9338.

Nicholas F. Brazeau, Robert Verity, Sarah Jenks, Han Fu, Charles Whittaker, Peter Winskill, Ilaria Dorigatti, Patrick G. T. Walker, Steven Riley, Rupert P. Schnekenberg, Henrique Hoeltgebaum, Thomas A. Mellan, Swapnil Mishra, H. Juliette T. Unwin, Oliver J. Watson, Zulma Cucunubá, Marc Baguelin, Lilith K. Whittles, Samir Bhatt, Azra C. Ghani, Neil M. Ferguson, and Lucy C. Okell. Covid-19 infection fatality ratio: Estimates from seroprevalence. Technical Report Report 34, Imperial College London, 2020.

Luis Felipe Céspedes, Roberto Chang, and Andrés Velasco. The macroeconomics of a pandemic: A minimalist framework. *Journal of International Money and Finance*, 127: 102668, 2022. doi: 10.1016/j.jimonfin.2022.102668.

Raj Chetty, John N. Friedman, Michael Stepner, and The Opportunity Insights Team. The economic impacts of COVID-19: Evidence from a new public database built using private sector data. Working Paper 27431, National Bureau of Economic Research, Cambridge, MA, June 2020. URL `https://www.nber.org/papers/w27431`.

Olivier Coibion, Yuriy Gorodnichenko, and Michael Weber. How did u.s. consumers use their stimulus payments? Working Paper 27693, National Bureau of Economic Research, Cambridge, MA, August 2020. URL `https://www.nber.org/papers/w27693`.

COVID-19 Forecasting Team. Variation in the COVID-19 infection-fatality ratio by age, time, and geography during the pre-vaccine era: A systematic analysis. *The Lancet*, 399(10334):1469–1488, 2022. doi: 10.1016/S0140-6736(21)02867-1.

James Cross, Shu Kit Ng, and Paul Scuffham. Trading health for wealth: The effect of covid-19 response stringency. *International Journal of Environmental Research and Public Health*, 17(23):8725, 2020. doi: 10.3390/ijerph17238725.

Nicholas G. Davies, Christopher I. Jarvis, W. John Edmunds, Nicholas P. Jewell, Karla Diaz-Ordaz, and Ruth H. Keogh. Increased mortality in community-tested cases of sars-cov-2 lineage b.1.1.7. *Nature*, 593:270–274, 2021. doi: 10.1038/s41586-021-03426-1.

Brechje de Gier, Stijn Andeweg, Jantien A. Backer, Susan J. M. Hahné, Susan van den Hof, Hester E. de Melker, and Mirjam J. Knol. Vaccine effectiveness against sars-cov-2 transmission and infections among household and other close contacts of confirmed cases, the netherlands, february to may 2021. *Eurosurveillance*, 26(31):2100640, 2021. doi: 10.2807/1560-7917.ES.2021.26.31.2100640.

Pragyan Deb, Davide Furceri, Jonathan D. Ostry, Nour Tawk, and Naihan Yang. The effects of fiscal measures during COVID-19. IMF Working Paper 2021/262, International Monetary Fund, 2021. URL `https://www.imf.org/en/Publications/WP/Issues/2021/11/05/The-Effects-of-Fiscal-Measures-During-COVID-19-504347`.

Martin S. Eichenbaum, Sergio Rebelo, and Mathias Trabandt. The macroeconomics of testing and quarantining. NBER Working Papers 27104, National Bureau of Economic Research, Inc, May 2020. URL `https://ideas.repec.org/p/nbr/nberwo/27104.html`.

Martin S Eichenbaum, Sergio Rebelo, and Mathias Trabandt. The macroeconomics of epidemics. *The Review of Financial Studies*, 34(11):5149–5187, 2021.

Christian Erikstrup, Alexander D. Laksafoss, Julie Gladov, Katrine A. Kaspersen, Susan Mikkelsen, Lise Hindhede, Jesper K. Boldsen, Søren W. Jørgensen, Steen Ethelberg, Dorte K. Holm, Søren D. Nielsen, and Henrik Ullum. Seroprevalence and infection fatality rate of the sars-cov-2 omicron variant in denmark: A nationwide serosurveillance study. *The Lancet Regional Health – Europe*, 21:100479, 2022. doi: 10.1016/j.lanepe.2022.100479.

European Parliament. Overview of How Major Economies Have Responded to the COVID-19 Pandemic. Technical Report PE 689.450, European Parliament, Policy Department for Economic, Scientific and Quality of Life Policies, February 2022. URL `https://www.europarl.europa.eu/RegData/etudes/STUD/2022/689450/IPOL_STU(2022)689450_EN.pdf`. Accessed: 2026-06-15.

European Systemic Risk Board. Monitoring the Financial Stability Implications of COVID-19 Support Measures. Technical report, European Systemic Risk Board, September 2021. URL `https://www.esrb.europa.eu/pub/pdf/reports/esrb.20210908.monitoring_the_financial_stability_implications_of_COVID-19_support_measures~3b86797376.en.pdf`. Accessed: 2026-06-15.

Miguel Faria-e Castro. Fiscal policy during a pandemic. *Journal of Economic Dynamics and Control*, 125:104088, 2021. doi: 10.1016/j.jedc.2021.104088.

Seth Flaxman, Swapnil Mishra, Axel Gandy, H. Juliette T. Unwin, Thomas A. Mellan, Helen Coupland, Charles Whittaker, Harrison Zhu, Tresnia Berah, Jeffrey W. Eaton, Mélodie Monod, Pablo N. Perez-Guzman, Nora Schmit, Lucia Cilloni, Kylie E. C. Ainslie, Marc Baguelin, Adhiratha Boonyasiri, Olivia Boyd, Lorenzo Cattarino, Laura V. Cooper, Zulma Cucunubá, Gina Cuomo-Dannenburg, Amy Dighe, Bimandra Djaafara, Ilaria Dorigatti, Sabine L. van Elsland, Richard G. FitzJohn, Katy A. M. Gaythorpe, Lily Geidelberg, Nicholas C. Grassly, William D. Green, Timothy Hallett, Arran Hamlet, Wes Hinsley, Ben Jeffrey, Edward Knock, Daniel J. Laydon, Gemma Nedjati-Gilani, Pierre Nouvellet, Kris V. Parag, Igor Siveroni, Hayley A. Thompson, Robert Verity, Erik Volz, Caroline E. Walters, Haowei Wang, Yuanrong Wang, Oliver J. Watson, Peter Winskill, Xiaoyue Xi, Patrick G. T. Walker, Azra C. Ghani, Christl A. Donnelly, Steven Riley, Michaela A. C. Vollmer, Neil M. Ferguson, Lucy C. Okell, Samir Bhatt, and Imperial College COVID-19 Response Team. Estimating the effects of non-pharmaceutical interventions on covid-19 in europe. *Nature*, 584 (7820):257–261, August 2020. ISSN 1476-4687. doi: 10.1038/s41586-020-2405-7. URL `https://doi.org/10.1038/s41586-020-2405-7`.

Luca Fornaro and Martin Wolf. Covid-19 coronavirus and macroeconomic policy. CEPR Discussion Papers 14529, C.E.P.R. Discussion Papers, 2020. URL `https://EconPapers.repec.org/RePEc:cpr:ceprdp:14529`.

G. Giordano, M. Colaneri, A. Di Filippo, F. Blanchini, P. Bolzern, G. De Nicolao, P. Sacchi, P. Colaneri, and R. Bruno. Modeling vaccination rollouts, sars-cov-2 variants and the requirement for non-pharmaceutical interventions in italy. *Nature Medicine*, 27(6):993–998, June 2021. doi: 10.1038/s41591-021-01334-5. URL `https://doi.org/10.1038/s41591-021-01334-5`. Epub 2021 Apr 16.

Austan Goolsbee and Chad Syverson. Fear, lockdown, and diversion: Comparing drivers of pandemic economic decline 2020. *Journal of Public Economics*, 193: 104311, 2021. ISSN 0047-2727. doi: 10.1016/j.jpubeco.2020.104311. URL `https://www.sciencedirect.com/science/article/pii/S0047272720301754`.

Daniel V. Gordon, R. Quentin Grafton, and Stein Ivar Steinshamn. Cross-country effects and policy responses to covid-19 in 2020: The nordic countries. *Economic Analysis and Policy*, 71:198–210, 2021. doi: 10.1016/j.eap.2021.04.015.

Michael Greenstone and Vishan Nigam. Does social distancing matter? Working Paper 2020-26, University of Chicago, Becker Friedman Institute for Economics, 2020.

Veronica Guerrieri, Guido Lorenzoni, Ludwig Straub, and Iván Werning. Macroeconomic implications of COVID-19: Can negative supply shocks cause demand shortages? *American Economic Review*, 112(5):1437–1474, 2022. doi: 10.1257/aer.20201063.

Thomas Hale, Noam Angrist, Rafael Goldszmidt, Beatriz Kira, Anna Petherick, Toby Phillips, Samuel Webster, Emily Cameron-Blake, Laura Hallas, Saptarshi Majumdar, and Helen Tatlow. A global panel database of pandemic policies (Oxford COVID-19 Government Response Tracker). *Nature Human Behaviour*, 5(4):529–538, 2021. doi: 10.1038/s41562-021-01079-8.

Robert E. Hall, Charles I. Jones, and Peter J. Klenow. Trading off consumption and COVID-19 deaths. *Federal Reserve Bank of Minneapolis Quarterly Review*, 42(1):1–14, 2020.

William S. Hart, Elizabeth Miller, Nick J. Andrews, Pauline Waight, Philip K. Maini, Sebastian Funk, and Robin N. Thompson. Generation time of the alpha and delta SARS-CoV-2 variants: An epidemiological analysis. *The Lancet Infectious Diseases*, 22 (5):603–610, 2022. doi: 10.1016/S1473-3099(22)00001-9.

International Monetary Fund. Fiscal monitor: Helping people bounce back. Technical report, International Monetary Fund, Washington, DC, April 2022.

Callum Jones, Thomas Philippon, and Venky Venkateswaran. Optimal mitigation policies in a pandemic: Social distancing and working from home. *The Review of Financial Studies*, 34(11):5188–5223, 09 2021. ISSN 0893-9454. doi: 10.1093/rfs/hhab076. URL `https://doi.org/10.1093/rfs/hhab076`.

Òscar Jordà and Alan M. Taylor. Local projections. *Journal of Economic Literature*, 63 (1):59–110, 2025. doi: 10.1257/jel.20241521.

Òscar Jordà, Sanjay R Singh, and Alan M Taylor. Longer-run effects of pandemics. *The Review of Economic Studies*, 89(1):166–195, 2022.

Greg Kaplan, Benjamin Moll, and Giovanni L. Violante. The great lockdown and the big stimulus: Tracing the pandemic possibility frontier for the u.s. NBER Working Paper 27794, National Bureau of Economic Research, 2020.

Michael König and Adalbert Winkler. Covid-19 and economic growth: Does good government performance pay off? *Intereconomics*, 55(4):224–231, 2020. doi: 10.1007/s10272-020-0906-0.

Eric Anthony Lacey, Joseph Paul Massad, and Robert Johann Utz. A review of fiscal policy responses to COVID-19. Equitable growth, finance and institutions insight, World Bank Group, Washington, DC, 2021. URL `https://hdl.handle.net/10986/35904`. World Bank report number 161128.

Andrew T. Levin, William P. Hanage, Nana Owusu-Boaitey, Kensington B. Cochran, Seamus P. Walsh, and Gideon Meyerowitz-Katz. Assessing the age specificity of infection fatality rates for COVID-19: Systematic review, meta-analysis, and public policy implications. *European Journal of Epidemiology*, 35(12):1123–1138, 2020a. doi: 10.1007/s10654-020-00698-1.

Andrew T. Levin, William P. Hanage, Nana Owusu-Boaitey, Kensington B. Cochran, Seamus P. Walsh, and Gideon Meyerowitz-Katz. Assessing the age specificity of infection fatality rates for COVID-19: Systematic review, meta-analysis, and public policy implications. *European Journal of Epidemiology*, 35(12):1123–1138, 2020b. doi: 10.1007/s10654-020-00698-1.

Weiwei Li and Emanuel Todorov. Iterative linear quadratic regulator design for nonlinear biological movement systems. In *Proceedings of the 1st International Conference on Informatics in Control, Automation and Robotics (ICINCO)*, pages 222–229, 2004.

Zachary J. Madewell, Yang Yang, Ira M. Longini, M. Elizabeth Halloran, Alessandro Vespignani, and Natalie E. Dean. Rapid review and meta-analysis of serial intervals for SARS-CoV-2 delta and omicron variants. *BMC Infectious Diseases*, 23(1):429, 2023. doi: 10.1186/s12879-023-08407-5.

Mattia Manica, Alfredo De Bellis, Giorgio Guzzetta, Pamela Mancuso, Massimo Vicentini, Francesco Venturelli, Alessandro Zerbini, Eufemia Bisaccia, Maria Litvinova, Francesco Menegale, Carla Molina Grané, Piero Poletti, Valentina Marziano, Agnese Zardini, Valeria d'Andrea, Filippo Trentini, Antonino Bella, Flavia Riccardo, Patrizio Pezzotti, Marco Ajelli, Paolo Giorgi Rossi, Stefano Merler, and Reggio Emilia COVID-19 Working Group. Intrinsic generation time of the SARS-CoV-2 omicron variant: An observational study of household transmission. *The Lancet Regional Health – Europe*, 19: 100446, 2022. doi: 10.1016/j.lanepe.2022.100446.

Valentina Marziano, Giorgio Guzzetta, Alessia Mammone, Flavia Riccardo, Piero Poletti, Filippo Trentini, Mattia Manica, Andrea Siddu, Paola Stefanelli, Antonino Bella, Marco Ajelli, Silvio Brusaferro, Giovanni Rezza, Stefano Merler, and Patrizio Pezzotti. Estimating sars-cov-2 infections and associated changes in covid-19 severity and fatality. *Influenza and Other Respiratory Viruses*, 17(8):e13181, 2023. doi: 10.1111/irv.13181.

Edouard Mathieu, Hannah Ritchie, Lucas Rodés-Guirao, Cameron Appel, Daniel Gavrilov, Charlie Giattino, Joe Hasell, Bobbie Macdonald, Saloni Dattani, Diana Beltekian, Esteban Ortiz-Ospina, and Max Roser. Excess mortality during the coronavirus pandemic (covid-19). *Our World in Data*, 2020. https://archive.ourworldindata.org/20260223-071105/excess-mortality-covid.html.

Edouard Mathieu, Hannah Ritchie, Lucas Rodés-Guirao, Cameron Appel, Charlie Giattino, Joe Hasell, Bobbie Macdonald, Saloni Dattani, Diana Beltekian, Esteban Ortiz-Ospina, and Max Roser. A global database of COVID-19 vaccinations. *Nature Human Behaviour*, 5(7):947–953, 2021. doi: 10.1038/s41562-021-01122-8.

Gideon Meyerowitz-Katz and Lea Merone. A systematic review and meta-analysis of published research data on covid-19 infection fatality rates. *International Journal of Infectious Diseases*, 101:138–148, 2020. doi: 10.1016/j.ijid.2020.09.1464.

W. Msemburi, A. Karlinsky, V. Knutson, S. Aleshin-Guendel, S. Chatterji, and J. Wakefield. The who estimates of excess mortality associated with the covid-19 pandemic. *Nature*, 613(7942):130–137, 2023. doi: 10.1038/s41586-022-05522-2. URL `https://doi.org/10.1038/s41586-022-05522-2`.

Robert A. Mundell. Capital mobility and stabilization policy under fixed and flexible exchange rates. *Canadian Journal of Economics and Political Science/Revue canadienne d'Economique et de Science politique*, 29(4):475–485, 1963. doi: 10.2307/139336.

Tommy Nyberg, Neil M. Ferguson, Sophie G. Nash, Harriet H. Webster, Seth Flaxman, Nick Andrews, Wes Hinsley, Jamie Lopez Bernal, Meaghan Kall, Samir Bhatt, Paula Blomquist, Asad Zaidi, Erik Volz, Nurin Abdul Aziz, Katie Harman, Sebastian Funk, Sam Abbott, Russell Hope, Andre Charlett, Meera Chand, Azra C. Ghani, Shaun R. Seaman, Gavin Dabrera, Daniela De Angelis, Anne M. Presanis, Simon Thelwall, and COVID-19 Genomics UK (COG-UK) Consortium. Comparative analysis of the risks of hospitalisation and death associated with SARS-CoV-2 omicron (B.1.1.529) and delta (B.1.617.2) variants in england: A cohort study. *The Lancet*, 399(10332):1303–1312, 2022. doi: 10.1016/S0140-6736(22)00462-7.

OECD. *OECD Economic Outlook, Volume 2020 Issue 2*. OECD Publishing, Paris, 2020. URL `https://www.oecd.org/content/dam/oecd/en/publications/reports/2020/12/oecd-economic-outlook-volume-2020-issue-2_2e595bb6/39a88ab1-en.pdf`. Accessed: 2026-06-12.

OECD. *Health at a Glance 2021: OECD Indicators*. OECD Publishing, Paris, 2021a. doi: 10.1787/ae3016b9-en. See Chapter 2: The health impact of COVID-19.

OECD. *OECD Sovereign Borrowing Outlook 2021*. OECD Publishing, Paris, 2021b. doi: 10.1787/48828791-en. URL `https://doi.org/10.1787/48828791-en`.

OECD. *OECD Sovereign Borrowing Outlook 2023*. OECD Publishing, Paris, 2023. doi: 10.1787/09b4cfba-en. URL https://doi.org/10.1787/09b4cfba-en.

OECD. Quarterly national accounts. OECD National Accounts Statistics (database), 2024. URL https://doi.org/10.1787/data-00017-en. Accessed: 2024.

Evi Pappa, Alejandro Ramos, and Eugenia Vella. Fiscal policy in the COVID-19 era: Evaluating short-term effectiveness. *Journal of International Money and Finance*, 124: 102614, 2022. doi: 10.1016/j.jimonfin.2022.102614.

Robert Price, Thai-Thanh Dang, and Jarmila Botev. Adjusting fiscal balances for the business cycle: New tax and expenditure elasticity estimates for oecd countries. OECD Economics Department Working Papers 1275, OECD Publishing, Dec 2015. URL https://ideas.repec.org/p/oec/ecoaaa/1275-en.html.

Valerie A. Ramey and Sarah Zubairy. Government spending multipliers in good times and in bad: Evidence from us historical data. *Journal of Political Economy*, 126(2): 850–901, 2018. doi: 10.1086/696277.

RECOVERY Collaborative Group. Dexamethasone in hospitalized patients with covid-19. *New England Journal of Medicine*, 384:693–704, 2021. doi: 10.1056/NEJMoa2021436.

Lisa A. Robinson, Ryan Sullivan, and Jason F. Shogren. Do the benefits of COVID-19 policies exceed the costs? exploring uncertainties in the age–VSL relationship. *Risk Analysis*, 41(5):761–770, 2021.

Thomas Ward, Martyn Fyles, Ian Hall, Thomas House, Lorenzo Pellis, Francesca Scarabel, Ellen Nixon, Shweta Bansal, Trevelyan J. McKinley, and Matt J. Keeling. The real-time infection hospitalisation and fatality risk across the covid-19 pandemic in england. *Nature Communications*, 15:4628, 2024. doi: 10.1038/s41467-024-47199-3.

World Bank. COVID-19 to Plunge Global Economy into Worst Recession since World War II. Press release, June 2020. URL https://www.worldbank.org/en/news/press-release/2020/06/08/covid-19-to-plunge-global-economy-into-worst-recession-since-world-war-ii. Accessed: 2026-06-12.

8 Appendix

A Data Appendix

All used materials and data are available on the corresponding GitHub page *Pandemic Trilemma*. The exact location of the following main elements are summarized in the following table:

Table 12: Replication Files and Documentation

Component	**Repository Location**
README	`README.md`
Fiscal Measures Codebook	`Files/text/codebook`
Data	`Files/data`
R Code	`Files/code/R`
MATLAB Code	`Files/code/matlab`
Figures and Tables	`Files/output`

B External Validation: The Trilemma in Comparative Perspective

To establish that the pandemic trilemma is not simply a generic feature of crisis management, I apply the four constitutive properties from Section 2 to four alternative crisis types: armed conflict, natural disasters, financial crises, and climate change. The purpose is diagnostic. A crisis satisfies the pandemic-style trilemma only if the threat is self-reinforcing, the suppression instrument is contractionary, which in turn creates the need for fiscal compensation. Table 13 summarizes the comparison.

Table 13: Diagnostic Comparison of Crisis Types

	Pandemic	**Armed conflict**	**Natural disaster**	**Financial crisis**	**Climate change**
Exogenous self-reinforcing threat	✓	Partial	×	×	✓
Contractionary suppression instrument	✓	×	n/a	×	Partial
Debt-generating fiscal compensation	✓	✓	✓	✓	✓
Endogenous fiscal feedback	✓	Partial	Partial	✓	✓
Pandemic-style trilemma	✓	×	×	×	Analogous, but not directly comparable

Notes: ✓ indicates that the property is satisfied; × indicates that the property is not satisfied; *Partial* indicates that the property is satisfied only in a restricted or non-structural sense; and *n/a* denotes that the property is not applicable. Climate change is structurally closest to the pandemic case but differs in its time scale, observability, and terminal condition.

Armed Conflict

Armed conflict shares some surface-level features with the pandemic. It can generate fiscal pressures through military expenditure and may trigger automatic stabilizers if economic activity is disrupted. However, it does not satisfy the full trilemma structure.

Property 1 is at most partially satisfied. While armed conflicts can escalate, this escalation is not governed by an autonomous biological growth process comparable to viral prevalence with $\rho_\theta > 1$. Military escalation is mediated by strategic decisions, military capacity, and political choices, rather than by a mechanical reproduction process that compounds independently of policy.

The main difference is Property 2. The instruments used to address the threat—military mobilization, defense production, and related public spending—are not inherently contractionary in the pandemic sense. They often increase demand, absorb idle resources, and raise employment. The instrument that addresses the threat therefore works with, rather than against, short-run economic stabilization. This breaks the circular mechanism that defines the pandemic trilemma. In addition, war often destroys physical capital, whereas the pandemic primarily suppresses economic flows while leaving much of the capital stock intact. The post-war problem is therefore a reconstruction problem; the pandemic problem is a stabilization problem under binding containment.

Natural Disasters

Natural disasters are the clearest example of an exogenous shock, but they fail the self-reinforcing-dynamics condition. An earthquake, flood, or storm strikes, causes damage, and then dissipates. The shock does not grow endogenously with a persistence parameter above unity. There is therefore no equivalent of viral prevalence that must be suppressed to prevent future growth.

As a result, Property 2 is not applicable. The planner does not need to impose a contractionary suppression instrument to stop the shock from reproducing. The relevant policy problem is reconstruction: restoring infrastructure, housing, and productive capacity after the event. Reconstruction spending may generate debt and output losses may activate automatic stabilizers, so Properties 3 and 4 can be present. But without a self-reinforcing threat and without a contractionary suppression instrument, the circular trilemma structure does not arise.

Financial Crises

Financial crises fail Property 1 because the shock is endogenous to the economic system. It emerges from leverage, balance-sheet fragility, asset-price misalignment, or institutional failure within the financial sector. Even when crises propagate internationally, the transmission operates through financial linkages and market expectations rather than through an external biological process.

Property 2 is also violated. The main crisis-fighting instruments—liquidity provision, bank recapitalization, lender-of-last-resort interventions, and asset purchases—are stabilizing or expansionary for the real economy. They are designed to prevent a collapse in credit, demand, and employment. This reverses the pandemic logic. In a pandemic, the instrument that suppresses the threat directly reduces output. In a financial crisis, the instrument that addresses the threat is intended to prevent output from falling further. Subsequent austerity or fiscal consolidation may be contractionary, but these are responses to the fiscal legacy of the crisis, not the crisis-fighting instruments themselves. The pandemic-style trilemma therefore does not emerge.

Climate Change

Climate change is the closest structural analogue to the pandemic trilemma, but it is not directly comparable.

Property 1 is broadly satisfied. Greenhouse gas accumulation is an external process with self-reinforcing features: emissions accumulate in the atmosphere, feedback loops can amplify warming, and tipping points may make the process harder to reverse. The health and welfare costs are also real, operating through heat stress, extreme weather, food-system disruption, and other climate-related damages.

Property 2 is partially satisfied. Climate mitigation can be contractionary for carbon-intensive sectors and may impose transitional costs on output, employment, and relative prices. However, unlike pandemic containment, climate policy does not mechanically shut down broad economic activity. Mitigation also induces substitution, innovation, and sectoral reallocation. The suppression instrument is therefore not contractionary in the same immediate and economy-wide sense as pandemic containment.

Properties 3 and 4 are present. The transition to a low-carbon economy can require substantial fiscal compensation, public investment, and support for affected households and firms. These measures can generate public liabilities. If mitigation or transition costs depress output, automatic stabilizers can further worsen the fiscal position. In this sense, climate change contains an analogous debt-output-policy tension.

The differences are nevertheless decisive. First, the relevant time scale is measured in decades rather than quarters. This turns the discount factor from a technical modeling choice into a normative question of intergenerational welfare. Second, the health consequences are diffuse, statistically attributed, and temporally remote, whereas pandemic mortality is immediate and politically salient. Third, there is no terminal condition analogous to vaccination. In the pandemic, vaccination reduces the fatality and transmission constraints and thereby dissolves the trilemma over a finite horizon. Climate change has no comparable exogenous terminal date; mitigation must be sustained over a much longer horizon.

Climate change therefore represents a potential analogue rather than a direct application of the pandemic framework. A climate trilemma would require different assumptions about the planning horizon, discounting, technological change, and terminal conditions.

The comparison shows that the pandemic case is structurally distinct. Armed conflict, natural disasters, and financial crises each activate parts of the fiscal trade-off, but they do not combine a self-reinforcing external threat with a contractionary suppression instrument. Climate change is the closest analogue, because mitigation can generate output costs and requires fiscal compensation, but its horizon, observability, and absence of a vaccine-like terminal condition make it a different planning problem.

The pandemic therefore provides a uniquely tractable setting in which all four properties bind simultaneously within a finite planning horizon. This is why the pandemic trilemma can be modeled as a dynamic allocation problem over health, output, and public debt, while the comparison cases either lack one of the constitutive properties or require a substantially different modeling framework.

C Further Descriptives

The key descriptive elements needed to understand the data are documented in the following section. You can either run `Analysis.R` directly, which reproduces most of the descriptives, or proceed in the intended order. In the latter case, start with `descriptives.R`. This file reports daily descriptives on the two policy instruments—containment and fiscal support—and on the health outcome, including the construction of $\hat{\theta}$, and ends by aggregating the data to the quarterly level, which in turn enables the quarterly descriptives covering the remaining two outcomes, the output gap and public debt.

C.1 Instrument: Containment

This section presents only the key elements. The code contains further descriptive analyses comparable to the more extensive descriptives provided in the original source of this index, Hale et al. (2021).

Table 14: Containment stringency by country, 2020–2022.

	Level and dispersion				*Tails*		
Country	**Mean**	**Median**	**SD**	**CV**	**%** $S > 60$	**%** $S < 20$	**Peak**
GRC	61.7	61.0	15.4	0.25	53.9	1.3	86.5
ITA	59.8	59.4	12.9	0.22	49.0	0.0	92.7

continued on next page

Table 14 continued

Country	**Mean**	**Median**	**SD**	**CV**	**%** $S > 60$	**%** $S < 20$	**Peak**
IRL	56.9	50.0	22.7	0.40	43.5	1.6	89.6
COL	56.8	54.2	18.3	0.32	37.1	1.6	89.6
PRT	55.9	57.3	15.8	0.28	33.4	1.3	86.5
CRI	54.5	54.2	11.5	0.21	31.1	1.3	79.2
NLD	53.6	57.3	18.1	0.34	38.7	1.6	80.2
ISR	52.5	47.5	19.4	0.37	32.8	1.9	93.8
TUR	51.8	59.4	18.0	0.35	42.6	2.2	75.0
GBR	51.7	54.2	19.7	0.38	39.0	2.8	86.5
SVN	51.5	44.8	19.4	0.38	33.8	2.2	88.5
AUT	51.1	45.4	17.6	0.34	36.7	1.5	80.2
CHL	51.0	56.2	16.7	0.33	33.1	10.3	69.8
DEU	50.8	54.2	18.1	0.36	43.1	7.6	80.2
FRA	50.5	45.8	18.4	0.36	25.8	0.0	86.5
POL	48.9	47.0	21.1	0.43	41.6	4.9	85.4
ESP	48.9	46.9	18.6	0.38	39.2	1.3	83.3
BEL	48.4	46.9	16.4	0.34	15.4	1.9	79.2
CZE	47.4	39.6	18.4	0.39	28.8	1.5	79.2
CAN	46.8	47.9	7.4	0.16	0.4	2.2	60.4
HUN	45.4	43.8	20.5	0.45	35.5	11.3	77.1
SWE	45.3	53.1	19.9	0.44	37.6	14.3	67.5
AUS	45.2	46.9	9.3	0.21	0.0	2.2	58.3
SVK	44.9	43.8	19.6	0.44	34.6	15.2	85.4
LTU	44.5	35.4	21.2	0.48	29.2	7.6	85.4
MEX	44.2	41.7	18.8	0.43	11.0	3.3	80.2
CHE	44.0	43.8	12.6	0.29	8.2	4.6	69.8
DNK	43.5	47.9	17.2	0.40	17.3	12.1	68.8
KOR	42.7	39.7	11.1	0.26	4.5	0.0	80.2
LVA	42.5	43.8	10.7	0.25	3.1	2.1	63.5
USA	42.1	44.3	6.4	0.15	0.0	1.5	51.0
LUX	42.0	40.6	13.6	0.32	9.4	5.1	77.1
NOR	41.9	43.8	17.9	0.43	20.6	11.9	77.1
JPN	34.6	38.5	7.8	0.23	0.0	7.7	44.8
ISL	33.9	33.3	11.4	0.34	2.8	7.7	61.5
EST	32.6	30.0	15.2	0.47	7.5	17.6	69.8
FIN	31.7	32.3	12.9	0.41	8.5	12.1	64.6
NZL	31.5	12.5	26.4	0.84	10.9	55.7	95.8
Mean	46.6	45.7	16.0	0.34	24.3	6.4	76.8

Notes: Statistics are computed on the daily Oxford COVID-19 Government Response Tracker containment and health index over 2020–2022; countries are sorted by their mean level. CV is the coefficient of variation (SD/Mean). The two tail columns report the share of days on which the index exceeds 60 (stringent containment) or falls below 20 (near-normal conditions); Peak is the maximum daily value. The table documents the cross-country heterogeneity that identifies the containment channel: mean stringency ranges from 31.5 (New Zealand) to 61.7 (Greece), and countries with similar means differ sharply in how they got there—New Zealand combines the lowest mean with the highest peak (95.8) and the most days near normal (55.7 percent), an elimination-and-release strategy, whereas the United States and Canada held near-constant intermediate levels (SD of 6.4 and 7.4) and never exceeded 60.

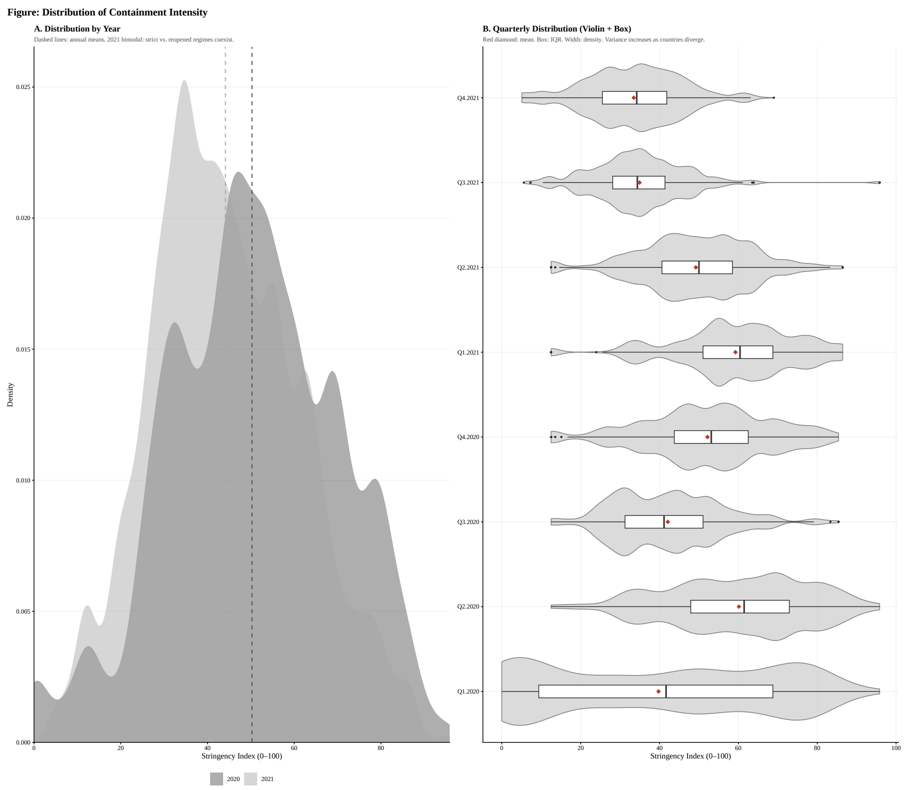


Figure 8: Distribution of StringencyIndex over Countries and Time. *Notes: Panel A shows the distribution of the index across countries in 2020 and 2021; Panel B reports the same distribution by quarter. Countries diverge markedly after Q2.2020.*

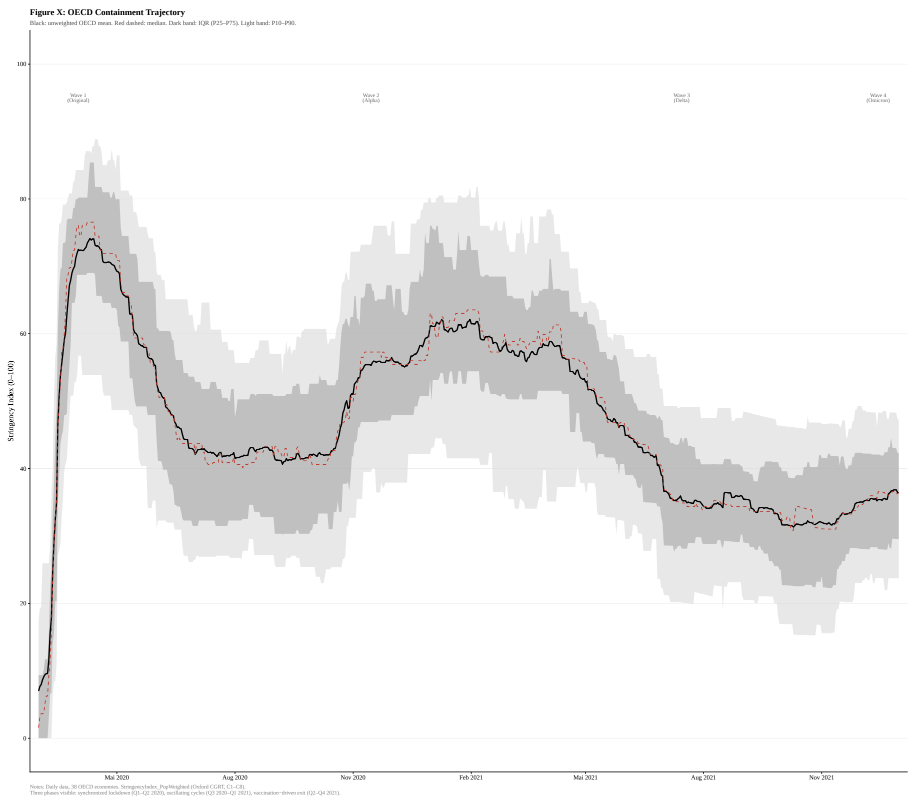


Figure 9: Containment stringency over time: OECD mean and dispersion. *Notes:* The successive waves are clearly visible. Beyond them, the series exhibits a downward trend, while the widening confidence bands indicate that countries diverged increasingly over the course of the pandemic.

C.2 Instrument: Fiscal Measures

Further descriptives are documented in the codebook accompanying the database as well as in the main analysis script.

Table 15: Fiscal Policy Sums by Country

Country	Capacity Preservation	Demand Injection	Health	Above	Loans	Guarantees
AUS	11.94	4.09	0.59	10.17	0.76	1.01
AUT	9.88	1.19	1.68	7.60	0.29	1.99
BEL	16.75	1.70	1.79	5.60	0.22	10.93
CAN	9.40	6.47	1.71	5.24	0.63	3.53
CHE	11.17	0.18	0.71	4.91	0.14	6.11
CHL	13.55	11.89	0.38	2.68	1.62	9.20
COL	3.92	2.64	0.50	1.18	0.54	2.20
CRI	7.65	1.43	0.06	0.39	4.77	2.48
CZE	20.34	2.54	1.38	4.93	0.09	15.32
DEU	27.61	2.63	1.70	7.12	3.10	17.39
DNK	22.40	4.77	0.83	5.47	13.26	3.67
ESP	19.74	1.89	0.08	3.96	2.80	12.98
EST	16.15	0.48	1.68	7.98	2.59	5.59
FIN	9.35	2.32	1.86	2.40	1.43	5.53
FRA	22.38	1.21	0.39	7.05	0.84	14.48
GBR	24.25	2.09	4.73	10.09	0.92	13.24
GRC	15.23	2.48	0.42	7.54	5.77	1.42
HUN	9.21	1.69	4.48	3.60	1.36	4.25
IRL	7.18	4.14	1.54	4.73	1.13	1.31
ISL	7.27	3.54	0.03	4.73	1.08	1.46
ISR	5.75	5.72	1.65	2.90	1.42	1.42
ITA	29.10	0.93	1.32	7.65	1.79	19.66
JPN	29.79	5.34	1.90	4.68	13.18	11.94
KOR	10.87	2.58	0.76	2.48	4.93	3.46
LTU	7.83	3.52	0.21	4.68	1.03	2.12
LUX	8.52	0.11	0.40	3.85	0.24	4.43
LVA	12.75	0.25	0.03	5.49	2.00	5.27
MEX	0.88	1.01	0.09	0.41	0.24	0.23
NLD	9.24	1.15	0.09	5.45	0.39	3.40
NOR	6.69	0.12	0.30	1.62	1.52	3.55
NZL	13.60	4.99	1.99	10.21	1.24	2.15
POL	10.14	1.90	0.33	6.60	0.29	3.25
PRT	12.49	1.41	0.00	2.93	1.50	8.05
SVK	7.64	0.13	0.08	5.11	2.45	0.08
SVN	16.97	0.72	2.31	11.62	0.63	4.73
SWE	7.76	0.70	0.75	2.87	1.72	3.16
TUR	10.87	0.50	0.45	2.56	0.02	8.29
USA	11.74	8.95	4.66	6.03	2.85	2.86

Note:Values are summed over the full sample period. CP = capacity preservation, DI = demand injection, H = health-related spending. Loans and guarantees refer to capacity-preservation below-the-line instruments.

Table 16: Summary Statistics by Transmission Channel

Channel	Measures	Countries	Total GDP (%)	Mean	Median	SD	Min	p25	p75	p90	p99	Max	Zeros	% Measures	% Volume
CP	1224	38	542.30	0.44	0.10	1.21	0	0.02	0.35	1.03	6.09	12.79	4	66.9	78.5
DI	433	38	103.82	0.24	0.06	0.53	0	0.02	0.25	0.70	1.99	7.95	2	23.7	15.0
H	172	38	44.90	0.26	0.11	0.39	0	0.04	0.31	0.79	1.80	2.20	1	9.4	6.5

Note: CP = capacity preservation, DI = demand injection, H = health-related spending. Loans and guarantees refer to capacity-preservation below-the-line instruments.

Table 17: Composition of the fiscal response by transmission channel over time.

	Volume				*Share of total (%)*		
Quarter	**CP**	**DI**	**H**	**Total**	**CP**	**DI**	**H**
Q1.2020	236.40	27.05	7.36	270.81	87.3	10.0	2.7
Q2.2020	188.81	26.97	12.41	228.19	82.7	11.8	5.4
Q3.2020	25.22	13.42	5.78	44.41	56.8	30.2	13.0
Q4.2020	29.34	9.11	6.87	45.32	64.7	20.1	15.2
Q1.2021	25.51	17.55	5.38	48.44	52.7	36.2	11.1
Q2.2021	11.50	5.29	1.63	18.42	62.4	28.7	8.9
Q3.2021	11.10	2.15	2.12	15.37	72.2	14.0	13.8
Q4.2021	14.44	2.28	3.35	20.07	71.9	11.4	16.7

Notes: CP denotes capacity preservation, DI demand injection, and H health-related spending; the classification is by transmission mechanism (Section 3.1). Volumes are in percent of 2019 GDP, summed across the 38 OECD economies; shares are each channel's percentage of the quarterly total and sum to 100 up to rounding. Quarters are dated by the measure's effective date. Two features stand out: the response is heavily front-loaded—the first two quarters account for roughly four fifths of the total volume—and its composition shifts markedly away from capacity preservation once the initial packages are committed, with the demand-injection share peaking in Q1.2021 at 36.2 percent.

C.3 Outcomes

The main elements are included. For further information, please refer to the Analysis.R file.

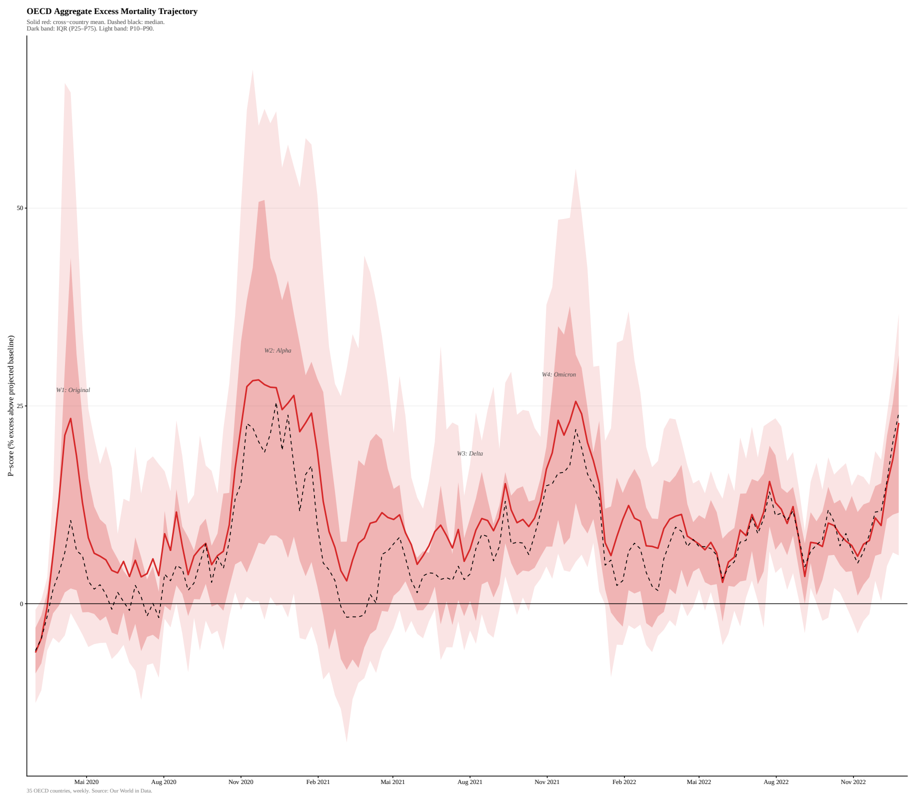


Figure 10: OECD Aggregated Excess Mortality Trajectory *Notes: Calculation based on the P-Score of Mathieu et al. (2020). For further information please reach out to their work.*

Table 18: Output gap by country, 2020–2022.

	Quarterly output gap (pp)				*Cumulative*	*Incidence*
Country	**Mean**	**Trough**	**Peak**	**SD**	**Cum. gap** (pp-quarters)	**% quarters below zero**
ESP	−9.82	−23.5	−5.97	4.88	−118.0	100.0
GBR	−7.33	−23.5	−3.20	5.97	−88.0	100.0
PRT	−7.92	−19.8	−4.53	4.53	−95.1	100.0
MEX	−8.29	−19.4	−4.95	4.31	−99.5	100.0
ITA	−4.13	−18.7	+1.27	5.74	−49.6	75.0
COL	−2.32	−18.0	+3.95	6.66	−27.9	58.3
FRA	−5.66	−17.9	−3.01	4.08	−67.9	100.0
HUN	−6.86	−16.5	−1.26	3.63	−82.4	100.0
GRC	−4.56	−16.2	+0.70	5.34	−54.7	83.3
AUT	−5.68	−15.7	−1.81	4.01	−68.1	100.0
ISL	−8.69	−15.2	−1.83	3.69	−104.0	100.0
CHL	−2.83	−15.1	+3.13	5.41	−33.9	66.7
SVN	−6.43	−14.7	−3.90	2.95	−77.2	100.0
CAN	−4.30	−14.0	−1.35	3.49	−51.6	100.0
CZE	−8.69	−13.7	−4.62	2.07	−104.0	100.0
BEL	−2.81	−13.5	+0.22	3.77	−33.7	83.3
LVA	−4.06	−13.2	−0.08	3.31	−48.7	100.0
NZL	−3.70	−13.0	−1.05	3.25	−44.4	100.0
TUR	+1.94	−12.7	+6.14	5.30	+23.3	16.7
DEU	−5.19	−12.5	−3.59	2.39	−62.3	100.0
ISR	−2.03	−12.0	+2.55	4.06	−24.4	58.3
SVK	−4.55	−11.6	−2.74	2.46	−54.5	100.0

continued on next page

Table 18 continued

Country	**Mean**	**Trough**	**Peak**	**SD**	**Cum. gap**	**% quarters**
POL	−5.35	−11.2	−1.19	2.71	−64.2	100.0
NLD	−2.97	−10.9	+0.73	3.35	−35.7	83.3
EST	−5.41	−10.5	−1.66	2.84	−64.9	100.0
CRI	−4.32	−10.4	−0.91	3.42	−51.9	100.0
JPN	−3.66	−9.83	−2.26	2.08	−43.9	100.0
USA	−2.23	−9.82	+0.03	2.57	−26.8	91.7
SWE	−2.53	−8.86	+0.24	2.36	−30.3	91.7
CHE	−1.43	−8.81	+1.10	2.99	−17.1	50.0
FIN	−4.68	−8.64	−1.80	1.70	−56.2	100.0
LUX	−1.13	−8.36	+2.62	2.80	−13.5	66.7
AUS	−2.15	−8.31	+0.10	2.60	−25.8	91.7
DNK	−2.32	−8.31	+2.99	2.71	−27.9	83.3
LTU	−3.31	−7.48	−1.26	1.86	−39.7	100.0
NOR	−1.49	−6.56	+2.18	2.38	−17.9	75.0
KOR	−3.00	−5.65	−2.03	1.08	−36.0	100.0
IRL*	+6.59	−4.56	+11.30	5.03	+79.1	8.3
Mean	−3.87	−12.9	−0.42	3.55	−46.4	84.6
Median	−4.10	−12.6	−1.22	3.34	−49.2	100.0

Notes: The output gap is the percentage deviation of real GDP from its HP-filtered trend, measured over Q1.2020–Q4.2022 and reported in percentage points; countries are sorted by the depth of their trough. The cumulative gap sums the quarterly gaps over the window and is the object the planner problem minimizes (Section 5). Two features of the table motivate the analysis. First, the dispersion is large and does not follow the depth of the initial shock: Spain and the United Kingdom share the deepest trough (−23.5 pp), yet Spain accumulated a third more cumulative loss, and Iceland and the Czech Republic reach identical cumulative losses (−104 pp-quarters) from very different troughs (−15.2 against −13.7) — the recovery path, not the impact, drives the cumulative cost. Second, Ireland and Türkiye record positive cumulative gaps over the window, reflecting the distortion of Irish GDP through multinational relocations and Türkiye's high-growth, high-inflation trajectory; both are treated as special cases in the cross-country analysis (Section 6.3). * Ireland: see notes on the GDP distortion.

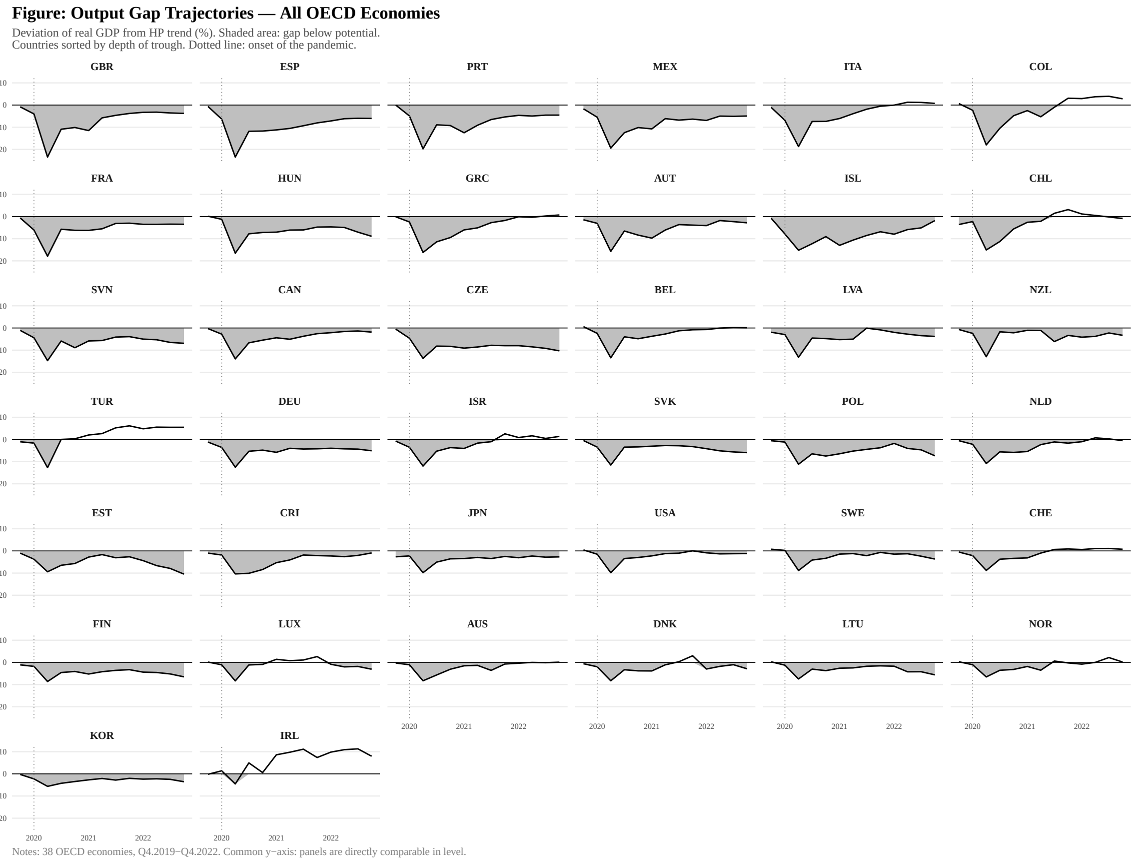


Figure 11: Output gap by country, 2020–2022. *Notes:* Each panel shows one country's output gap, the percentage deviation of real GDP from its HP-filtered trend; panels share a common vertical scale and are sorted by the depth of the trough. In nearly every economy the gap remains below zero for most of the window. Ireland and Türkiye are the exceptions: both record positive cumulative gaps, reflecting the distortion of Irish GDP through multinational relocations and Türkiye's high-growth, high-inflation trajectory. Neither drives the results (Appendix F).

For further descriptive statistics on debt, such as debt levels at the beginning of the pandemic, please refer to the appendix to the calibration section, where they are reported (see Appendix G).

D Identification Strategy

D.1 Identification Concerns and Dynamic-Panel Interpretation

Three features of the pandemic setting complicate the causal interpretation of the estimated transition parameters. First, containment and fiscal support may both respond to pandemic severity: countries experiencing worse outbreaks may impose stricter containment measures and deploy larger fiscal packages, generating a correlation between policy variables and unobserved shocks to output. Second, containment S_{ik} and fiscal support are implemented simultaneously in the same country-quarter units. This creates a multiple-treatment setting in which estimating the conditional contribution of one instrument requires conditioning on the other. Third, treatment intensity is continuous, time-varying, and global, which rules out standard 2×2 difference-in-differences designs.

The maintained assumption is therefore not strict exogeneity, but conditional sequential exogeneity in a dynamic panel setting. The output equation includes the lagged dependent variable $y_{i,k-1}$, which is required by the transition-system interpretation of the model. With a lagged dependent variable, strict exogeneity is generally not available because innovations to current output enter future values of the dependent variable. The relevant assumption is instead that, conditional on country fixed effects, lagged economic states, observed pandemic conditions, and the timing structure implied by the model, the remaining output innovation is not systematically correlated with the predetermined policy variables entering the equation.

This interpretation is consistent with the timing of the fiscal variables. Above-the-line capacity preservation enters with a two-quarter lag, $F^{\text{above}}_{i,k-2}$, while demand injection enters with a one-quarter lag, $F^{\text{DI}}_{i,k-1}$. These measures were typically announced and implemented before the corresponding quarterly GDP outcomes were observed, and often before those outcomes were publicly available. Below-the-line support enters through the cumulative liquidity stock K^{below}_{ik}, reflecting the availability of liquidity rather than a one-period expenditure flow. The lag structure therefore aligns the empirical specification with the economic transmission channels derived in the theoretical framework and limits, although it cannot fully eliminate, reverse causality from realized output to fiscal deployment.

The data provide useful variation for this parameterization strategy. Containment intensity and fiscal deployment are only weakly correlated over the sample period (0.04), and fiscal responses vary substantially more across countries than containment intensity (see next Section D.2 for the full discussion). This mitigates, but does not eliminate, the concern that stricter containment mechanically induces larger fiscal compensation. Countries facing broadly similar pandemic conditions adopted markedly different fiscal strategies, particularly in the composition of support across demand injection, above-the-line capacity preservation, and below-the-line liquidity instruments. This compositional heterogeneity is the main source of variation used to discipline the channel-specific transition parameters.

A direct way to assess whether cross-country selection on crisis severity contaminates the estimates is to decompose the fiscal coefficients into their within- and between-country components (Mundlak Regression). The within and between estimators differ in sign for the key fiscal channels: the between estimator is dominated by the fact that countries with deeper contractions deployed larger fiscal packages, so that the cross-country covariation between fiscal support and output is contaminated by selection on crisis severity. The within estimator, which exploits only the timing and composition of fiscal deployment within a country over the pandemic trajectory, is free of this contamination. This sign reversal is precisely the pattern that motivates the fixed-effects specification: it demonstrates that the identifying variation must come from within-country differences in fiscal composition rather than from cross-country comparisons of total support. Full decomposition results can be found in the R Code in the replication package (see Data Appendix A.

Several remaining concerns are addressed directly in the robustness check section. First, voluntary reductions in economic activity may not be fully captured by mandated stringency (S_{ik}). I therefore add specification where I include excess mortality (d_{ik}) to capture the fear channel implied by the theoretical framework, namely behavioral responses to observed mortality. The coefficient on containment is consequently interpreted as the conditional output association of mandated restrictions given the observed health state. Empirically, the results are not sensitive to this choice: (d_{ik}) is not statistically significant, and excluding it leaves the coefficients of interest largely unchanged. I therefore retain it in the baseline for theoretical consistency and report specifications with the fear term as robustness checks.

Second, vaccination may confound the estimated relationships if countries that vaccinated faster also adjusted fiscal composition differently. I therefore control for vaccination dynamics where relevant and report robustness checks restricting the sample to the pre-vaccination period. Third, the classification of fiscal measures into demand injection and capacity preservation involves judgment. The classification protocol mitigates this concern by assigning measures according to their primary transmission mechanism, and robustness checks under alternative classification rules assess whether the results are driven by borderline cases (see Data Appendix A). Fourth, cross-country spillovers through trade and supply chains may generate correlated shocks across countries. Common shocks are addressed through robustness specifications with alternative time controls, while asymmetric spillovers remain a maintained limitation. Robustness checks controlling for trade openness assess the quantitative relevance of this concern (see Section F).

Inference is based on cluster-robust HC1 standard errors clustered at the country level, allowing for heteroskedasticity and serial correlation within countries over time. Since the number of country clusters is moderate ($N = 38$), I additionally report wild cluster bootstrap (p)-values clustered at the country level as a small-sample robustness check F.

The inclusion of the lagged output gap follows directly from the transition-system interpretation of the model. Pandemic output losses were persistent, and the planner problem requires an estimate of this persistence because current output gaps affect future states. Omitting $y_{i,k-1}$ would force the fiscal and containment variables to absorb part of this persistence and could therefore distort the parameterization of the output transition. Since the output equation (12) includes a lagged dependent variable together with country fixed effects in a short panel, the within estimator of ρ_y is subject to Nickell bias of order $1/T$. I therefore report bias-corrected dynamic-panel estimates in Section F. The persistence-channel results are robust to these corrections.

The estimates should therefore be interpreted as conditional transition parameters under the maintained timing and sequential-exogeneity assumptions rather than as randomized treatment effects. Their role is to discipline the model quantitatively and to assess whether the signs, timing, and relative magnitudes implied by the transition system are consistent with the data.

Moving from this interpretation to causal treatment effects would require exogenous variation in the policy instruments, and it is worth stating explicitly what such an agenda would involve—and why it is not the design chosen here. First, an instrumental-variables strategy would need country-time variation in containment and fiscal deployment that is orthogonal to output innovations. Plausible candidates exist on the fiscal side—pre-pandemic fiscal rules and constitutional debt brakes, supranational programs such as the EU-level guarantee facilities that were assigned on criteria predetermined at the country-quarter level, or the generosity of pre-existing short-time-work systems that mechanically scaled support—and on the containment side, court rulings striking down restrictions or policy spillovers from neighboring jurisdictions. The difficulty is the exclusion restriction: in a global crisis in which a single common shock moves outcomes, policies, and expectations simultaneously, few of these instruments affect output only through the instrumented policy. Second, the measure-level database itself opens a narrative path: because each of the 2,301 measures carries an announcement date and a legislative origin, measures could be classified into those responding to contemporaneous macroeconomic conditions and those predetermined by institutional triggers or supranational programs, with only the latter used for identification. This is, in my view, the most promising extension the database permits. Third, local projections (Jordà and Taylor, 2025)—in their LP-IV form or the heterogeneity-robust LP-DiD implementation—would estimate the dynamic output responses to fiscal impulses without imposing the first-order structure of the transition system. Local projections cannot, however, replace that system here: the planner problem requires a state-space law of motion, which impulse responses alone do not deliver. Their natural role is instead external validation—comparing the model-implied impulse responses to LP estimates of the same shocks—which I regard as the most direct test the framework could face in future work. For the retrospective, episode-specific evaluation conducted in this paper, the conditional transition parameters, disciplined by the timing structure, the within-country identifying variation documented above, and the robustness battery of Appendix F, remain the appropriate object.

D.2 Correlation Matrix and Variation Decomposition

Table 19: Within vs. Between Variation Decomposition

Variable	**Overall SD**	**Between SD**	**Within SD**	**Within (%)**	**Between (%)**
y_{ik} (pp of potential)	4.766	3.018	3.688	77.4	63.3
S_{ik} (0–100)	18.696	6.143	17.658	94.4	32.9
F_{ik}^{CP} (pp GDP)	3.032	0.691	2.953	97.4	22.8
F_{ik}^{DI} (pp GDP)	0.705	0.251	0.659	93.5	35.6
F_{ik}^{H} (pp GDP)	0.381	0.123	0.361	94.7	32.3
d_{ik} (P-score, %)	14.873	7.603	12.783	85.9	51.1
θ_{ik} (% of pop.)	1.889	0.533	1.813	95.9	28.2
Vax_{ik} (% of pop.)	33.038	4.212	32.768	99.2	12.7

Notes: 38 OECD countries, Q1 2020–Q2 2022 ($N = 380$). Overall SD: total standard deviation. Between SD: standard deviation of country means. Within SD: standard deviation of deviations from country means. Within (%) = Within SD / Overall SD × 100. Country fixed effects absorb between-variation; within-variation identifies the coefficients. All policy variables exhibit within-shares above 93%, confirming that the identifying variation survives absorption of country fixed effects.

Table 20: Pairwise Correlation Matrix

	y_{ik}	S_{ik}	F_{ik}^{CP}	F_{ik}^{DI}	d_{ik}	θ_{ik}	Vax_{ik}
y_{ik}	1						
S_{ik}	−0.43	1					
F_{ik}^{CP}	−0.25	0.13	1				
F_{ik}^{DI}	−0.11	0.12	0.25	1			
d_{ik}	−0.09	0.20	−0.15	−0.04	1		
θ_{ik}	0.20	−0.39	−0.19	−0.17	0.32	1	
Vax_{ik}	0.43	−0.56	−0.36	−0.27	0.04	0.53	1

Notes: Pearson correlations, estimation sample (38 countries, Q1 2020–Q2 2022, $N = 380$). Lower triangle reported. The correlations $r(S, F^{CP}) = 0.13$ and $r(S, F^{DI}) = 0.12$ confirm that fiscal composition is only weakly correlated with containment intensity—the key identifying condition for the $S \times F^{CP}$ interaction term.

E Health-Side Estimates

The baseline calibration of the health block takes the infection parameters and the wave-specific mortality mapping from the epidemiological literature. Each parameter is anchored separately: the pre-vaccination transmission potential ρ_θ to the estimated reproduction numbers and generation times of the successive variants (Hart et al., 2022; Manica et al., 2022; Madewell et al., 2023), and its post-vaccination decline to the vaccine-effectiveness evidence on onward transmission (de Gier et al., 2021); the suppression effectiveness ϕ_S to the non-pharmaceutical-intervention studies of Brauner et al. (2021) and Flaxman et al. (2020); and the wave-specific mortality mapping δ_θ to the effective-IFR literature—age-stratified meta-analyses for the ancestral phase (Levin et al., 2020b; Meyerowitz-Katz and Merone, 2020; Brazeau et al., 2020), treatment improvements (RECOVERY Collaborative Group, 2021), variant severity (Davies et al., 2021; Nyberg et al., 2022), and by-variant effective rates for the Delta and Omicron phases (Marziano et al., 2023; Ward et al., 2024; Erikstrup et al., 2022)—as set out in Appendix G. This appendix reports my own reduced-form estimates of the same two equations. They are not used in the baseline; their purpose is to verify that the panel points in the same direction as the external evidence.

Both equations are estimated at weekly frequency on the pre-Omicron window (through 2021Q4), where the infection–mortality link is still operative. Costa Rica, Japan, and Türkiye lack weekly excess-mortality data and drop out, leaving 35 countries and 3,325 country-weeks; the calibration itself covers all 38 economies, using the monthly-interpolated series for the three excluded countries. Containment is the weekly mean of the daily index on a $[0, 1]$ scale. The infection transition $\theta_{k+1} = \rho_\theta(1 - \phi_S S_k)\theta_k$ is estimated as $\theta_{k+1} = \beta_1\theta_k + \beta_2 S_k\theta_k + \mu_i$, so that $\rho_\theta = \beta_1$ and $\phi_S = -\beta_2/\beta_1$; the mortality mapping $d_{k+1} = \delta_\theta\theta_k$ is estimated directly, with d in excess deaths per million per week.

Table 21: Reduced-form estimates of the health block.

	Infection transition			*Mortality mapping*		
	Pooled	Country FE	Country			
+ wave FE	Pooled	Country FE	Country			
+ wave FE						
θ_k	1.044***	1.035***	1.029***	3,448***	3,211***	3,397***
	(0.025)	(0.027)	(0.033)	(135.7)	(137.2)	(153.8)
$S_k\,\theta_k$	−0.290***	−0.325***	−0.336***			
	(0.054)	(0.060)	(0.065)			
Implied structural parameters						
ρ_θ (weekly)	1.044	1.035	1.029	—	—	—
ϕ_S	0.278	0.314	0.326	—	—	—
	(0.046)	(0.051)	(0.054)			
Implied effective IFR (%)	—	—	—	0.345	0.321	0.340
Observations	3,325	3,325	3,325	3,325	3,325	3,325
Within R^2	—	0.823	0.796	—	0.485	0.488

Notes: Weekly panel, 35 OECD countries, 2020Q1–2021Q4 (N = 3,325). Standard errors clustered by country in parentheses; $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. θ is the imputed infection prevalence (share of population), S the weekly mean containment index on a $[0, 1]$ scale, and d excess deaths per million per week. $\phi_S = -\beta_2/\beta_1$ with standard errors by the delta method. The implied effective IFR is $\delta_\theta/10^6$.

The estimates line up with the calibration on the two dimensions that matter for the planner. First, prevalence is explosive before vaccination: the weekly persistence of 1.03–1.04 compounds to 1.45–1.75 over a thirteen-week quarter, bracketing the calibrated quarterly value of $\rho_\theta = 1.50$. Wave-specific slopes (1.05, 1.19, 1.08) have overlapping confidence intervals, which supports the parsimonious two-regime specification rather than a fully wave-varying ρ_θ. Second, the mortality mapping implies an effective infection fatality rate of 0.32–0.35 percent, squarely inside the calibrated wave-specific range and close to its sample-window average of 0.4 percent (Table 24); the coefficient is stable across specifications and dependent variables.

Containment enters the infection transition with the expected sign and is significant throughout, but its magnitude is not directly comparable across frequencies: suppression compounds within the quarter, so a weekly $\phi_S \approx 0.31$ implies a far stronger quarterly suppression factor than the calibrated quarterly $\phi_S = 0.80$ at the sample-mean containment level. The calibration is therefore conservative on this margin, and it is anchored to the non-pharmaceutical-intervention literature (Brauner et al., 2021; Flaxman et al., 2020), since containment is chosen in response to prevalence.

F Robustness

The following table summarizes the robustness checks for the two main outcome equations. The full output is reproduced by running `Analysis.R`, where each check is documented and commented.

F.1 Robustness Analysis Output

This appendix documents the full battery of checks behind the preferred output-gap specification (eq. 12) estimated on 38 OECD countries over $t \in [4, 14]$ (Q4.2019–Q2.2022, $N = 418$), with standard errors clustered by country (HC1). The exercises span six concerns: (i) identification and collinearity, (ii) estimator and dynamic-panel bias, (iii) heterogeneous-treatment / negative-weight contamination, (iv) time effects and sample composition, (v) instrument construction and functional form, and (vi) small-cluster inference. Table 22 lists each check, what it probes, and its outcome.

Table 22: Robustness battery for the V14 output-gap specification

Check	**What it probes**	**Outcome**
A. Identification & collinearity		
S–composition orthogonality	Fiscal mix endogenous to severity?	$r(\bar{S}, \bar{F}^{CP}) \approx 0.13$; composition driven by institutions/safety nets, not contemporaneous severity.
B. Estimator & dynamic-panel bias		
`feols` vs. `plm`	Estimator invariance	Identical point estimates.
Nickell-bias assessment	Short-T dynamic-panel bias	Bias $<$ 0.7pp, negligible vs. SEs; corroborated by Arellano–Bond GMM.
Mundlak / Chamberlain CRE	Fixed vs. random effects	Within-effects = V14; joint test $\chi^2 = 25.7$, $p < 0.001 \Rightarrow$ FE required.
C. Heterogeneous-treatment robustness		
de–Chaisemartin–D'Haultfœuille	Negative TWFE weights	Negative-weight shares: Above 8.7%, DI 2.7% (benign); Below 26.8% (flags its fragility).
D. Time effects & sample composition		
Year-FE / linear trend / Quarter-FE	Common time variation	Above-Flow & DI$\times S$ trend-robust; Below-Stock trend-confounded (sign flips under Year-FE absorbing the same variation).
Start/end restrictions ($t \geq 5, \geq 6, \leq 13$)	Pre-pandemic anchor	Above/DI$\times S$ stable; Below inflates to 5.87 at $t \geq 6$ — the Q4.2019 zero-stock anchor is essential.
Estimation–window (Narrow/Wide/Only-2020)	Window choice	Above 0.37–0.73, DI$\times S$ -0.04 to -0.05 stable; Below significant only in baseline.
Outlier exclusion (TUR, IRL)	Inflation- / MNC-driven GDP	Core unchanged: Above 0.45*, DI$\times S$ -0.035*.
E. Instrument construction & functional form		
Take-up grid (loans 40/60/80%, guar. 25/35/50%)	Below-stock take-up assumptions	Above (0.54) & DI$\times S$ (-0.04) invariant; Below 0.13–0.26 (baseline largest).
Lag selection (Above 0–3, DI 0–2)	Disbursement timing	Only Above lag-2 and DI lag-1 yield sign-correct, theory-consistent estimates.
Alternative containment (peak S, transit mobility)	Measurement of S	Above and push-on-string survive both.
F. Mechanism & heterogeneity		
Tightening vs. loosening asymmetry	Hysteresis / ratchet	$\Delta S^{+} = -0.125^{***}$ vs. $\Delta S^{-} = -0.005$ ($\approx 26\times$): structural support for AR(1) persistence.
Sample splits (S, income, pre-debt, safety net)	Where channels operate	Above strongest in high-S/high-debt; DI push-on-string in all splits; Below only in high-S & weak-safety-net groups what makes sense.
G. Small-cluster inference ($N_{\text{cl}} = 38$)		
Wild-cluster restricted bootstrap-t (CGM 2008, Rademacher)	CRV1 over-rejection	Above-Flow $p = 0.008$; Below & DI$\times S$ marginal ($p \approx 0.06$); qualitative pattern unchanged.
Cluster-jackknife (leave-one-country-out)	Influential clusters	Above-Flow stable; no single country drives the result.

Notes: Dependent variable: output gap y_{ik} (pp of potential GDP). All fiscal variables in pp of 2019 GDP; S on a 0–100 scale. All models include country fixed effects with country-clustered standard errors. Significance: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Above $= F^{CP,\text{above}}_{i,k-2}$ (above-the-line flow); Below $= K^{CP,\text{below}}_{ik}$ (below-the-line accumulated stock); DI$\times S$ = the demand-injection $\times$ stringency interaction.

Taken together, the robustness checks show that the paper's two main transition-system claims do not depend on any single modeling choice, sample window, construction assumption, or inference procedure: capacity preservation has a positive lagged effect on output, and the effect of demand injection is state-dependent, becoming ineffective under tight containment. The below-the-line stock channel is the weakest element of the specification, since the cumulative liquidity stock is mechanically trending and therefore harder to separate from common pandemic time dynamics once richer time controls are introduced.

F.2 Robustness Analysis: Debt Equation

This appendix documents the robustness battery for the preferred debt-transition specification (eq. 13). The dependent variable is the interest-adjusted change in the real debt-to-GDP ratio, which nets out the mechanical debt-service component. It is estimated on 38 OECD countries over $t \in [4, 16]$ (Q4.2019–Q4.2022, $N = 494$), with standard errors clustered by country. Below-the-line commitments enter at the central take-up calibration (loans 60%, guarantees 25%). Table 23 lists every check.

Table 23: Robustness battery for the V15 debt-transition specification

Check	What it probes	Outcome
A. Outcome & output-state definition		
Unadjusted debt change (drop debt-service netting)	DV = Δb instead of $\Delta\tilde{b}$	Near-identical: $\kappa_{\text{loans}} = 0.79^{***}$, $\kappa_{\text{above}} = 0.57^{*}$, $\kappa_H = 1.03^{*}$; netting is not driving results.
Output-gap timing	$y_{i,k-1}$ (main) vs. y_{ik} vs. $y_{i,k-2}$	Fiscal channels stable; y_{ik} absorbs more ($\gamma = -0.20^{***}$), $y_{i,k-2}$ null (-0.01). Predetermined $y_{i,k-1}$ is the conservative choice.
Nominal outcome	DV = nominal debt change	$\kappa_{\text{loans}} = 0.91^{***}$ stable
C. Instrument construction & timing		
Take-up grid (loans 40/60/80%, guar. 25/35/50%)	Below-the-line take-up assumptions	Effective pass-through invariant ($0.40\times 1.22 = 0.60 \times 0.81 \approx 0.49$ for loans); coefficient scales inversely.
Lag below-the-line 1Q	Loans/guarantees lagged	Lagged terms insignificant (0.55, 0.12) $\Rightarrow$ below-the-line affects debt *contemporaneously* (booked at announcement).
Health-spending timing	F^H vs. F^H_{k-1} vs. both	Contemporaneous F^H null (-0.08); $F^H_{k-1} = 1.08^{*}$ $\Rightarrow$ lag-1 (delayed budgetary recording) is correct.
D. Time effects & sample composition		
Estimation window ($t \geq 5$; ≤ 14; ≤ 15)	Inclusion of late pandemic quarters	All coefficients stable (κ_{loans} 0.79–0.82***, κ_H 0.99–1.08*).
Linear trend / Year-FE / Quarter-FE	Common debt drift & deployment cycle	κ_{loans} & κ_H survive throughout; κ_{above} weakens under Year-FE (0.39, n.s.) and Quarter-FE (0.26, n.s.). Quarter-FE over-absorbs the deployment cycle.
Outlier exclusion (TUR, IRL)	Inflation- / MNC-driven debt & GDP	Core unchanged.
E. Small-cluster inference ($N_{\text{cl}} = 38$)		
Wild-cluster restricted bootstrap-t (CGM 2008, Rademacher)	CRV1 over-rejection	γ_y ($p = 0.005$), κ_{above} (0.014), κ_{loans} (0.036), κ_H (0.014) remain significant; κ_{guar} (0.13) and κ_{DI} (0.38) not.
Leave-one-country-out jackknife	Influential clusters	$\kappa_{\text{loans}} \in [0.74, 1.03]$, $\kappa_{\text{above}} \in [0.52, 0.77]$, $\gamma_y \in [-0.14, -0.11]$; no single country overturns the result (most influential: JPN).

Notes: Dependent variable: interest-adjusted real debt change $\Delta\tilde{b}_{ik}$ (pp of 2019 GDP). All fiscal variables in pp of 2019 GDP. All models include country fixed effects with country-clustered standard errors; small-sample correction applied. Significance: $^{*}p < 0.1$, $^{**}p < 0.05$, $^{***}p < 0.01$. Loans = L_{ik} (60% take-up), guarantees = G_{ik} (25% take-up), F^H_{k-1} lagged one quarter. Effective pass-through = coefficient × assumed take-up rate.

Overall, the debt equation's central message—that realized debt accumulation is driven mechanically by above-the-line and below-the-line loans (near one-for-one) and by health outlays recorded with a one-quarter delay, partly offset by automatic stabilizers, while guarantees and demand injection contribute less—it is invariant to the outcome definition, the timing of the output control, instrument-construction assumptions, sample window, and small-cluster inference. Those parameters are robust except under the most aggressive time-fixed-effects controls, which absorb the common debt drift that also identifies it.

G Calibration Details and Additional Validation

This appendix reports additional details on the calibration exercise described in Section 4.3. The purpose of the calibration is to assess whether the estimated and calibrated model can reproduce the main pandemic dynamics under observed policy paths. The validation is conducted on the full panel of 38 OECD economies over Q4 2019–Q4 2022. Output is evaluated over Q4 2019–Q2 2022, corresponding to the main pandemic macroeconomic window used in the output transition, while debt is evaluated through Q4 2022 to capture the fiscal legacy of the pandemic response.

G.1 Health Calibration

Country-level infection states are matched using calibrated innovations, reflecting the fact that observed infection dynamics were strongly affected by variant arrivals, testing regimes, vaccination, and other epidemiological shocks that are not fully captured by the parsimonious transition system. The mortality mapping is wave-specific and converts the infection state into excess deaths using effective population IFRs. These effective IFRs incorporate intrinsic variant severity, treatment improvements, vaccination coverage, and prior immunity. The pre-vaccine IFR calibration is anchored in age-specific and population-level IFR estimates from early meta-analyses and global evidence (Levin et al., 2020b; Meyerowitz-Katz and Merone, 2020; Brazeau et al., 2020; COVID-19 Forecasting Team, 2022). Treatment-related declines are motivated by evidence on dexamethasone (RECOVERY Collaborative Group, 2021), while variant-specific severity and the decline in fatality risk over Delta and Omicron phases are based on evidence from Alpha severity studies and later variant-period IFR estimates (Davies et al., 2021; Marziano et al., 2023; Ward et al., 2024; Erikstrup et al., 2022). Vaccination and prior-immunity effects enter through the effective IFR interpretation and through the post-vaccination reduction in infection persistence (de Gier et al., 2021).

Concretely, the pre-vaccination infection persistence is set to $\rho_\theta = 1.50$; from Q3 2021 onward—the first quarter in which majority vaccination coverage is reached across the OECD—it drops to $\rho_\theta = 0.75$, so the infection state is stable without intervention in the post-vaccination regime (cf. Eq. (31)).

Table 24: Wave-specific effective population IFRs used to map the infection state into excess deaths. Effective IFRs incorporate variant severity, treatment improvements, vaccination coverage, and prior immunity.

Wave	**Interpretation**	**Effective IFR**	**Quarters**
W1	First wave, pre-treatment, no vaccination	0.50%	Q4.19–Q1.20
W1-summer	Lower overload and treatment improvement	0.20%	Q2.20
W2-wildtype	Autumn/winter 2020–21 wildtype phase	0.70%	Q3.20
Alpha	Alpha phase, partly offset by early vaccination	0.40%	Q4.20–Q1.21
Delta	Delta phase with increasing vaccination coverage	0.20%	Q2.21
Early Omicron	Omicron transition phase	0.20%	Q3.21–Q1.22
Late Omicron	High immunity, lower fatality phase	0.02%	Q2.22–Q4.22

Notes: The quarter assignment shown is the one used in the country-level (median) validation. The representative-economy roll used in the planner exercise assigns Q1 2021 to the Delta-phase value (0.20%) and Q1 2022 to the late-Omicron value (0.02%), advancing the fatality decline by one quarter at each of these two wave transitions; the two assignments are identical in all other quarters.

G.2 Representative mean OECD Economy

The planner is solved for a representative OECD economy constructed using the cross-country **means** of observed policy inputs and initial conditions. Its inputs are as follows: the country intercepts are set to zero; the initial debt stock is set to the OECD mean of the pre-pandemic levels; the onset shock is set to the cross-country mean of the country-specific Q2 2020 output shocks (Table 26, last column); and the epidemiological innovations are set to the cross-country mean innovation path rather than being fixed as in the median forward roll. It is therefore important to validate the same object used in the optimization exercise. Under the observed mean policy path, the representative-economy rollout reproduces the aggregate pandemic dynamics reasonably well. The output RMSE is 1.24 percentage points, the quarterly debt-change RMSE is 0.85 percentage points, and the excess-mortality RMSE is 2.87 deaths per million per week, all tighter than the corresponding median-roll fit.

This supports the use of the representative OECD economy as the baseline object for the counterfactual planner exercise.

Table 25: Representative OECD economy validation under observed mean policy inputs.

Moment	**Horizon**	**RMSE**
Output gap	Q4.19–Q2.22	1.24 pp
Debt change	Q4.19–Q4.22	0.85 pp
Debt level	Q4.19–Q4.22	2.96 pp
Infection state θ	Q4.19–Q2.22	0.0024
Excess mortality d	Q4.19–Q2.22	2.87

Figure 12 reports the graphical interpretation of the mean forward roll. The output-gap trajectory is closely aligned with the observed mean path, and the fit for excess mortality is even stronger than in the country-level panel validation. The debt trajectory is also well matched until around 2021Q2. Afterwards, however, the model starts to overestimate the debt level, reaching a gap of roughly 5 percentage points by the end of 2022. This difference is not large relative to the level of public debt, but it should be stated explicitly. Since the simulated output gap remains negative, and since the transition system does not include an additional consolidation, repayment, or stock-flow adjustment mechanism, there is no model-implied force that brings debt back down after the main fiscal expansion. This limitation does not undermine the efficiency comparison in the planner exercise, because the observed and counterfactual policy paths are evaluated within the same representative-economy transition system. The comparison therefore remains internally consistent.

Figure 13 shows the corresponding validation for quarterly debt changes. The model tracks the quarterly debt-change dynamics reasonably well, with the main deviation occurring in the first pandemic quarter. This is consistent with the construction of the fiscal variables: loans and guarantees enter the model as take-up-adjusted flows, while the observed debt data partly reflect the announcement and accounting recognition of large fiscal envelopes. The forward roll therefore underestimates the initial jump in debt, but captures the subsequent debt-change dynamics more closely.

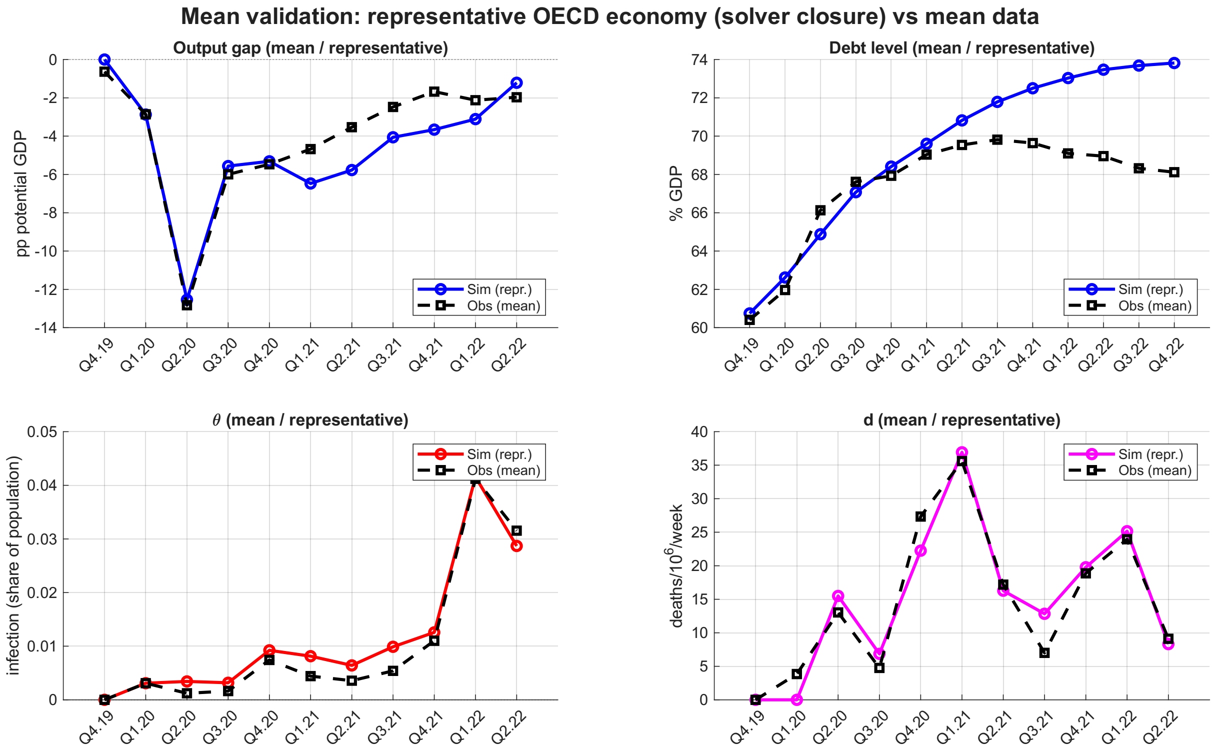


Figure 12: Representative OECD economy validation under observed mean policy inputs. The figure compares the simulated representative-economy path to the observed OECD mean path.

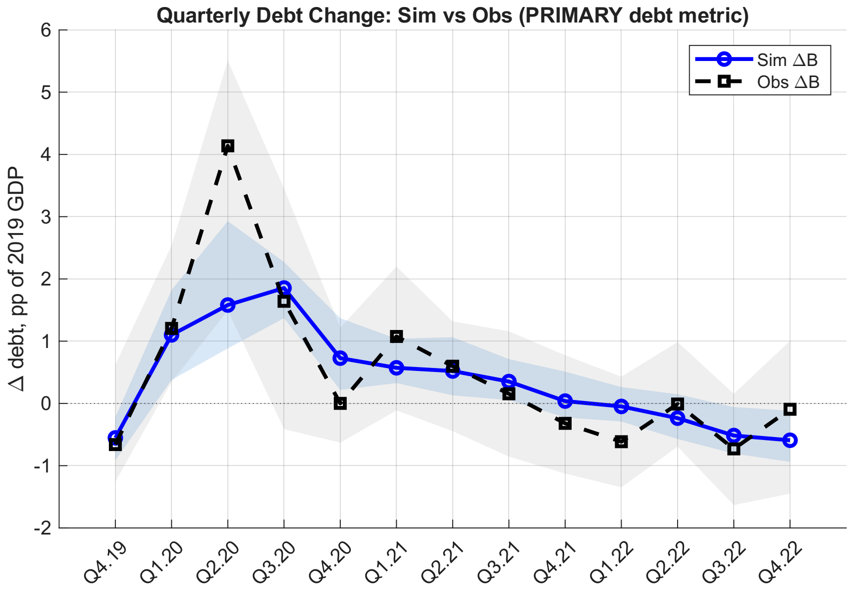


Figure 13: Quarterly debt-change validation under observed policy paths. Debt changes are the primary debt validation object and the weakest fitted component of the transition system.

G.3 Calibration Fit per Country

Table 27 reports the endpoint fit underlying the aggregate moments of Section 4.3: for each country, the observed and simulated values of the Q2.2022 output gap, excess mortality, and terminal debt, together with their residuals. Debt is matched tightly and near-uniformly (median residual +1.4 pp on a mean level of 68 percent of GDP), and the output endpoints show no systematic bias (mean residual −1.0 pp against a cross-country dispersion of 3.6). The mortality endpoints are noisier, and deliberately so: the common wave-specific IFR mapping is not re-fitted to individual countries, so countries whose age structure, immunity profile, or reporting conventions deviate most from the OECD-wide calibration—Chile, Colombia, Mexico, and Türkiye with overpredicted, Finland and Portugal with underpredicted deaths—carry the largest residuals, which are level errors symmetric around zero (mean +0.2) rather than a directional bias. The frontier-distance measure of Section 6.3 is invariant to exactly this country-specific mortality-level misfit, and the ranking is rank-stable to excluding the worst-fitting quartile (Appendix J.1).

G.4 Country-specific Inputs

Table 26: Country-specific inputs to the calibration.

Country	$B_{i,2019Q4}$ (% GDP)	$\mu_{y,i}$ (pp)	$\mu_{b,i}$ (pp)	$\varepsilon^{y}_{i,2020Q2}$ (pp)
AUS	37.6	+1.106	-0.853	-3.62
AUT	69.7	-1.040	-0.600	-8.55
BEL	77.9	+0.301	-0.873	-6.46
CAN	44.2	-0.098	-0.265	-7.79
CHE	16.3	+1.299	-0.431	-4.73
CHL	29.6	+1.325	-1.029	-10.10
COL	47.7	+1.989	+0.549	-11.20
CRI	53.2	+0.147	+0.121	-3.75
CZE	30.2	-3.562	-0.657	-5.10
DEU	34.0	-1.738	-1.095	-5.98
DNK	34.1	-0.083	-1.661	-1.95
ESP	86.0	-4.896	-0.962	-9.57
EST	11.8	-1.833	-0.481	-2.43
FIN	54.5	-1.483	-0.631	-1.97
FRA	85.4	-1.869	-0.967	-8.49
GBR	107.0	-3.327	-1.372	-12.50
GRC	205.0	+0.291	-1.304	-9.94
HUN	52.1	-2.015	-0.967	-8.99
IRL	54.3	+8.319	-0.270	-1.10
ISL	69.9	-4.849	-0.817	-4.70
ISR	56.9	+2.367	-0.342	-0.66
ITA	122.0	-0.501	-1.193	-10.90
JPN	199.0	-1.956	-1.657	-6.82
KOR	35.1	+0.614	-0.514	-1.83
LTU	29.3	+0.700	-0.482	-0.03
LUX	26.6	+2.357	+0.119	-5.31
LVA	39.7	-0.683	+0.048	-7.43
MEX	7.2	-3.206	-0.908	-7.97
NLD	43.9	+1.060	-0.427	-5.17
NOR	16.6	+1.222	-0.109	-2.45
NZL	73.0	-1.062	+0.039	-5.49
POL	45.6	-0.597	-0.693	-1.80
PRT	111.0	-2.775	-1.724	-9.75
SVK	48.4	-0.258	-0.131	-4.87
SVN	72.1	-1.828	-0.991	-6.78
SWE	27.7	+1.035	-0.511	-4.70
TUR	54.0	+4.066	-0.315	-10.40
USA	97.3	+1.099	-0.225	-4.79

Notes: $B_{i,2019Q4}$ is the observed pre-pandemic public debt level (% of 2019 GDP), used as the initial debt stock. $\mu_{y,i}$ and $\mu_{b,i}$ are the estimated country fixed effects in the output and debt equations. $\varepsilon^{y}_{i,2020Q2}$ is the country-specific initial pandemic output shock in 2020Q2. The fixed effects and initial shock are estimated on the panel; the debt baseline is taken from observed data.

Table 27: Country-level endpoint fit: observed, simulated, and residuals.

	Output gap (pp)			Mortality (d)			Debt (% GDP)		
Country	Obs	Sim	Resid	Obs	Sim	Resid	Obs	Sim	Resid
AUS	-0.02	-0.94	+0.92	11.45	9.41	+2.04	41.28	40.39	+0.89
AUT	-1.81	-4.77	+2.96	10.45	7.10	+3.35	80.58	79.71	+0.87
BEL	-0.06	+0.89	-0.95	7.99	5.07	+2.92	81.56	80.19	+1.37
CAN	-1.58	-2.08	+0.50	8.92	5.76	+3.16	58.08	57.62	+0.46
CHE	+1.07	+1.32	-0.26	2.82	4.06	-1.24	18.44	18.58	-0.13
CHL	+0.48	+0.10	+0.39	8.10	19.46	-11.36	32.13	32.81	-0.68
COL	+3.75	+0.02	+3.73	0.62	10.87	-10.25	63.81	64.16	-0.35
CRI	-2.60	+0.30	-2.90	3.28	5.46	-2.18	67.23	65.44	+1.80
CZE	-8.51	-4.63	-3.88	5.46	3.56	+1.90	42.23	41.42	+0.81
DEU	-4.24	-1.46	-2.78	11.74	2.61	+9.13	39.77	38.95	+0.82
DNK	-1.80	+2.53	-4.33	11.27	3.32	+7.95	30.99	29.06	+1.93
ESP	-6.16	-6.59	+0.42	9.08	2.82	+6.26	97.40	95.92	+1.48
EST	-6.63	-3.43	-3.19	8.63	13.41	-4.77	21.99	20.95	+1.04
FIN	-4.57	-2.44	-2.13	25.70	12.36	+13.34	60.66	58.77	+1.89
FRA	-3.52	-2.04	-1.49	8.50	4.76	+3.75	92.23	90.72	+1.52
GBR	-3.20	-3.87	+0.67	10.53	0.42	+10.11	118.13	116.70	+1.43
GRC	-0.33	-2.46	+2.13	20.89	21.13	-0.24	212.81	208.10	+4.71
HUN	-4.93	-2.37	-2.57	10.86	4.78	+6.09	59.00	58.24	+0.76
IRL	+10.90	+10.73	+0.17	11.65	3.34	+8.30	52.38	50.78	+1.60
ISL	-5.89	-6.30	+0.42	13.32	21.23	-7.91	80.58	78.70	+1.88
ISR	+1.65	+2.06	-0.41	3.79	11.21	-7.41	65.61	63.44	+2.17
ITA	+1.27	-1.15	+2.43	12.03	9.42	+2.61	128.11	126.00	+2.11
JPN	-2.30	-2.25	-0.05	1.85	8.11	-6.26	206.86	201.18	+5.68
KOR	-2.24	+0.34	-2.58	13.86	21.01	-7.16	40.57	38.92	+1.65
LTU	-4.25	+0.60	-4.85	8.98	22.89	-13.92	34.30	32.84	+1.45
LUX	-2.03	+1.06	-3.09	7.71	0.83	+6.87	34.12	33.89	+0.24
LVA	-2.78	-1.18	-1.60	11.58	17.16	-5.58	53.30	52.56	+0.73
MEX	-4.98	-6.16	+1.18	1.74	15.06	-13.32	8.19	9.05	-0.85
NLD	+0.73	+0.18	+0.55	2.02	4.80	-2.78	49.45	48.75	+0.70
NOR	-0.00	+1.68	-1.68	13.48	7.61	+5.87	20.57	20.48	+0.09
NZL	-3.78	-3.05	-0.73	11.55	5.29	+6.26	94.19	92.32	+1.86
POL	-4.05	-1.70	-2.35	3.29	7.03	-3.74	51.71	50.13	+1.58
PRT	-4.95	-3.70	-1.25	18.30	3.59	+14.70	108.34	106.38	+1.96
SVK	-5.14	-0.36	-4.78	12.29	11.54	+0.75	59.60	57.97	+1.63
SVN	-5.31	-2.88	-2.42	6.34	5.79	+0.56	82.17	80.26	+1.91
SWE	-1.28	+1.45	-2.73	6.73	2.92	+3.81	29.55	28.89	+0.66
TUR	+5.55	+5.20	+0.35	0.81	10.90	-10.09	53.97	53.62	+0.34
USA	-1.33	-0.34	-0.99	8.43	11.58	-3.15	116.57	113.89	+2.67
Mean	-1.97	-0.99	-0.98	9.11	8.89	+0.22	68.12	66.78	+1.33
Median	-2.27	-1.16	-0.97	8.95	7.06	+1.32	58.54	57.80	+1.44
SD	+3.64	+3.24	+2.11	5.41	6.22	+7.33	44.88	43.85	+1.24

Notes: Output gap: Q2.2022 endpoint in pp of potential GDP. Mortality: excess deaths per million per week, end-of-window level. Debt: terminal level in percent of 2019 GDP (Q4.2022). Residual = observed − simulated. Bottom rows report cross-country moments of each column.

G.5 Additional Fit Diagnostics

The calibration also matches several non-targeted moments. The simulated output series has a mean cross-country SD ratio of 1.154, indicating that the model does not flatten the observed dispersion. Output persistence is close to the data, with an observed mean

AC(1) of 0.244 and a simulated mean AC(1) of 0.155. The endpoint residuals are small: the median output-gap residual is −0.97 pp in Q2 2022, while the mean and median terminal debt residuals are +1.33 pp and +1.44 pp, respectively. The weakest component remains the quarter-to-quarter debt change, consistent with the main text.

Table 28: Additional calibration diagnostics.

Diagnostic	**Value**
Output SD ratio	1.154
Output AC(1), observed	0.244
Output AC(1), simulated	0.155
Output AC(1) gap	0.089
Output ICC, observed	0.356
Output ICC, simulated	0.420
Debt ICC, observed	1.018
Debt ICC, simulated	1.020
Terminal debt residual, mean	+1.33 pp
Terminal debt residual, median	+1.44 pp
Output endpoint residual, mean	-0.98 pp
Output endpoint residual, median	-0.97 pp

G.6 Fiscal Channel Decomposition

The fiscal channel decomposition confirms that the signs of the estimated policy channels are economically consistent. Over the main 11-quarter validation window, the median cumulative contribution of above-the-line support is +3.45 percentage points, while the median contribution of below-the-line liquidity support is +6.35 percentage points. Demand injection contributes negatively at the median, −1.21 percentage points, reflecting the estimated interaction with containment. Total fiscal support is positive in 33 of 38 countries and equals +8.05 percentage points at the median while it is only negative in the DI heavy countries.

Table 29: Fiscal channel decomposition over the main validation window.

Channel	**Median contribution**	**Positive countries**	**Negative countries**
Above-line fiscal support	+3.45 pp	38	0
Below-line liquidity support	+6.35 pp	38	0
Demand injection	-1.21 pp	6	32
Total fiscal support	+8.05 pp	33	5
Health drag	+0.00 pp	0	0

H Solution Method and Augmented State Space

H.1 Why the Policy Trade-off Is Unavoidable

Mortality is predetermined within the quarter: from Eq. (11), $d_{k+1} = \delta_\theta(k)\,\theta_k$ depends only on prevalence already realized. The planner controls future mortality by suppressing current transmission. Combining the infection and mortality transitions (absent innovations),

$$d_{k+2} = \delta_\theta(k+1)\,\rho_\theta(k)\big(1 - \phi_S\,S_k/100\big)\,\theta_k, \tag{19}$$

so the marginal effect of containment on mortality two quarters out is

$$\frac{\partial d_{k+2}}{\partial S_k} = -\,\delta_\theta(k+1)\,\rho_\theta(k)\,\frac{\phi_S}{100}\,\theta_k. \tag{20}$$

The benefit of containment vanishes when the variant is non-lethal ($\delta_\theta = 0$): suppression then saves no lives and the planner sets $S_k = 0$ to protect output. It is strictly positive only when the variant is simultaneously lethal ($\delta_\theta > 0$) and transmissible ($\rho_\theta > 0$) during an active outbreak ($\theta_k > 0$). The planner accepts the output cost of containment solely because of this conjunction. This is the mechanism behind the post-vaccination unwinding of containment in Section 6: once vaccination drives δ_θ toward zero, Eq. (20) collapses and the optimal S_k falls to zero even while prevalence remains high.

H.2 Augmented State Space

The transition system of Section 2, Eqs. (8)–(11), is not first-order Markov: direct income support disburses with a one-quarter delay, entering both output and the budget through F^{DI}_{k-1}, and the two capacity-preservation channels operate through persistent stocks of accumulated fiscal support. So I augment the four economic states $(y_k, b_k, \theta_k, d_k)$ with three bookkeeping states that carry the lag and the two stocks:

$$\boldsymbol{x}_k = \big(y_k,\ b_k,\ \theta_k,\ d_k,\ z_k,\ L_k,\ K_k\big)' \in \mathbb{R}^7, \tag{21}$$

where $z_k = F^{\mathrm{DI}}_{k-1}$, L_k is the below-the-line liquidity stock, and K_k the above-the-line capacity-preservation stock. The control vector is $\boldsymbol{u}_k = (S_k, F^{\mathrm{above}}_k, F^{\mathrm{loans}}_k, F^{\mathrm{guar}}_k, F^{\mathrm{DI}}_k)' \in \mathbb{R}^5$. The bookkeeping recursions are $z_{k+1} = F^{\mathrm{DI}}_k$ and the stock accumulation equations of Eq. (14),

Loan and guarantee flows are take-up-adjusted at the data-construction stage (realization rates $c_{\mathrm{lo}} = 0.6$ and $c_{\mathrm{gu}} = 0.25$), so the flows entering Eq. (14) and the debt transition are already effective values and no take-up factor appears inside the dynamics.

Above-the-line support is therefore both a flow and a stock: the flow is absorbed contemporaneously into K_{k+1}, which then governs output directly and raises the effectiveness of outstanding liquidity. This capacity-stock channel replaces the lag-2 flow specification used in estimation; the coefficients are calibrated in Section H.3. The fiscal timing embedded in the augmentation reflects the institutional sequence.

H.3 Planner-Specific Parameters

Beyond the estimated transition coefficients of Table 7, the planner problem requires four calibrated parameters—the two stock decay rates, the direct capacity coefficient, and the complementarity parameter—and two objects taken directly from the data, the saturation scale $\bar{K}$ and the liquidity coefficient.

The liquidity coefficient requires no reinterpretation: below-the-line support enters the estimation as a cumulative stock, so the estimated coefficient carries over directly, $\alpha_L = \alpha_{\text{below}} = 0.261$. The liquidity stock decays at $\delta_L = 0.10$ per quarter (half-life of roughly 6.6 quarters), and the capacity stock at $\delta_K = 1 - 0.5^{1/6} \approx 0.109$, imposing a half-life of six quarters—the persistence horizon over which preserved employment relationships and firm structures plausibly retain their value without renewed support. The direct capacity coefficient is set to $\alpha_K = 0.30\,\alpha_{\text{above}} \approx 0.163$, reinterpreting the estimated lag-2 flow effect as a persistent stock margin and deliberately discounting it to avoid double-counting persistence already implicit in the estimated lag structure. The complementarity parameter is $\chi = 0.50$ in the baseline, so preserved capacity can raise the effectiveness of outstanding liquidity by at most 50 percent; the frontier results under $\chi \in \{0, 0.25, 0.50\}$ are reported in Table 31, and alternative and zero decay rates are examined in Appendix I. The saturation scale $\bar{K}$ is set to the above-the-line instrument cap (the in-support 99th percentile), so the multiplier attains half of its maximal uplift when the outstanding capacity stock equals one cap-sized program.

The fear channel is switched off in the planner baseline, $\beta_d = 0$: the coefficient on lagged mortality is statistically indistinguishable from zero and not robustly identified in the panel (Appendix D) on quarterly level, so the baseline omits a behavioral feedback the data do not support; it is retained in the estimating equation for theoretical consistency. Solving the model with a calibrated parameter does not change the results.

The control-cost matrix is $\boldsymbol{R} = 10^{-8}\,\boldsymbol{D}$ with $\boldsymbol{D} = \operatorname{diag}(\bar{u}_1^{-2}, \ldots, \bar{u}_5^{-2})$, a vanishing penalty on normalized instrument use that keeps the problem well-posed; its contribution to the objective is below one percent at every reported solution.

The epidemiological innovations ε_θ are calibrated as residuals of the infection transition under the observed containment path and are held fixed across all policy counterfactuals. Before the vaccination break, negative innovation realizations are floored at zero to guarantee the non-negativity of the prevalence state: early-pandemic prevalence is close to zero, and along counterfactual containment paths a negative innovation could otherwise push the state below its natural bound. After the break, prevalence is far from the bound and the innovations enter unrestricted.

H.4 Jacobian Matrices

Linearizing around a nominal trajectory $(\bar{\boldsymbol{x}}_k, \bar{\boldsymbol{u}}_k)$ gives $\delta\boldsymbol{x}_{k+1} \approx \boldsymbol{A}_k\,\delta\boldsymbol{x}_k + \boldsymbol{B}_k\,\delta\boldsymbol{u}_k$. The system is nonlinear through the bilinear terms $\alpha_{S,\mathrm{DI}}\,S_k F^{\mathrm{DI}}_{k-1}$ in output and $\phi_S\,S_k\theta_k$ in infection, and through the complementarity multiplier $m(K_{k+1})$, so $\boldsymbol{A}_k$ and $\boldsymbol{B}_k$ depend on the nominal point. Write $\bar{m}_k = m(\bar{K}_{k+1})$ and $\bar{m}'_k = \chi\bar{K}/(\bar{K}_{k+1}+\bar{K})^2$ for the multiplier and its derivative evaluated at the nominal end-of-period capacity stock. With states ordered as in Eq. (21) and controls as $(S, F^{\mathrm{above}}, F^{\mathrm{loans}}, F^{\mathrm{guar}}, F^{\mathrm{DI}})$, the nonzero entries are

$$
\begin{aligned}
\boldsymbol{A}_k:\quad & A_{yy} = \rho_y, \quad A_{yd} = -\beta_d, \quad A_{yz} = \alpha_{\mathrm{DI}} + \alpha_{S,\mathrm{DI}}\,\bar{S}_k, \\
& A_{yL} = \alpha_L\,\bar{m}_k\,(1-\delta_L), \quad A_{yK} = \big(\alpha_K + \alpha_L\,\bar{L}_{k+1}\,\bar{m}'_k\big)(1-\delta_K), \\
& A_{by} = -\gamma_y, \quad A_{bb} = 1 + r, \quad A_{bz} = \kappa_{\mathrm{DI}}, \\
& A_{\theta\theta} = \rho_\theta(k)\big(1 - \phi_S\,\bar{S}_k/100\big), \quad A_{d\theta} = \delta_\theta(k), \\
& A_{LL} = 1 - \delta_L, \quad A_{KK} = 1 - \delta_K,
\end{aligned}
\tag{22}
$$

$$
\begin{aligned}
\boldsymbol{B}_k:\quad & B_{yS} = \alpha_S + \alpha_{S,\mathrm{DI}}\,\bar{z}_k, \quad B_{y,\mathrm{ab}} = \alpha_K + \alpha_L\,\bar{L}_{k+1}\,\bar{m}'_k, \\
& B_{y,\mathrm{lo}} = B_{y,\mathrm{gu}} = \alpha_L\,\bar{m}_k, \\
& B_{b,\mathrm{ab}} = \kappa_{\mathrm{above}}, \quad B_{b,\mathrm{lo}} = \kappa_{\mathrm{loans}}, \quad B_{b,\mathrm{gu}} = \kappa_{\mathrm{guar}}, \\
& B_{\theta S} = -\rho_\theta(k)\,\phi_S\,\bar{\theta}_k/100, \quad B_{z,\mathrm{DI}} = 1, \\
& B_{L,\mathrm{lo}} = B_{L,\mathrm{gu}} = 1, \quad B_{K,\mathrm{ab}} = 1.
\end{aligned}
\tag{23}
$$

The nonlinear couplings appear in three places. First, A_{yz} carries $\alpha_{S,\mathrm{DI}}\,\bar{S}_k$ and B_{yS} carries $\alpha_{S,\mathrm{DI}}\,\bar{z}_k$, the two partial derivatives of the $S_k F^{\mathrm{DI}}_{k-1}$ term. Second, $B_{\theta S}$ carries the derivative of $S_k\theta_k$; the stringency entries scale by $1/100$ because containment enters the infection equation in fractional units while entering output in points. Third, the capacity entries A_{yK} and $B_{y,\mathrm{ab}}$ combine the direct effect α_K with the indirect effect through the multiplier, $\alpha_L\bar{L}_{k+1}\bar{m}'_k$: because m is bounded, $\bar{m}'_k$ shrinks as capacity saturates, so the marginal return to above-the-line support declines in the outstanding stock.

H.5 The Algorithm

I solve the problem with the iterative Linear-Quadratic Regulator (iLQR) of Li and Todorov (2004), which successively linearizes the dynamics around the current trajectory and solves the resulting time-varying linear-quadratic (LQ) subproblem in closed form. Abhijeet and Chakravorty (2025) show iLQR is equivalent to Sequential Quadratic

Programming on the discrete-time control problem, with cost descent at each iteration; the notation follows Bertsekas (2017, Chapters 3–4).

Backward pass. Let V_x, V_{xx} denote the gradient and Hessian of the cost-to-go. Starting from the terminal condition (Section H.6), the recursion at each k is

$$\begin{aligned} \boldsymbol{Q}_x &= \boldsymbol{A}_k' \, V_x', & \boldsymbol{Q}_u &= \boldsymbol{l}_u + \boldsymbol{B}_k' \, V_x', \\ \boldsymbol{Q}_{xx} &= \boldsymbol{A}_k' \, V_{xx}' \, \boldsymbol{A}_k, & \boldsymbol{Q}_{uu} &= \boldsymbol{l}_{uu} + \boldsymbol{B}_k' \, V_{xx}' \, \boldsymbol{B}_k + \mu \, \boldsymbol{D}, \\ \boldsymbol{Q}_{ux} &= \boldsymbol{B}_k' \, V_{xx}' \, \boldsymbol{A}_k, \end{aligned} \tag{24}$$

where $V_x' = \boldsymbol{l}_x + V_x$ and $V_{xx}' = \boldsymbol{l}_{xx} + V_{xx}$ fold in the running cost, and $\boldsymbol{l}_x, \boldsymbol{l}_u, \boldsymbol{l}_{xx}, \boldsymbol{l}_{uu}$ are the discounted gradients and Hessians of the stage loss. Because the stage cost is separable in $(\boldsymbol{x}, \boldsymbol{u})$, there is no cost cross term ($\boldsymbol{l}_{ux} = 0$), which keeps $\boldsymbol{Q}_{uu}$ well-conditioned. The feedforward and feedback gains are $\boldsymbol{k}_k = -\boldsymbol{Q}_{uu}^{-1}\boldsymbol{Q}_u$ and $\boldsymbol{K}_k = -\boldsymbol{Q}_{uu}^{-1}\boldsymbol{Q}_{ux}$.

Regularization. The explosive eigenvalues $1 + r > 1$ and $\rho_\theta(k)(1 - \phi_S \bar{S}_k/100) > 1$ in early waves inflate V_{xx} and can render $\boldsymbol{Q}_{uu}$ indefinite. I therefore add Levenberg–Marquardt damping $\mu \, \boldsymbol{D}$ to $\boldsymbol{Q}_{uu}$, with $\boldsymbol{D} = \operatorname{diag}(\bar{u}_1^{-2}, \ldots, \bar{u}_5^{-2})$ scaled by the instrument caps so the damping is uniform in normalized control units; the damping enters $\boldsymbol{Q}_{uu}$ only, leaving $\boldsymbol{Q}_{ux}$ and hence the gain pair consistent with a single quadratic model of the control Hessian. The parameter μ is increased tenfold when the Cholesky factorization of $\boldsymbol{Q}_{uu}$ fails or the forward pass finds no cost-reducing step, and decreased geometrically after each accepted iteration.

Forward pass. Given the gains, the updated controls are applied to the nonlinear dynamics with line-search step $\alpha \in (0, 1]$:

$$\boldsymbol{u}_k^{\text{new}} = \boldsymbol{u}_k + \alpha \, \boldsymbol{k}_k + \boldsymbol{K}_k \big(\boldsymbol{x}_k^{\text{new}} - \boldsymbol{x}_k\big), \tag{25}$$

starting from $\boldsymbol{x}_0^{\text{new}} = \boldsymbol{x}_0$. The step α is halved from unity until total cost decreases; if no decrease is found, μ is increased and the backward pass repeats.

Constraints. Stringency and the four fiscal instruments are bounded below at zero and above at the 99th percentile of the corresponding in-support observed values across the OECD panel; relaxing the stringency cap to the full index scale $[0, 100]$ is examined as a robustness exercise. The Q4.2019 controls are locked at zero. The bounds are enforced by projection (clipping) in the forward pass. In addition, quarter-on-quarter changes in stringency are capped at $|S_k - S_{k-1}| \leq \Delta S_{\max}$, where $\Delta S_{\max}$ is the 95th percentile of observed $|\Delta S|$ across the panel, enforced by clamping S_k relative to the previously updated S_{k-1} within the same forward pass—containment cannot ramp faster than governments managed in-sample. The speed limit is a hard constraint rather than a quadratic penalty: a ΔS penalty would require augmenting the state with S_{k-1} and introduce a cost cross term ($\boldsymbol{l}_{ux} \neq 0$) that, combined with the bilinear dynamics, ill-conditions $\boldsymbol{Q}_{uu}$.

Certainty equivalence. At each iteration the subproblem is linear-quadratic, so certainty equivalence holds (Bertsekas, 2017, Chapter 3.1): the feedback gains are identical to the deterministic case, and the epidemiological innovations ε_θ raise expected cost without altering the optimal policy rule. The realized innovation path is fed through the nonlinear forward pass.

Convergence and global optimality. The algorithm terminates when the relative cost reduction falls below 10^{-9}. The bilinear terms and the complementarity multiplier make the problem non-convex, so iLQR converges to a local optimum. I therefore restart the solver from four economically distinct initializations—the observed average policy, the zero-control path, and two corner allocations tilted toward above- and below-the-line support—and report the solution with the lowest cost. Besides tolerance convergence, the solver terminates when no cost-reducing step is found over forty consecutive damping increases or when the damping reaches its ceiling without further descent; such trajectories are accepted as converged under maximal damping, and the multistart comparison guards against reporting an inferior local solution. In the baseline configuration this safeguard is not needed: all four starts converge to the same optimum, with a relative spread of the terminal objective values below 2×10^{-4}, so the reported solution is not an artifact of the initialization.

H.6 Terminal Condition

The terminal cost weights public debt alone. By the end of the horizon, vaccination has severed the link from infection to death and removed the output cost of containment, so the only state that persists as a welfare-relevant legacy is accumulated debt. The terminal weight capitalizes the discounted stream of future debt-loss deviations under a stationary post-pandemic continuation, $Q_N^{(b,b)} = M\,\tau_b\,w_b$ with $M = 1/(1-\beta\,\lambda^2) \approx 34$, where $\lambda = 0.99$ is the assumed post-horizon persistence of the terminal debt deviation. All other terminal weights are zero.

H.7 Computation of the Constrained Policy Frontier

The frontier problems of Section 6.2 are solved as exact nonlinearly constrained programs over the full control path $\boldsymbol{u} \in \mathbb{R}^{5\times N}$, using sequential quadratic programming. In each scenario, one of the three welfare-relevant outcomes—discounted output loss, cumulative excess mortality, or terminal debt—is minimized subject to the other two being no worse than their model-implied values under the observed policy, evaluated through the full nonlinear transition system. Formally, with quarters indexed $k = 0$ (Q4.2019, locked) through N (Q4.2022), the three outcome functionals of a control path $\boldsymbol{u}$ are the discounted quadratic output loss $\mathcal{L}_y(\boldsymbol{u}) = \sum_{k=1}^{N} \beta^k y_k^2$, cumulative excess mortality $D(\boldsymbol{u}) = 13 \sum_{k=1}^{N} d_k$—the factor 13 converting the weekly mortality state into quarterly cumulative deaths per million—and terminal debt $b_N(\boldsymbol{u})$. Feasibility of each reported solution is verified ex post against the observed thresholds at a tolerance of 10^{-4}; the SQP subproblems are solved to optimality and constraint tolerances of 10^{-8}.

The box bounds and the ramping constraint of Section H.5 are imposed as bounds and nonlinear inequality constraints, respectively; a vanishing tie-breaking penalty on normalized instrument use (10^{-8}) selects among cost-equivalent policies and carries no welfare interpretation. Because the constrained problems inherit the non-convexity of the dynamics, each scenario is solved from five starting policies—the observed policy, the weighted-planner solution, the zero-control path, a corner allocation with above-the-line support at its cap under the observed containment path, and the observed policy with above-the-line support raised to its cap—and the best feasible local solution across starts is reported.

H.8 Eigenvalue Structure and Fiscal Self-Financing

Eigenvalues

The augmented system matrix $\boldsymbol{A}_k$ in Eq. (22) is block-triangular, so its eigenvalues are its diagonal entries. Two are zero (mortality d and the lagged control z carry no own-persistence), and five govern the dynamics:

$$\lambda_y = \rho_y < 1 \qquad \text{(output: self-stabilizing)}, \tag{26}$$

$$\lambda_b = 1 + r > 1 \qquad \text{(debt: unconditionally unstable)}, \tag{27}$$

$$\lambda_\theta = \rho_\theta(k)\big(1 - \phi_S\, S_k/100\big) \qquad \text{(infection: conditionally unstable)}, \tag{28}$$

$$\lambda_L = 1 - \delta_L < 1 \qquad \text{(liquidity stock: stable)}, \tag{29}$$

$$\lambda_K = 1 - \delta_K < 1 \qquad \text{(capacity stock: stable)}. \tag{30}$$

Two features drive the planner's problem. First, the debt eigenvalue exceeds unity for any positive interest rate and is invariant to policy: no control enters A_{bb}, so debt cannot be stabilized by any available instrument. During the crisis both forces on debt are explosive—the automatic stabilizer $-\gamma_y\, y_k > 0$ when output is below potential, and discretionary support $\kappa \cdot F_k > 0$. Second, the infection eigenvalue exceeds unity, and the epidemic grows, whenever stringency falls short of the threshold

$$\lambda_\theta \geq 1 \quad \Longleftrightarrow \quad S_k < \underline{S}(k) \equiv \frac{100}{\phi_S}\Big(1 - \tfrac{1}{\rho_\theta(k)}\Big). \tag{31}$$

The threshold is wave-specific through $\rho_\theta(k)$ but independent of current prevalence. At the pre-vaccination calibration ($\rho_\theta = 1.5$, $\phi_S = 0.8$), $\underline{S} \approx 42$ points—of the order of the average stringency actually observed during the acute waves; once vaccination lowers ρ_θ below unity, $\underline{S} \leq 0$ and the infection state is stable without intervention.

The planner therefore faces a cascade: epidemiological instability requires containment; containment depresses output through α_S; the output gap widens the deficit through $-\gamma_y\, y_k$; and the resulting debt cannot be stabilized by any control. Each stabilizing action on one dimension feeds the instability of another. This is the trilemma as a dynamical-systems property.

Fiscal self-financing

A fiscal instrument is self-financing if the automatic-stabilizer savings from the output it generates outweigh its direct budgetary cost. Consider demand injection, which is disbursement-based on both margins. Authorized in quarter 0, it reaches the budget and output at disbursement one quarter later: $\kappa_{\mathrm{DI}}\, z_1 = \kappa_{\mathrm{DI}}\, F_0^{\mathrm{DI}}$ enters b_2 and $\alpha_{\mathrm{DI}}\, z_1$ enters y_2. The higher y_2 feeds back into the budget through the automatic stabilizer, reducing b_3 by $\gamma_y\, \alpha_{\mathrm{DI}}$. The net debt effect is

$$\frac{\partial b_3}{\partial F_0^{\mathrm{DI}}} = (1+r)\, \kappa_{\mathrm{DI}} - \gamma_y\, \alpha_{\mathrm{DI}}, \tag{32}$$

so demand injection is self-financing when $\gamma_y\, \alpha_{\mathrm{DI}} \geq (1+r)\, \kappa_{\mathrm{DI}}$. At the estimated parameters, $\gamma_y\, \alpha_{\mathrm{DI}} = 0.17$ against $(1+r)\, \kappa_{\mathrm{DI}} = 0.53$.

Above-the-line support operates through the persistent capacity stock, so the relevant comparison credits the entire output stream the stock generates. Authorized in quarter 0, the flow commits the budget contemporaneously ($\kappa_{\mathrm{above}} F_0^{\mathrm{above}}$ in b_1) and raises output through $\alpha_K K_{k+1}$ in every subsequent quarter as the stock decays. Ignoring discounting and interest compounding—both of which work against self-financing—the cumulative stabilizer saving per unit of flow is bounded by $\gamma_y\, \alpha_K \sum_{j\geq 0}(1-\delta_K)^j = \gamma_y\, \alpha_K/\delta_K \approx 0.18$, against a budgetary cost of $\kappa_{\mathrm{above}} = 0.66$. The same bound for below-the-line liquidity, credited with the full complementarity uplift $m \leq 1+\chi$, gives $\gamma_y\, \alpha_L(1+\chi)/\delta_L \approx 0.46$ against $\kappa_{\mathrm{loans}} = 0.84$ and $\kappa_{\mathrm{guar}} = 0.54$: still short of self-financing, but by the smallest margin of any instrument—a first indication of why the frontier recomposes support toward the below-the-line channel.

Every instrument therefore leaves a positive net debt footprint even under assumptions deliberately tilted in its favor. This is the analytical counterpart to the paper's central result. Because no instrument repays itself through output and debt cannot be stabilized ($\lambda_b > 1$), fiscal policy cannot resolve the trilemma—it can only redistribute the burden across instruments and time. The normative content of the model is therefore debt efficiency at given output and health outcomes, not output stabilization through fiscal expansion.

I Complementary Results

I.1 Preference and VSL Sensitivity of the Weighted Benchmark

Table 30 re-solves the weighted planner problem of Section 5 under seven preference configurations: the baseline, an output-oriented and a balanced variant, a precautionary variant that doubles the debt weight, and the VSL grid of Section 5, which re-anchors λ_d to statistical life values of \$5, \$9, and \$11 million. For each configuration, the observed policy is re-evaluated under the same weights, so the reported gain is internally consistent; objective levels are not comparable across rows.

Two results stand out. First, the compositional core of the paper is invariant to preferences: demand injection is exactly zero in all seven configurations, and every planner recomposes support toward liquidity—between 15.5 and 21.5 percent of GDP in effective below-the-line support, against 2.7 observed—complemented by an above-the-line envelope of 4.6 to 6.7 percent. What the weights govern is the position on the trilemma, not the strategy: raising the mortality weight from a \$5 to an \$11 million VSL tightens average stringency from 47.1 to 53.8, saves 76 deaths per million, and costs 2.0 percentage points of terminal debt and 3.3 pp-quarters of output; doubling the debt weight (precautionary) delivers the lowest terminal debt (77.3 percent) at the deepest output loss (-30.2 pp-quarters). Second, the welfare gain over the observed policy is large under every preference configuration, ranging from 30 to 46 percent of the observed loss—the dominance of the recomposed strategy does not hinge on the baseline weights.

Table 30: Weighted planner under alternative preferences and VSL anchors.

Scenario	τ_b	λ_d	*Outcomes* Cum. gap (pp-qtrs)	Deaths (per mio.)	Debt (% GDP)	*Policy* Avg. S	Above (% GDP)	Below (% GDP)	DI	Gain (%)
Output-oriented	0.025	25	−8.1	2,013	82.0	48.6	6.7	21.5	0	46.2
Balanced	0.05	50	−15.1	1,987	79.5	46.3	5.8	18.7	0	33.9
Baseline	0.05	75	−16.5	1,938	80.3	48.7	6.0	19.3	0	34.6
Precautionary	0.10	150	−30.2	1,944	77.3	48.1	4.6	15.5	0	30.2
VSL \$5M	0.05	58	−15.5	1,965	79.8	47.1	5.9	18.9	0	34.1
VSL \$9M	0.05	104	−17.8	1,907	81.3	51.9	6.2	20.5	0	35.3
VSL \$11M	0.05	127	−18.8	1,889	81.8	53.8	6.4	21.1	0	35.8

Notes: Each row re-solves the planner problem of Section 5 under the stated weights; all other parameters, caps, and the ramping constraint are held at baseline. Evaluation window Q1.2020–Q4.2022; fiscal totals in percent of 2019 GDP, below-the-line in take-up-adjusted effective terms. The gain is $100\,(J^{\text{obs}} - J^*)/J^{\text{obs}}$, with the observed policy re-evaluated under each row's weights; objective levels are not comparable across rows. The baseline row reproduces the benchmark of Table 9. VSL anchoring of λ_d: Section 5.

The gain is carried by the two dimensions the planner deliberately prioritizes and paid for by the one it deliberately sacrifices: lower mortality contributes 73 percent of the total gain and the improved output path 56 percent, while higher running debt and the larger terminal debt burden offset 29 percent. This is the weighted counterpart of the frontier logic—the same liquidity-led recomposition that the frontier identifies without weights, here pushed beyond the observed outcome levels: unlike the frontier solutions, the weighted optimum is not itself a Pareto improvement, since plausible preferences find it worthwhile to buy additional lives and output at a debt level above the observed one. The frontier establishes that improvement required no sacrifice; the weighted benchmark shows where welfare-maximizing preferences would go beyond it.

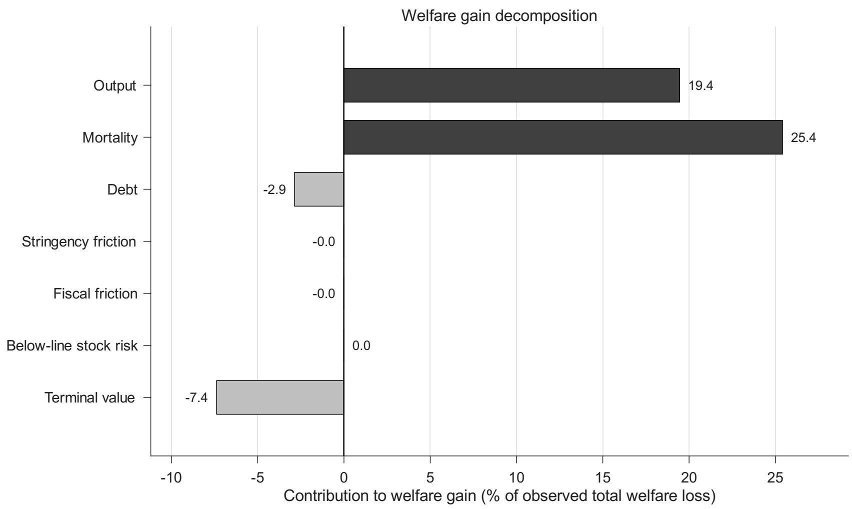


Figure 14: Decomposition of the baseline welfare gain by loss component. *Notes:*

I.2 Sensitivity to the Capacity–Liquidity Complementarity

The complementarity parameter χ is the one planner-specific parameter that shapes the interaction between the two capacity-preservation channels (Section H.3). Table 31 re-solves the three frontier programs and the weighted planner under $\chi \in \{0, 0.25, 0.50\}$, holding all other parameters, caps, and constraints at baseline. Because χ enters the transition system, the model-implied outcomes of the observed policy shift slightly across rows; each row's improvements are therefore computed against its own observed benchmark, keeping the comparison internally consistent.

The dominance result does not depend on the complementarity. Shutting the channel down entirely ($\chi = 0$) leaves the mortality and debt improvements essentially unchanged and reduces the output improvement by five percentage points—from 78.7 to 73.6 percent—while the weighted gain falls from 34.6 to 31.3 percent. The role of χ is compositional rather than quantitative: at $\chi = 0$ the planner abandons above-the-line support altogether and shifts entirely into liquidity instruments, whereas at the baseline value the optimal package combines 6.0 percent of GDP in above-the-line support with 19.3 percent in effective liquidity. Wage-subsidy-type instruments thus earn their place in the optimum through their interaction with the liquidity channel, not through their stand-alone output effect—consistent with the institutional reading of Section 4.

Table 31: Frontier and planner results under alternative complementarity values.

		Frontier improvement (%)			*Planner package (% GDP)*		
χ	$\mathcal{L}_y^{\text{obs}}$	**A** Output	**B** Mortality	**C** Debt	**Above**	**Below**	**Gain** (%)
0	295.7	73.6	22.3	9.0	0.0	22.8	31.3
0.25	290.7	74.7	22.2	8.8	4.4	20.5	32.3
0.50	285.7	78.7	22.3	8.6	6.0	19.3	34.6

Notes: Each row re-solves the three constrained frontier programs (Section 6.2) and the weighted planner under the stated complementarity value; all other parameters at baseline. Because χ enters the transition system, the model-implied observed output loss $\mathcal{L}_y^{\text{obs}}$ varies across rows (observed mortality and terminal debt are essentially unaffected); improvements are computed against each row's own observed benchmark. Above and Below report the weighted planner's cumulative above-the-line and take-up-adjusted below-the-line deployment. The $\chi = 0.50$ row reproduces the baseline of Tables 9 and 10. Demand injection is zero in every solution. Weighted gain as defined in Table 30.

I.3 The Value of Foresight: A No-Anticipation Benchmark

The baseline planner solves under perfect foresight of the realized shock paths and therefore pre-positions policy in Q1.2020, one quarter before the onset shock materializes. To price this foresight, the no-anticipation variant locks the Q1.2020 controls at their observed values—containment and all four fiscal instruments—and lets the planner re-optimize only from Q2.2020 onward, facing the identical shock path, menu, and ramping constraint. The exercise removes exactly one advantage over the governments in the data:

acting before the shock arrives.

Foresight is valuable but far from decisive. The weighted planner's welfare gain falls from 34.6 to 28.8 percent of the observed loss, so 5.9 percentage points—roughly one sixth of the gain—reflect anticipation; the Q2.2020 output trough deepens from -7.1 to -8.9 percentage points, against -11.8 under the observed policy. The dominance result is unaffected: all three frontier improvements remain strictly positive without anticipation—69.4 percent of the discounted output loss, 18.2 percent of excess mortality, and 6.9 percent of terminal debt (5.1 percentage points of GDP), against 78.7, 22.3, and 8.6 percent with foresight—and the composition of the frontier solutions is unchanged: liquidity-led support with no discretionary demand injection.[24]

Two readings follow. Quantitatively, roughly one sixth of what the benchmark planner achieves requires knowing the shock in advance and is therefore unattainable for a government acting in real time; the remaining five sixths come from composition and within-crisis timing alone. Qualitatively, the recomposition result of Section 6 is not an artifact of the pre-positioning: a planner that starts one quarter late, from the observed starting position, still selects the same strategy and still dominates the observed policy on every margin.

I.4 Relaxing the Containment Cap to the Full Index Scale

The baseline menu caps stringency at the in-support 99th percentile ($S \leq 78$), so the planner cannot exceed what OECD governments demonstrably implemented. Relaxing this cap to the full index scale ($S \leq 100$), with all other constraints unchanged, tests whether the results are limited by that choice. They are not. The frontier improvements rise moderately—to 81.3 percent of the discounted output loss, 26.0 percent of excess mortality, and 9.1 percent of terminal debt, against 78.7, 22.3, and 8.6 at baseline—and the weighted planner's gain increases from 34.6 to 36.5 percent, saving an additional 155 deaths per million through harder suppression in the acute waves (average stringency 51.3 against 48.7). The composition is unchanged: demand injection remains exactly zero, and support stays liquidity-led. Full-scale containment thus shifts the attainable frontier outward at the margin but alters neither the dominance result nor the strategy; the in-support cap is a conservative choice, not a binding driver of the findings.

[24] In the no-anticipation frontier solutions, demand injection equals its observed Q1.2020 value of 0.66 percent of GDP by construction: the locked pre-shock quarter carries the observed disbursement. No demand injection is chosen in any later quarter.

J Cross-Country Analysis: Implementation and Diagnostics

J.1 Country-Level Implementation

For each country i, the three frontier programs of Eq. (18) are solved with the common structural parameters of Table 7 and the common policy menu—the in-support caps and the ramping limit $\Delta S_{\max}$ of Section 5, both computed from the pooled OECD panel—while five ingredients are country-specific: the epidemiological innovation path $\varepsilon_{\theta,i}$, backed out from the country's own estimated prevalence series under the common transmission parameters; the onset shock $\varepsilon_{y,i}$ in Q2.2020; the country intercepts $\mu_{y,i}$ and $\mu_{b,i}$ from the panel fixed effects; and the initial debt stock $b_{0,i}$, which also serves as the country's debt target. The constraint thresholds $(\mathcal{L}_y^{\mathrm{obs},i}, D^{\mathrm{obs},i}, b_N^{\mathrm{obs},i})$ are the model-implied outcomes of the country's *unconstrained* observed policy: the menu restricts only the counterfactual, never the evaluation of the country itself. Debt is reported throughout as the change relative to $b_{0,i}$, since initial levels ranging from 7 to 205 percent of GDP are not comparable across countries. Each program is solved by SQP from up to five starting policies, including the country's weighted-planner solution, with feasibility verified ex post against all constraints (outcome thresholds, box bounds, and ramping); solutions failing any ex-post check are discarded (Appendix H.7).

J.2 Invariance of Frontier Distances to Mortality-Level Misfit

The common infection-fatality sequence δ_q over- or under-predicts cumulative deaths for individual countries; the multiplicative scale factor s_i that matches model deaths to the raw data has a median of 0.89 and a range of [0.38, 1.59]. The following result shows that this misfit cannot contaminate the frontier distances.

Under the maintained $\beta_d = 0$, the feasible sets and optimal policies of all three frontier programs are invariant to the country-specific rescaling $\delta_{q,i} = s_i\,\delta_q$, $s_i > 0$. Reported death levels scale by s_i; the percentage mortality improvement and the output and debt improvements are unchanged.

Proof. Deaths are a linear readout of prevalence, $d_{k+1} = \delta_q(k)\,\theta_k$, and the prevalence dynamics do not involve δ_q. With $\beta_d = 0$, neither the output nor the debt transition involves d_k. Hence for any policy path $\boldsymbol{u}$, rescaling δ_q by s_i scales the entire death path, and thus $D(\boldsymbol{u})$, by s_i, while $\mathcal{L}_y(\boldsymbol{u})$ and $b_N(\boldsymbol{u})$ are unchanged. In scenarios A and C the mortality constraint becomes $s_i D(\boldsymbol{u}) \le s_i D^{\mathrm{obs}}$, which defines the same feasible set; objectives are unaffected. In scenario B the objective is scaled by the positive constant s_i, leaving the minimizer unchanged, and the percentage improvement $(s_i D^{\mathrm{obs}} - s_i D^*)/(s_i D^{\mathrm{obs}})$ is invariant. $\square$

J.3 Fit Diagnostics and Ranking Stability

Model fit is summarized by a continuous score see Table 35. The worst-fitting quartile comprises Belgium, Switzerland, Colombia, Spain, Estonia, Iceland, Lithuania, the Netherlands, and Türkiye; the dominant pattern is over-prediction of deaths under the common infection-fatality sequence, which by the proposition above does not affect the frontier distances. Output misfit, which is not covered by the invariance, is material for three countries (COL, EST, TUR). The ranking is insensitive to fit-based exclusion: re-ranking after dropping the five (ten) worst-fitting countries yields a rank correlation of 0.998 (0.997) with the baseline on the surviving sample. Country-specific rescaling of the infection-fatality sequence is thus unnecessary for the ranking; extending the calibration to country-specific mortality profiles remains a natural possible refinement for the reported death levels.

J.4 Demand Injection by Containment State

The exchange-rate gradient reported in Section 6.3 is sharpened by decomposing each country's demand injection according to the containment state in which it becomes effective. Because demand injection acts with a one-quarter lag, the relevant state for support committed in quarter k is stringency in $k+1$, where the multiplier $\alpha_{\mathrm{DI}} + \alpha_{S,\mathrm{DI}}\, S_{k+1}$ applies; the decomposition therefore splits each package's demand injection into the share deployed above and below the threshold $S^* = \alpha_{\mathrm{DI}}/|\alpha_{S,\mathrm{DI}}| \approx 36$ at the effective quarter. Both components vary substantially across countries: the below-threshold share ranges from zero to 0.28 of the total package, with ten countries above 0.05. The model's prediction is that only the above-threshold component should be costly, and the cross-section confirms it exactly: the realized fiscal exchange rate correlates at -0.89 with the above-threshold component—identical to the correlation with the total demand-injection share—and at $+0.03$ with the below-threshold component. Finland illustrates the split: it deployed one of the larger demand-injection shares of the frontier-near group, but over four fifths of it below the threshold, and realized an above-median exchange rate. The decomposition also speaks against a country-characteristics confound: an institutional or informality account of the exchange-rate gradient would burden both components alike, whereas the data assign the entire penalty to the component deployed in the state where the estimated multiplier is negative.

J.5 Special Cases: Off-Menu Policies and the Output Ceiling

Seven countries committed at least one instrument beyond the common caps in at least one quarter: concentrated wage-subsidy commitments in Q1.2020 (Australia, New Zealand), a demand-injection spike in Q1.2021 (Chile), a loan-program spike in Q2.2020 (Denmark), and sustained near-maximal containment (Colombia in Q2.2020; Ireland and the United Kingdom in Q1.2021); Table 32 reports the instrument, quarter, and magnitude. All seven are nonetheless dominated from within the common menu, so their off-menu intensity was dispensable for their realized outcomes. Italy is the converse case: sixteen countries exceeded the ramping limit $\Delta S_{\max}$ in Q2.2020, but only for Italy is this speed pivotal—without it, the mortality threshold is unattainable from the menu and the frontier programs are infeasible. Finally, seven countries reach the output ceiling (ImpA ≥ 99 percent): for Australia, Denmark, Israel, Japan, Korea, New Zealand, and the United States, the frontier eliminates the policy-amenable output loss entirely, so their output distances are bounded by the transition technology rather than by fiscal space, and their positions in the ranking should be read accordingly.

Table 32: Observed policies beyond the common menu: instrument, quarter, and magnitude.

	Instrument	**Quarter**	**Exceedance**
Australia	Above-the-line (wage subsidy)	Q1.2020	3.53
New Zealand	Above-the-line (wage subsidy)	Q1.2020	2.04
Chile	Demand injection	Q1.2021	4.65
Denmark	Loans	Q2.2020	0.67
Colombia	Stringency	Q2.2020	7.61
Ireland	Stringency	Q1.2021	6.27
United Kingdom	Stringency	Q1.2021	1.61

Notes: Largest exceedance of the common in-support caps per country: fiscal instruments in percent of 2019 GDP above the cap (effective, take-up-adjusted terms), stringency in index points above the cap of 78.0. Fiscal exceedances reflect commitments in the quarter of announcement. All seven countries are nonetheless dominated from within the common menu (Section 6.3).

J.6 Full Country Tables

Table 33: Distance to the country-specific frontier: all countries.

	Frontier distance	*Ranking dimensions*		*Context*
	(avg. rank, 1–37)	**Output impr.** (% of own $\mathcal{L}_y$)	**Debt impr.** (pp of GDP)	**Deaths impr.** (% of own deaths)
Mexico	1.0	35.3	2.2	16.6
Sweden	4.0	58.5	3.7	21.3
Portugal	5.0	48.7	5.0	16.2
Finland	6.0	62.3	3.7	13.3
France	7.0	52.1	5.6	13.3
Norway	7.5	77.9	3.0	12.5
Iceland^{+}	9.5	60.8	5.6	28.5
Belgium^{+}	11.0	61.3	5.6	22.0
Luxembourg	11.5	83.9	4.0	22.0
Switzerland^{+}	12.0	82.1	4.5	23.5
Colombia$^{\#+}$	12.5	80.7	5.1	19.9
Netherlands^{+}	13.0	69.2	5.7	17.1
Hungary	14.0	80.5	5.6	17.0
Spain^{+}	14.5	59.0	7.7	22.2
Austria	15.0	67.9	6.3	16.4
Costa Rica	16.0	89.2	5.3	21.9
Germany	16.5	76.7	6.6	15.6
United Kingdom$^{\#}$	17.0	59.7	8.2	21.9
Türkiye^{+}	17.5	96.0	4.0	21.7
Czechia	18.5	80.3	7.6	17.6
Latvia	19.0	89.1	5.9	18.9
Slovak Republic	19.0	88.9	6.0	18.1
Estonia^{+}	21.0	90.6	6.1	20.7
Lithuania^{+}	22.0	94.8	5.9	20.2
Ireland$^{\#}$	23.0	90.9	7.0	16.3
Slovenia	25.0	87.8	8.6	15.0
Greece	28.5	93.2	8.6	14.2
Korea*	28.5	99.9	7.8	24.0
Poland	28.5	96.3	7.9	19.6
Canada	29.0	91.8	9.2	27.0
Israel*	30.5	100.0	7.8	27.5
Chile$^{\#}$	31.5	92.9	11.7	27.4
Denmark$^{*\#}$	32.0	100.0	8.3	20.6
Australia$^{*\#}$	33.5	100.0	8.9	28.6
Japan*	33.5	99.6	10.8	21.2
New Zealand$^{*\#}$	34.5	100.0	10.2	38.7
United States*	35.0	100.0	10.6	34.6
Italy†	—	—	—	—

Notes: Countries in ascending order of frontier distance, the average of the within-sample ranks in the output and debt improvements (37 ranked countries; lower = closer to the own frontier). Each improvement column reports a separate frontier program: the output and debt improvements hold the country's mortality (and the respective other dimension) at its observed level; the deaths improvement is from the program that instead minimizes mortality at the observed output loss and debt—it is contextual and does not enter the distance. Debt improvements in absolute pp of GDP; the distance uses within-column ranks and is invariant to any monotone choice of units. * Output ceiling reached (ImpA $\geq$ 99 percent). $^{\#}$ Observed policy committed at least one instrument beyond the common menu (Table 32). $^{+}$ Worst-fitting quartile of the continuous fit score (Appendix J.3); by the invariance result of Appendix J.2, mortality-level misfit does not affect the reported distances. † Italy: observed policy not replicable within the common menu (first-wave ramp-up exceeds $\Delta S_{\max}$); the output and debt programs are infeasible and the mortality program cannot improve on the observed outcome, so no distance is defined.

Table 34: Frontier ranking and realized outcomes in the raw data.

	Frontier distance (avg. rank, 1–37)	**Cum. output gap** (pp-quarters, raw)	**Excess deaths** (per million, raw)	**Debt change** (pp of GDP, raw)
Mexico	1.0	−101.2	2,630	+0.9
Sweden	4.0	−29.6	2,169	+1.9
Portugal	5.0	−95.0	2,491	−2.7
Finland	6.0	−57.3	1,418	+6.2
France	7.0	−68.5	2,407	+6.8
Norway	7.5	−17.6	869	+4.0
Iceland^{+}	9.5	−105.1	766	+10.7
Belgium^{+}	11.0	−33.0	2,949	+3.7
Luxembourg	11.5	−13.4	1,925	+7.5
Switzerland^{+}	12.0	−17.6	1,730	+2.1
Colombia$^{\#+}$	12.5	−27.2	2,899	+16.1
Netherlands^{+}	13.0	−36.2	1,391	+5.5
Hungary	14.0	−82.2	5,002	+6.9
Spain^{+}	14.5	−118.5	2,746	+11.4
Austria	15.0	−69.6	2,435	+10.9
Costa Rica	16.0	−52.9	1,784	+14.0
Germany	16.5	−63.5	1,962	+5.8
United Kingdom$^{\#}$	17.0	−88.8	3,320	+11.1
Türkiye^{+}	17.5	+22.3	1,226	−0.0
Czechia	18.5	−104.7	3,988	+12.0
Latvia	19.0	−50.7	3,229	+13.6
Slovak Republic	19.0	−55.0	3,810	+11.2
Estonia^{+}	21.0	−65.9	2,150	+10.2
Lithuania^{+}	22.0	−39.5	3,376	+5.0
Ireland$^{\#}$	23.0	+78.9	1,668	−1.9
Slovenia	25.0	−78.3	3,347	+10.1
Greece	28.5	−54.9	3,240	+7.8
Korea*	28.5	−36.2	620	+5.5
Poland	28.5	−64.8	3,186	+6.1
Canada	29.0	−51.9	1,304	+13.9
Israel*	30.5	−25.1	1,300	+8.7
Chile$^{\#}$	31.5	−37.5	3,272	+2.5
Denmark$^{*\#}$	32.0	−28.5	1,344	−3.1
Australia$^{*\#}$	33.5	−26.1	661	+3.7
Japan*	33.5	−46.6	456	+7.9
New Zealand$^{*\#}$	34.5	−45.1	460	+21.2
United States*	35.0	−26.3	3,307	+19.3
Italy†	—	−50.6	3,379	+6.1

Notes: Countries in ascending order of frontier distance (Table 33); outcomes are the *raw* realized data over Q1.2020–Q4.2022—cumulative output gap from the estimation panel, cumulative excess mortality from the weekly mortality data, and the change in the debt-to-GDP ratio from reported levels—not the model-implied values on which the frontier thresholds are based (closure gaps are documented in Table 35). The table is not an outcome ranking: frontier distance measures conversion efficiency at the country's own realized position, so frontier-near countries need not have favorable outcomes (Mexico) and frontier-far countries need not have unfavorable ones (Denmark). What the raw data do display is the efficiency pattern itself: Sweden and the United States realized nearly identical output losses, but Sweden with fewer excess deaths and one tenth of the debt increase; Norway and Switzerland reached small losses at low fiscal cost, while the furthest group paid the largest debt increases of the panel (United States +19.3, New Zealand +21.2 pp) for moderate output outcomes. Türkiye and Ireland recorded positive cumulative output gaps (growth over the window); flags as in Table 33.

Table 35: Observed policy composition, realized fiscal exchange rate, and fit diagnostics.

	Observed composition			*Counterfactuals*		*Diagnostics*
	Liquidity share	**DI share**	**Front-loading**	**Fiscal exch. rate**	**Deaths avoided /M**	**Fit score**
Mexico	0.12	0.62	1.00	−0.14	9,027	1.7
Sweden	0.34	0.13	0.96	3.12	7,065	0.8
Portugal	0.40	0.19	0.82	2.81	6,342	1.2
Finland	0.32	0.33	0.84	2.95	1,018	1.1
France	0.33	0.10	0.61	3.74	10,289	0.8
Norway	0.51	0.03	0.83	3.74	1,183	1.7
Iceland^{+}	0.11	0.38	0.46	1.71	1,834	4.4
Belgium^{+}	0.28	0.17	0.88	2.94	17,181	3.0
Luxembourg	0.24	0.02	0.69	2.93	3,594	0.6
Switzerland^{+}	0.24	0.03	0.80	3.57	7,361	4.4
Colombia$^{\#+}$	0.19	0.56	0.73	−0.02	4,906	3.0
Netherlands^{+}	0.14	0.15	0.72	2.17	12,006	5.4
Hungary	0.26	0.24	0.53	2.29	5,453	1.6
Spain^{+}	0.46	0.18	0.74	2.82	24,359	4.5
Austria	0.07	0.13	0.66	2.42	4,826	1.3
Costa Rica	0.66	0.27	0.62	1.88	1,519	2.1
Germany	0.39	0.16	0.82	3.62	3,443	1.0
United Kingdom$^{\#}$	0.24	0.13	0.84	2.92	12,511	1.2
Türkiye^{+}	0.41	0.10	0.86	4.09	2,664	2.7
Czechia	0.34	0.22	0.66	2.96	6,377	1.2
Latvia	0.31	0.03	0.75	3.65	3,118	1.1
Slovak Republic	0.22	0.02	0.63	2.72	5,319	1.1
Estonia^{+}	0.26	0.04	0.86	3.52	1,629	2.6
Lithuania^{+}	0.12	0.38	0.78	1.40	8,014	3.2
Ireland$^{\#}$	0.10	0.42	0.62	0.30	5,821	0.7
Slovenia	0.11	0.05	0.64	2.86	5,722	1.3
Greece	0.28	0.18	0.58	2.08	2,913	1.8
Korea*	0.43	0.29	0.81	2.40	348	0.9
Poland	0.10	0.20	0.77	1.77	5,368	1.7
Canada	0.10	0.50	0.74	0.90	2,020	0.8
Israel*	0.12	0.58	0.92	−0.50	1,858	1.2
Chile$^{\#}$	0.18	0.67	0.40	−0.18	4,250	2.6
Denmark$^{*\#}$	0.46	0.25	0.97	2.81	923	2.1
Australia$^{*\#}$	0.05	0.27	0.71	1.76	796	1.0
Japan*	0.52	0.26	0.70	3.30	230	2.6
New Zealand$^{*\#}$	0.08	0.30	0.83	2.32	242	0.5
United States*	0.14	0.51	0.54	1.12	6,036	0.3
Italy†	0.41	0.06	0.74	3.88	20,994	2.3

Notes: Countries in the order of Table 33 (ascending frontier distance). Liquidity and DI shares of the total observed fiscal package (take-up-adjusted effective terms); front-loading is the share committed in Q1–Q3.2020. The fiscal exchange rate is the cumulative output gain per pp of fiscally induced debt against a counterfactual that keeps the observed containment path and sets fiscal instruments to zero; deaths avoided (per million) are computed against a full-zero counterfactual (no containment, no fiscal support) and thus reflect the entire policy package, containment included. Fit score as defined in Appendix J.3 (higher = worse fit); by the invariance result of Appendix J.2, mortality-level misfit does not affect the frontier distances. * output ceiling; $^{\#}$ instrument committed beyond the common menu; $^{+}$ worst-fitting quartile; † Italy: not ranked (see Table 33); its composition, exchange rate, and counterfactuals remain well-defined.

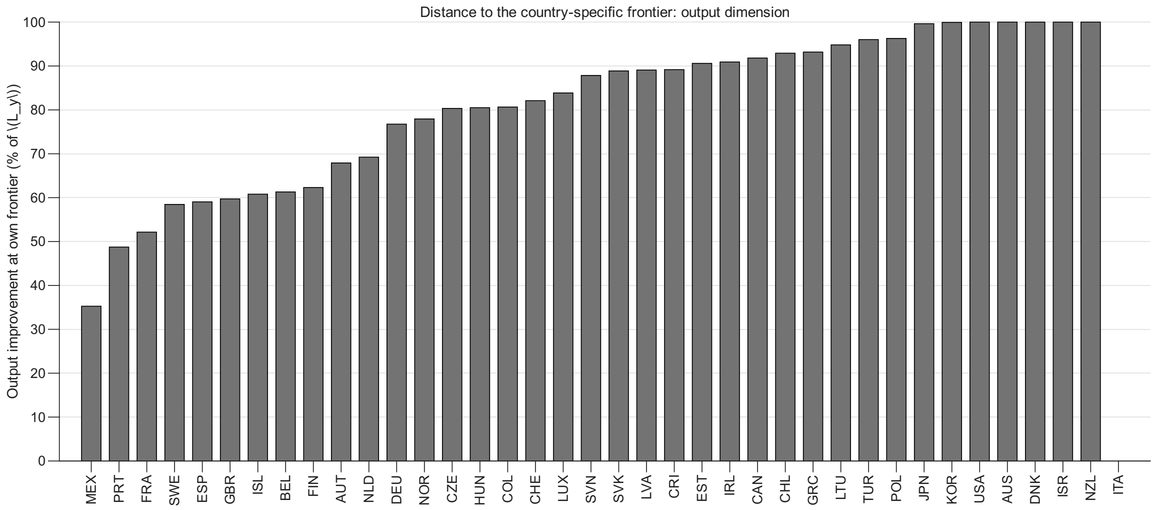


Figure 15: Output improvement at the country-specific frontier, by country. *Notes:* Improvement in the discounted output loss $\mathcal{L}_y$ at the country's own realized mortality and terminal debt, in percent, sorted ascending. The cluster at 100 percent marks the output-ceiling countries, for which the frontier eliminates the policy-amenable output loss entirely within the estimated linear transition technology. Italy omitted (Appendix J.5).

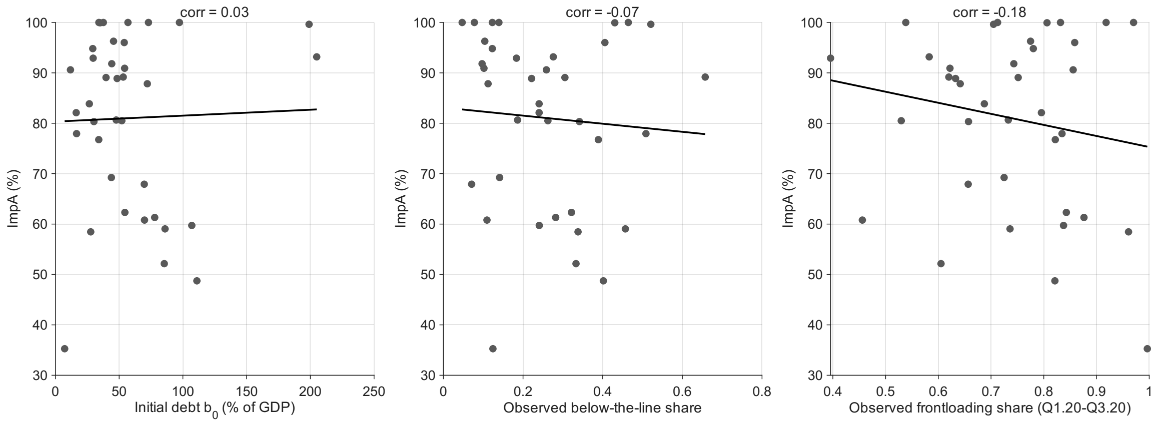


Figure 16: Frontier distance and observed policy characteristics. *Notes:* Output improvement at the own frontier against initial debt, the observed liquidity share, and the observed front-loading share; solid lines are linear fits. The linear cross-sectional associations are weak—attenuated by the output-ceiling countries and the conditional nature of the distance measure—which is why the main text reports the composition–distance link as a group pattern (Section 6.3) rather than a regression.

J.7 Full Frontier Plot

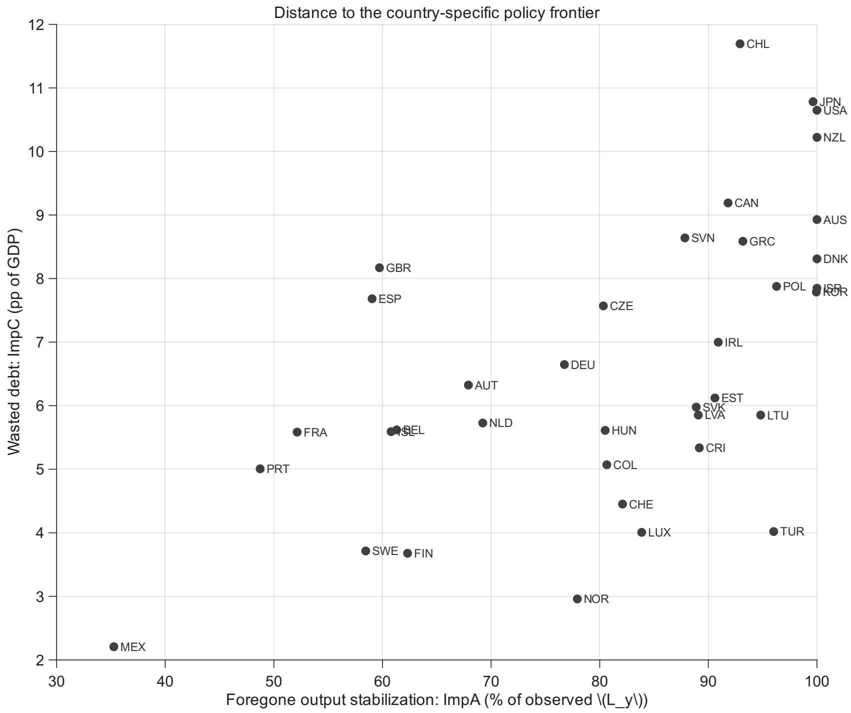


Figure 17: Distance to the country-specific policy frontier.